

MX17 Pair Simulation Analysis

X17 signal (5,000,686 events)

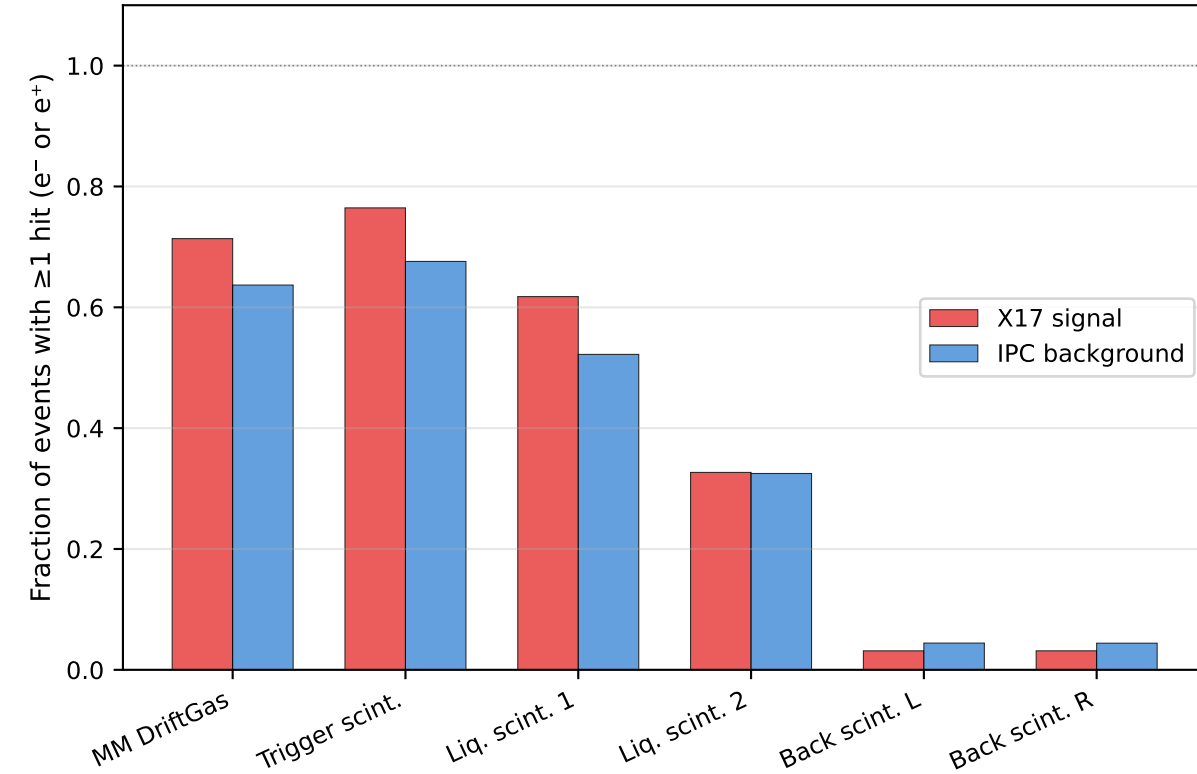
MM Double (both tracks)	:	19.6%
LiqScint_1 reached	:	61.8%
LiqScint_2 reached	:	32.7%
Single trigger (all evt)	:	61.8%
Double trigger (all evt)	:	12.4%
Single MM Double	:	96.8%
Double MM Double	:	61.6%

IPC background (4,999,314 events)

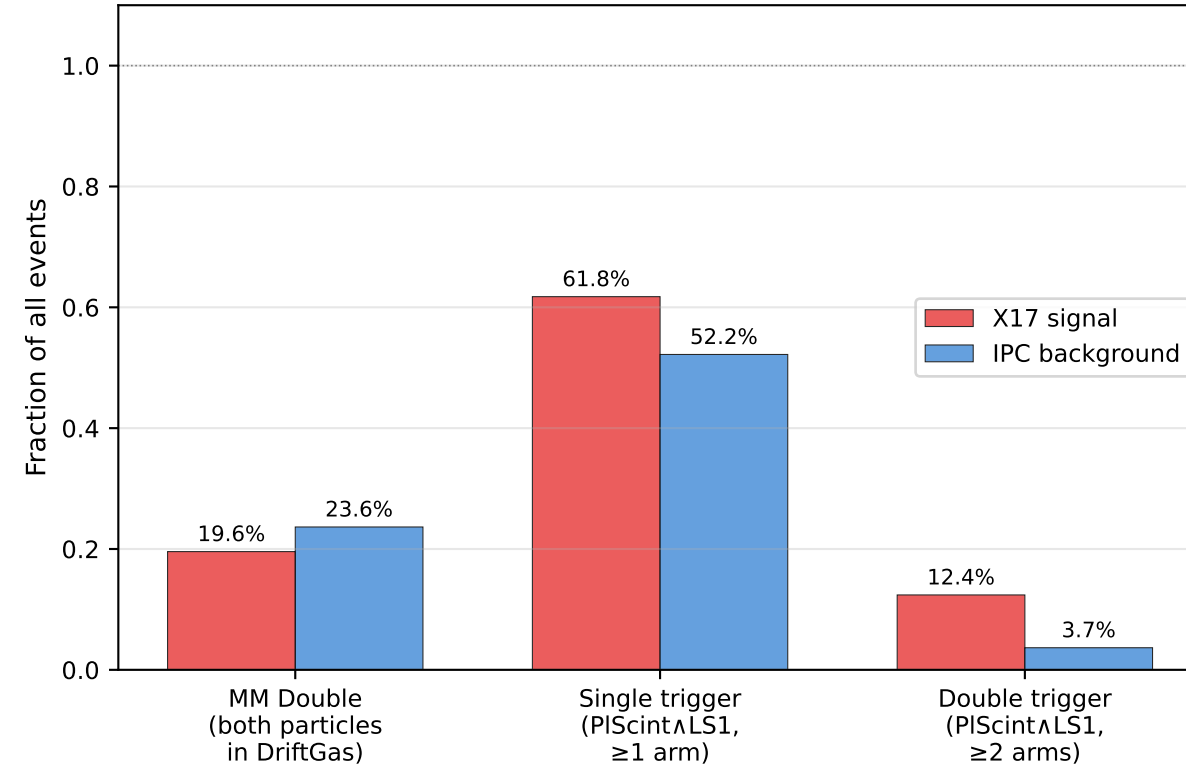
MM Double (both tracks)	:	23.6%
LiqScint_1 reached	:	52.2%
LiqScint_2 reached	:	32.5%
Single trigger (all evt)	:	52.2%
Double trigger (all evt)	:	3.7%
Single MM Double	:	96.0%
Double MM Double	:	15.1%

Detection Acceptance

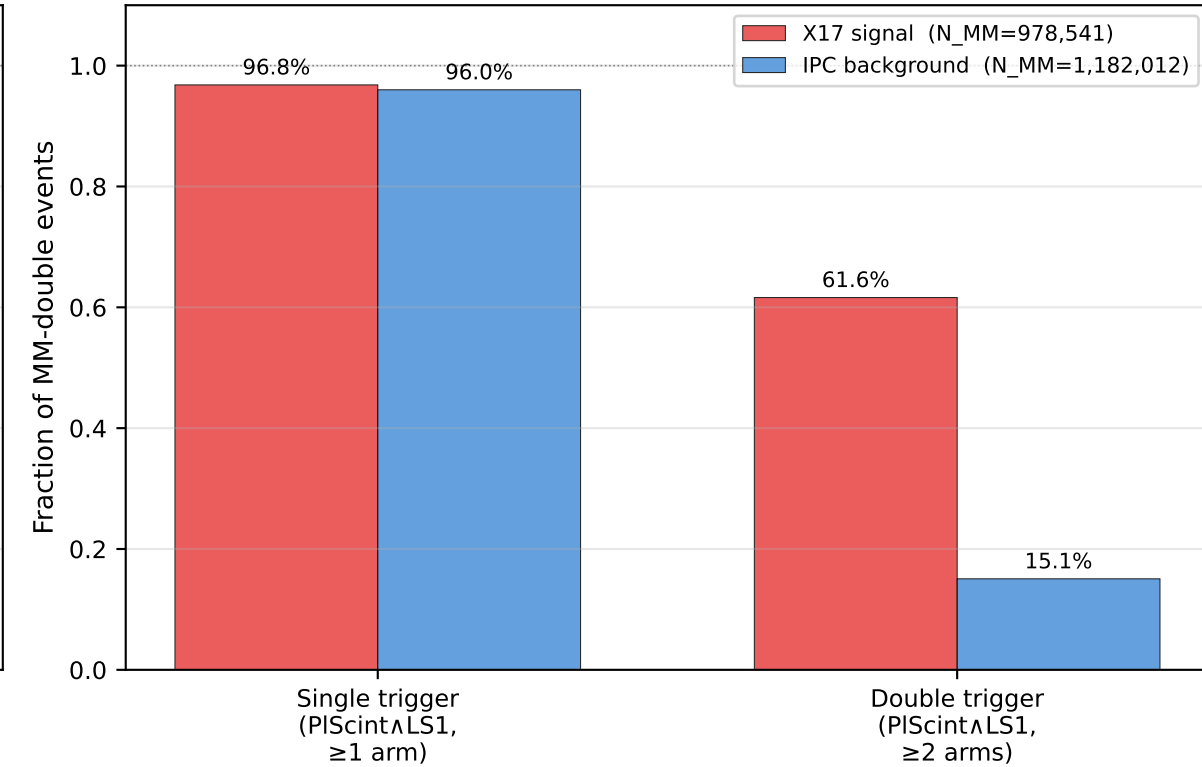
Layer acceptance



Trigger rates (fraction of generated events)



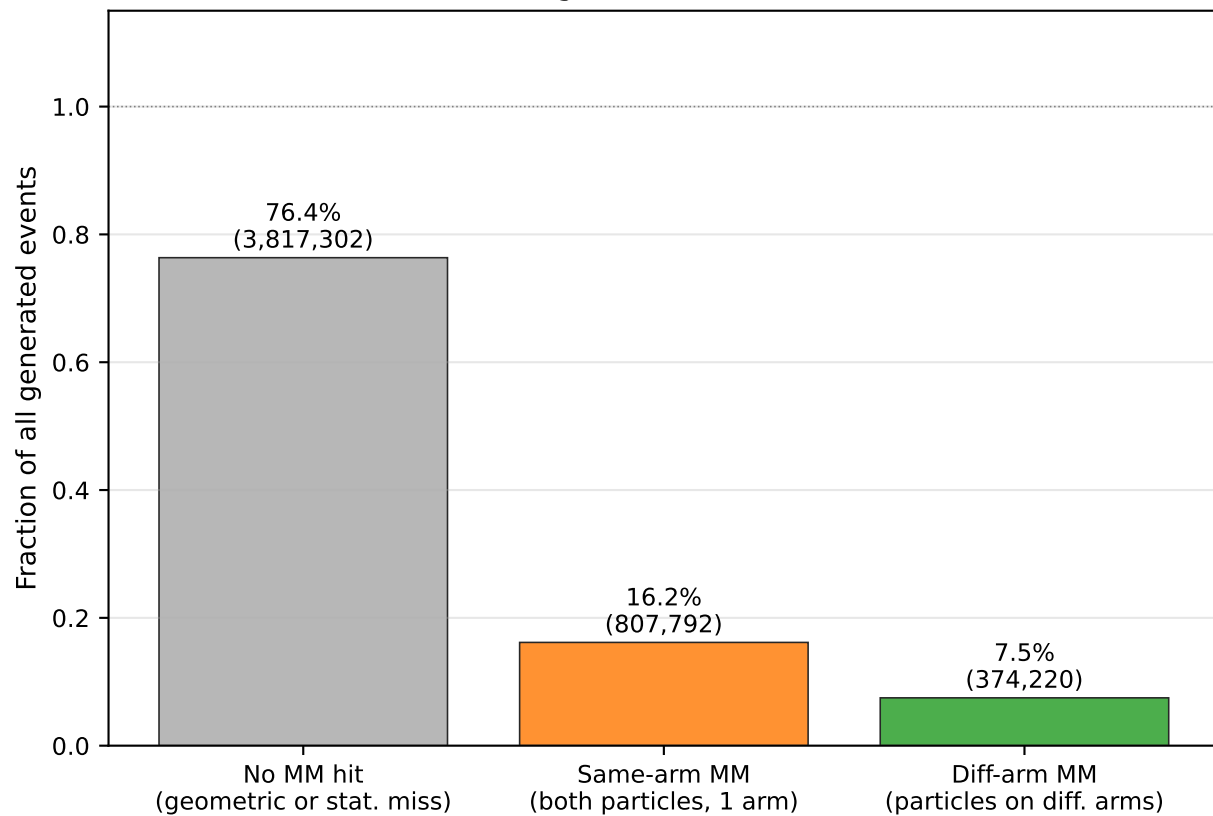
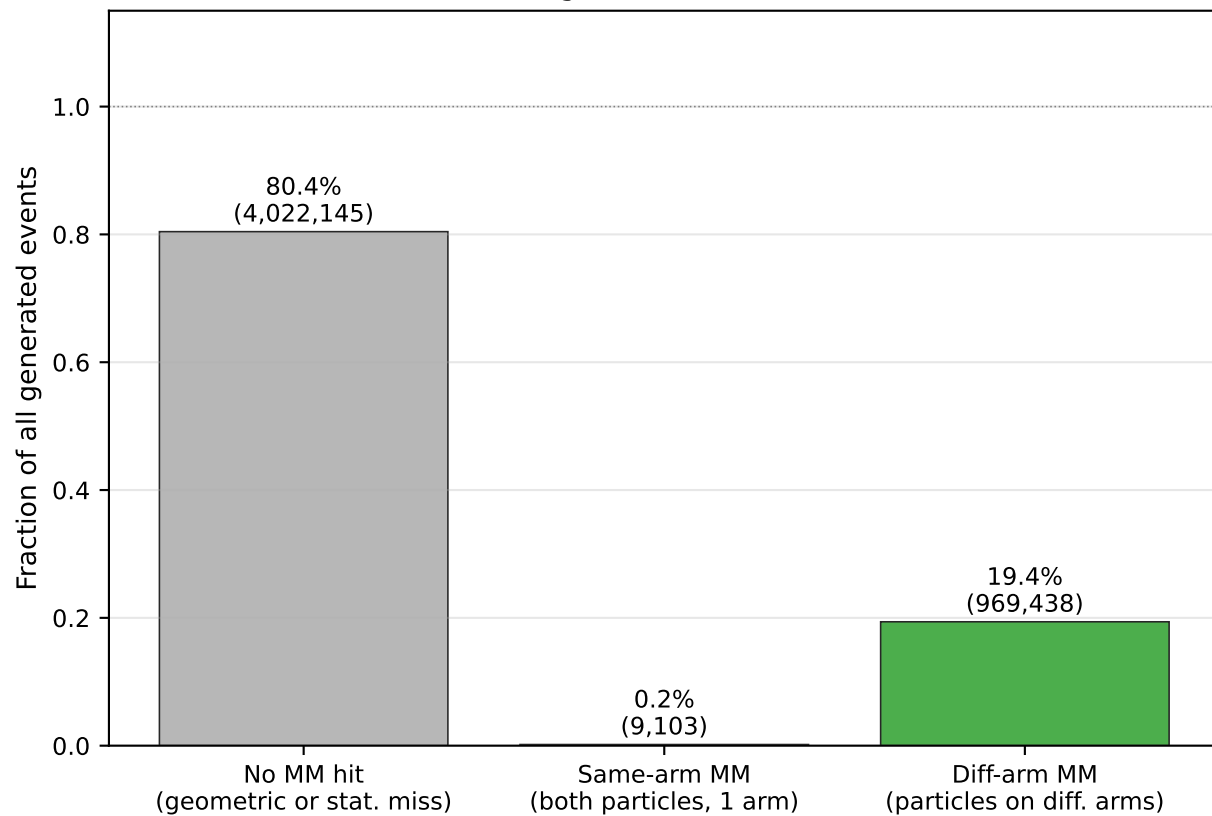
Scintillator trigger efficiency given MM Double (both tracks present)



MM Detector Topology — Event Classification

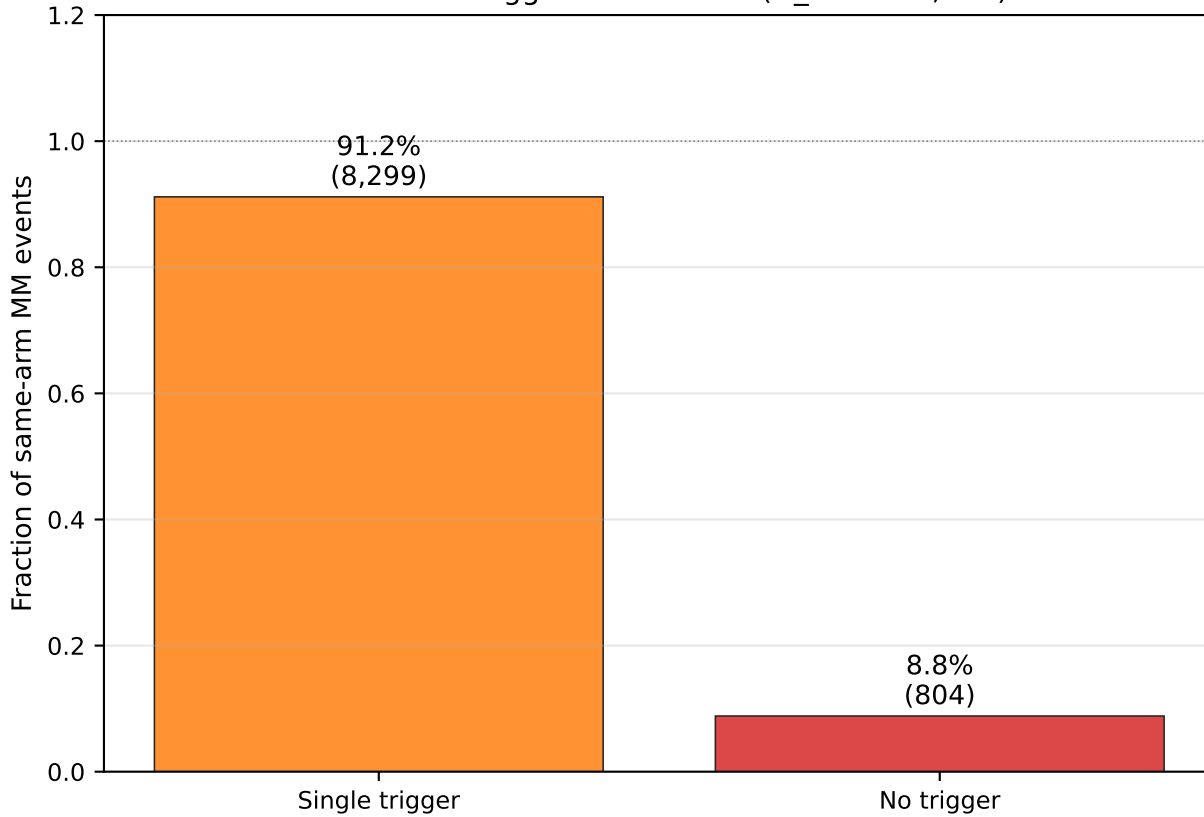
X17 signal (N=5,000,686)

IPC background (N=4,999,314)

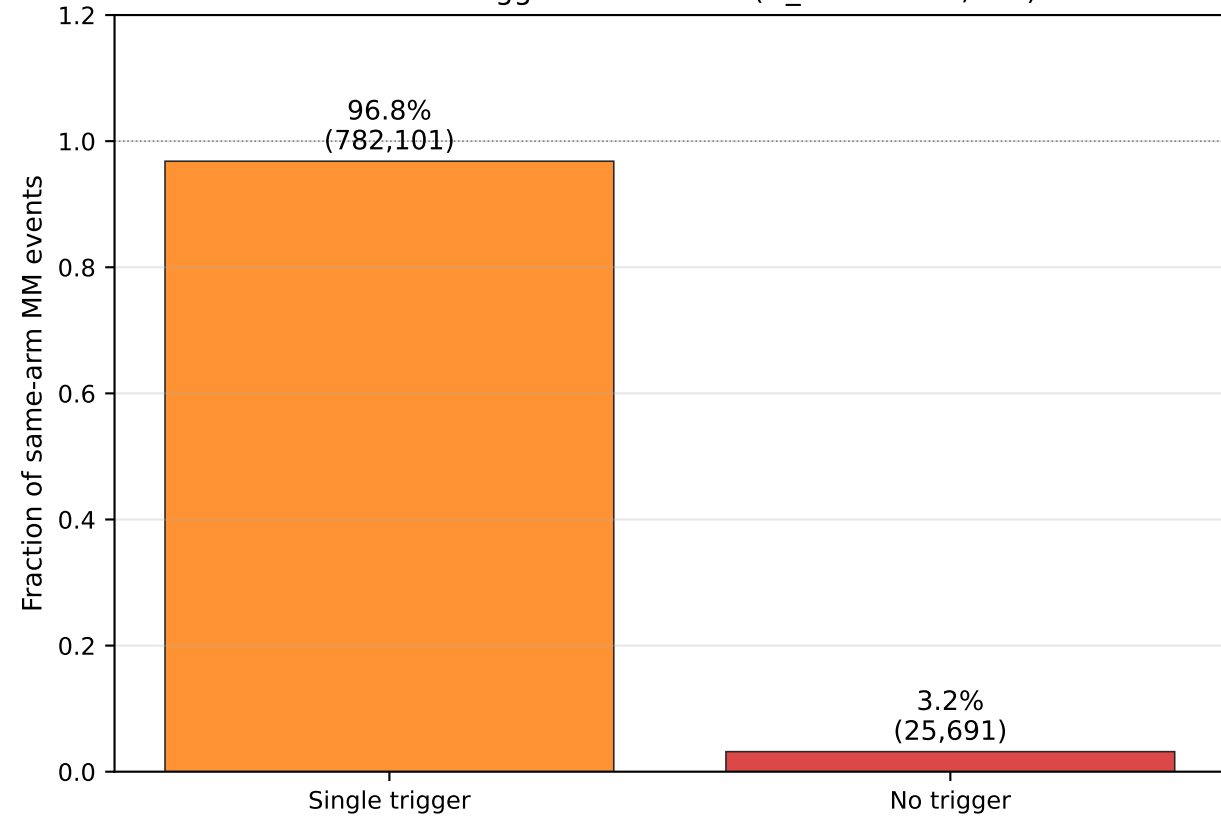


Same-arm MM Events: Trigger Efficiency
(can only fire Singles — if not triggered, event is lost)

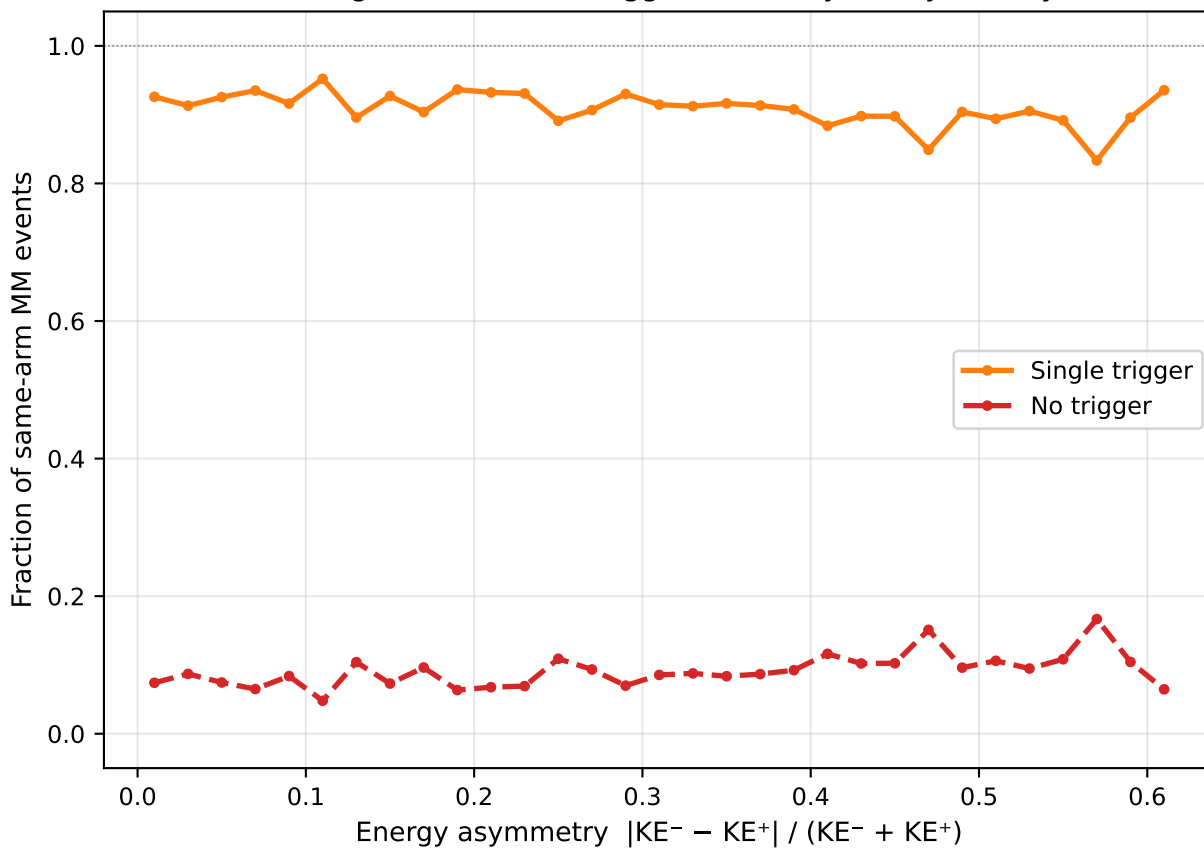
X17 signal
Same-arm trigger breakdown (N_same=9,103)



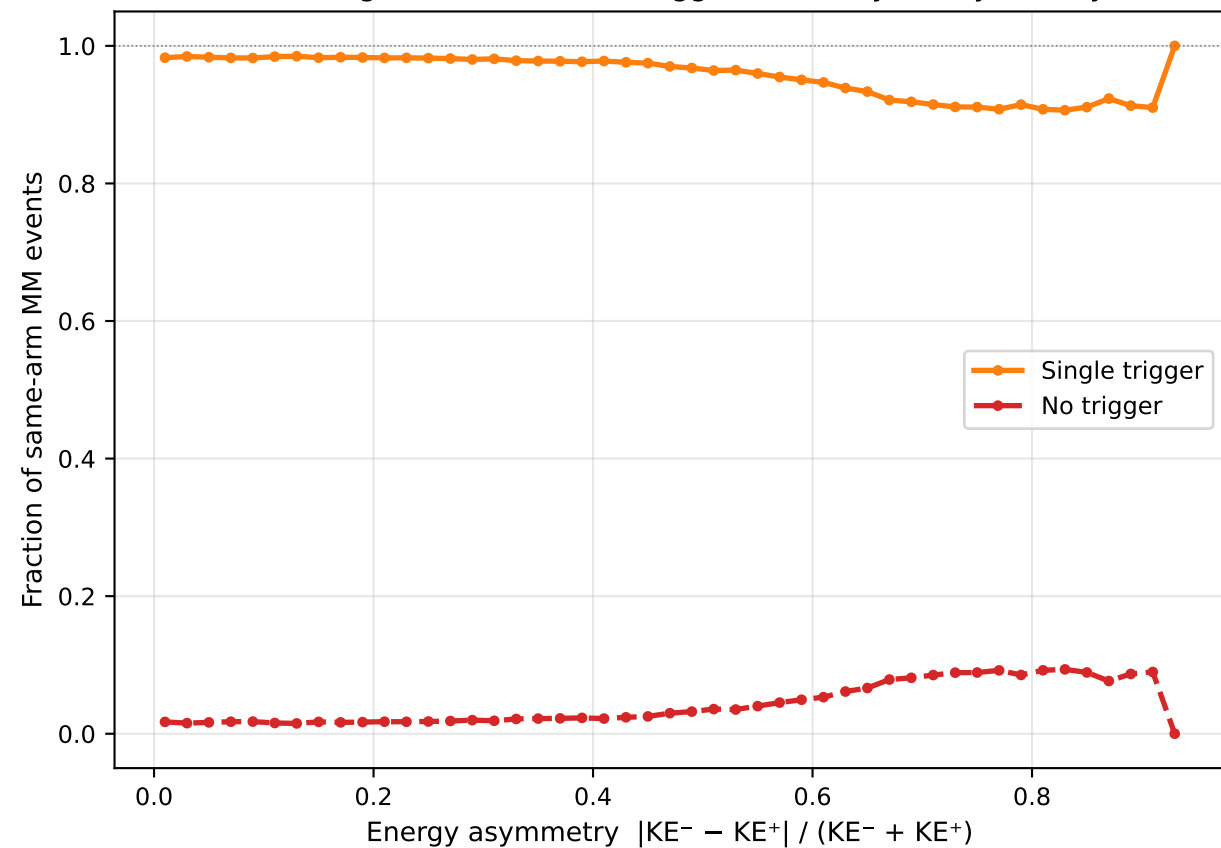
IPC background
Same-arm trigger breakdown (N_same=807,792)



X17 signal: same-arm trigger efficiency vs asymmetry

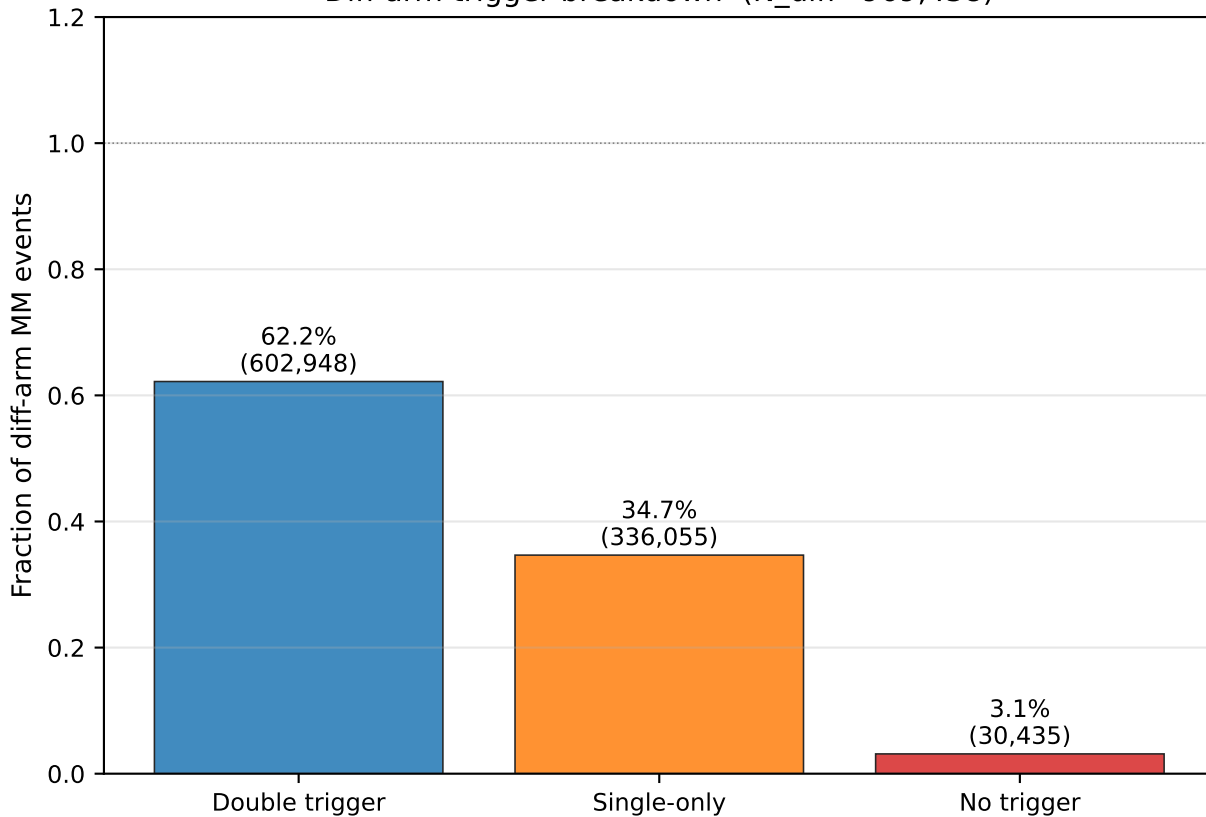


IPC background: same-arm trigger efficiency vs asymmetry

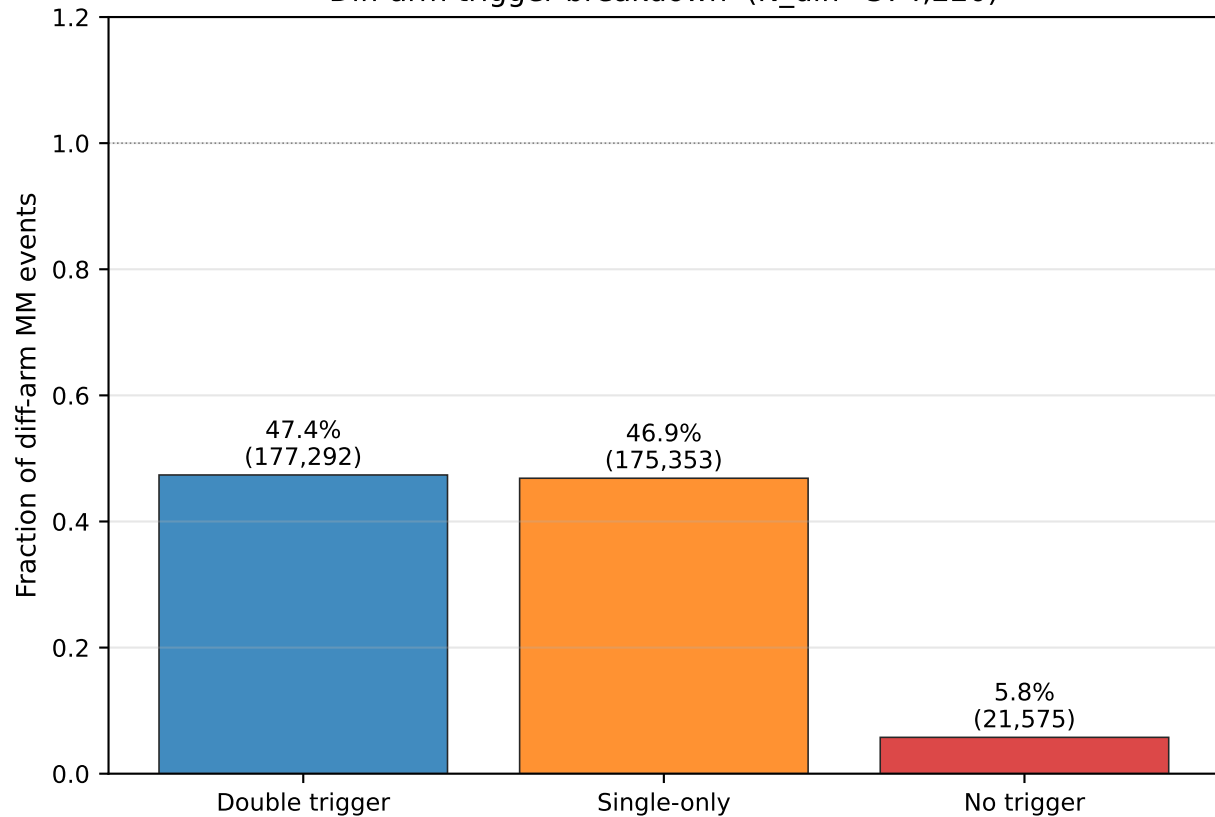


Diff-arm MM Events: Trigger Breakdown
(ideal topology — can fire Singles or Doubles)

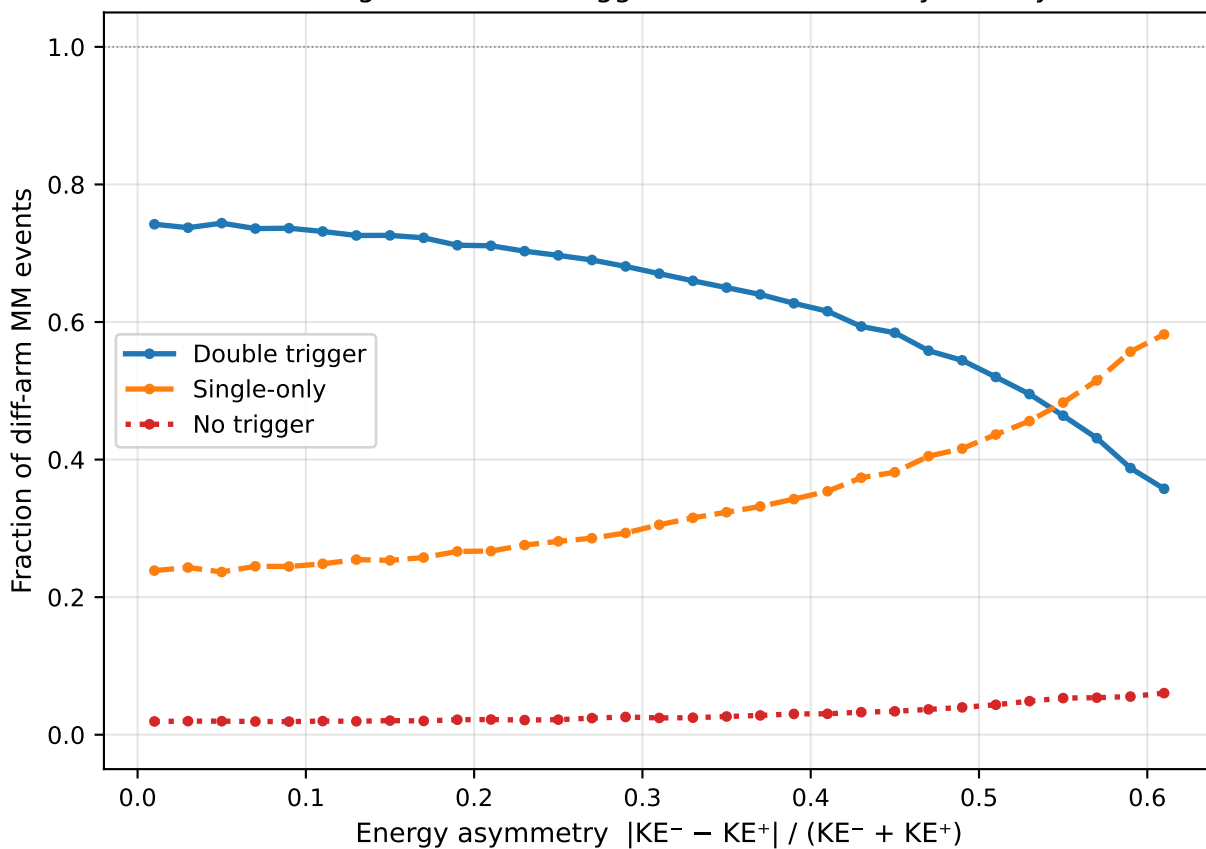
X17 signal
Diff-arm trigger breakdown (N_diff=969,438)



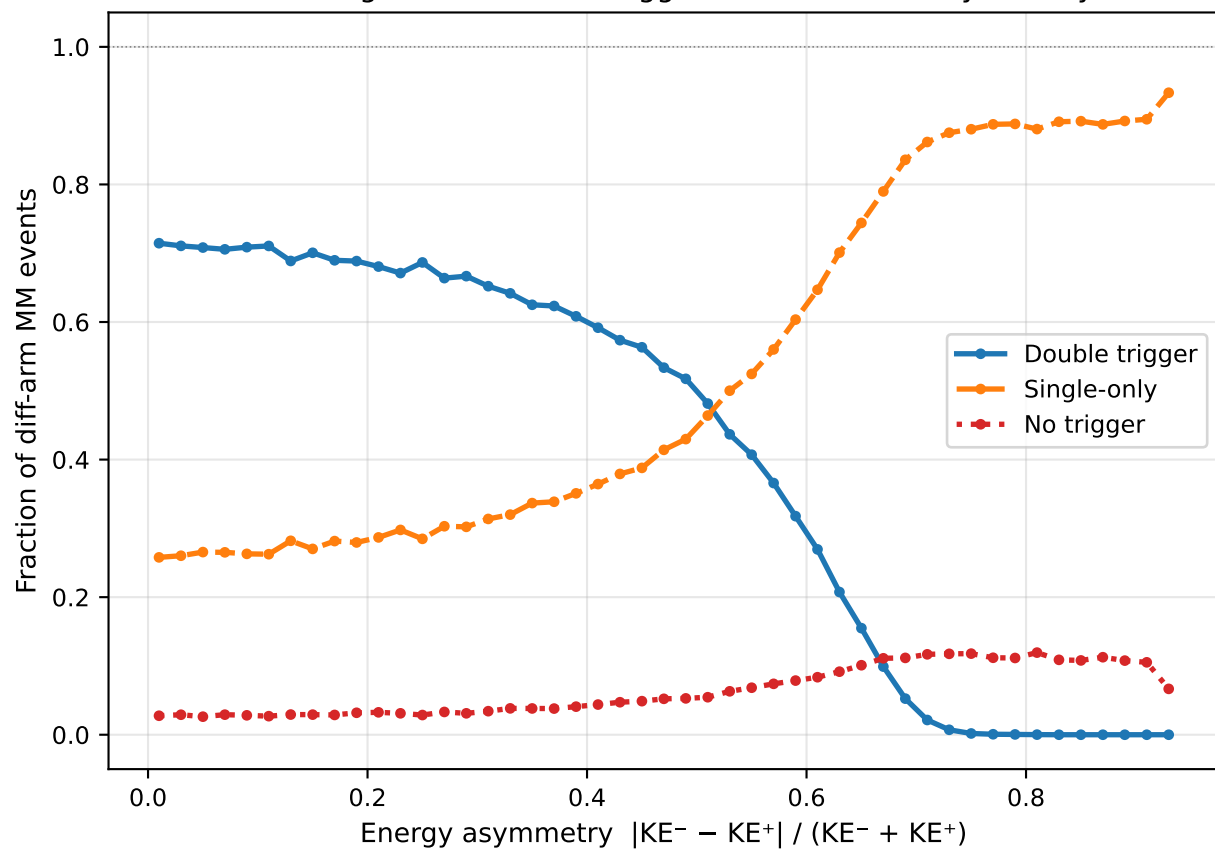
IPC background
Diff-arm trigger breakdown (N_diff=374,220)



X17 signal: diff-arm trigger breakdown vs asymmetry

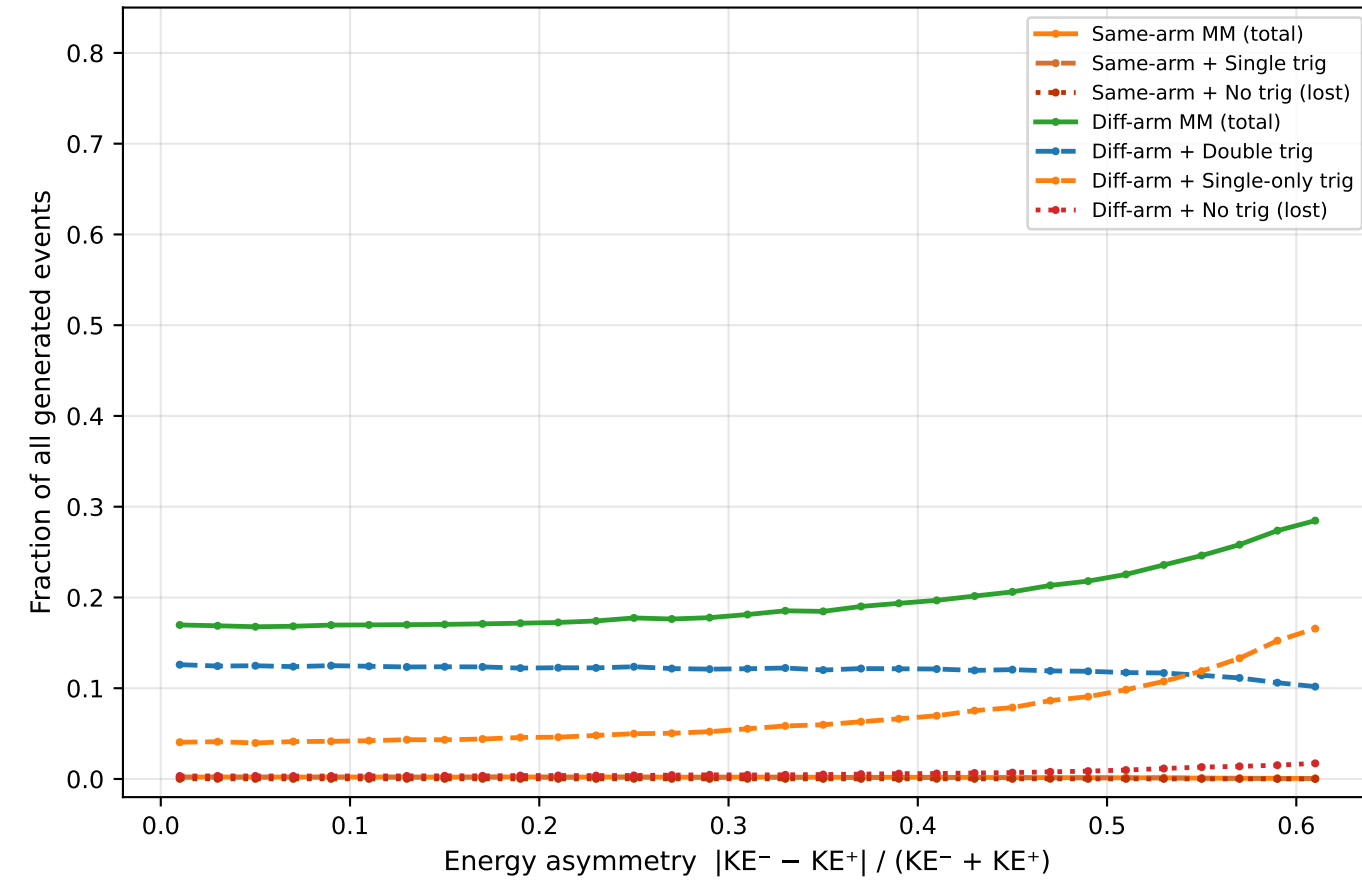


IPC background: diff-arm trigger breakdown vs asymmetry

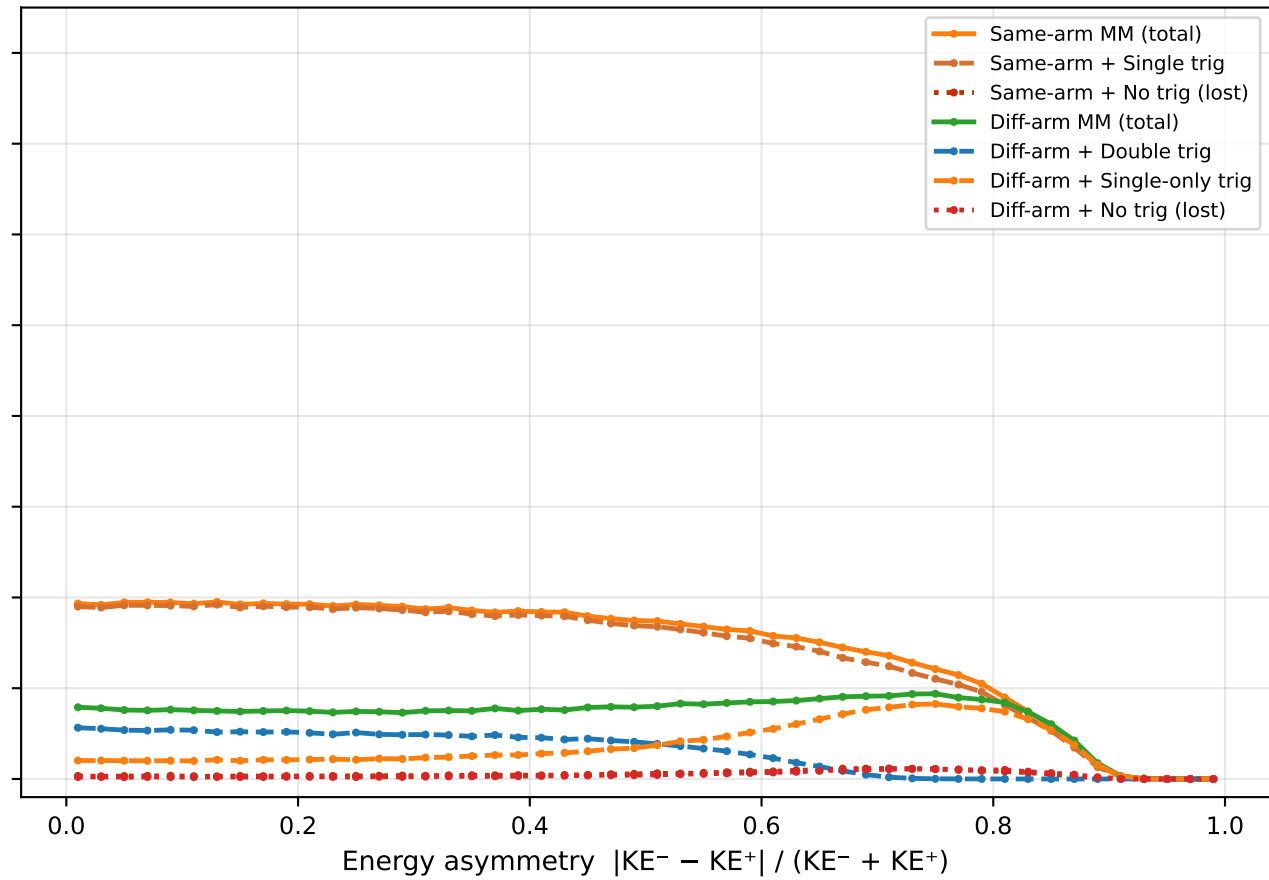


MM Topology Classes vs Energy Asymmetry (all classes normalised to total generated events)

X17 signal

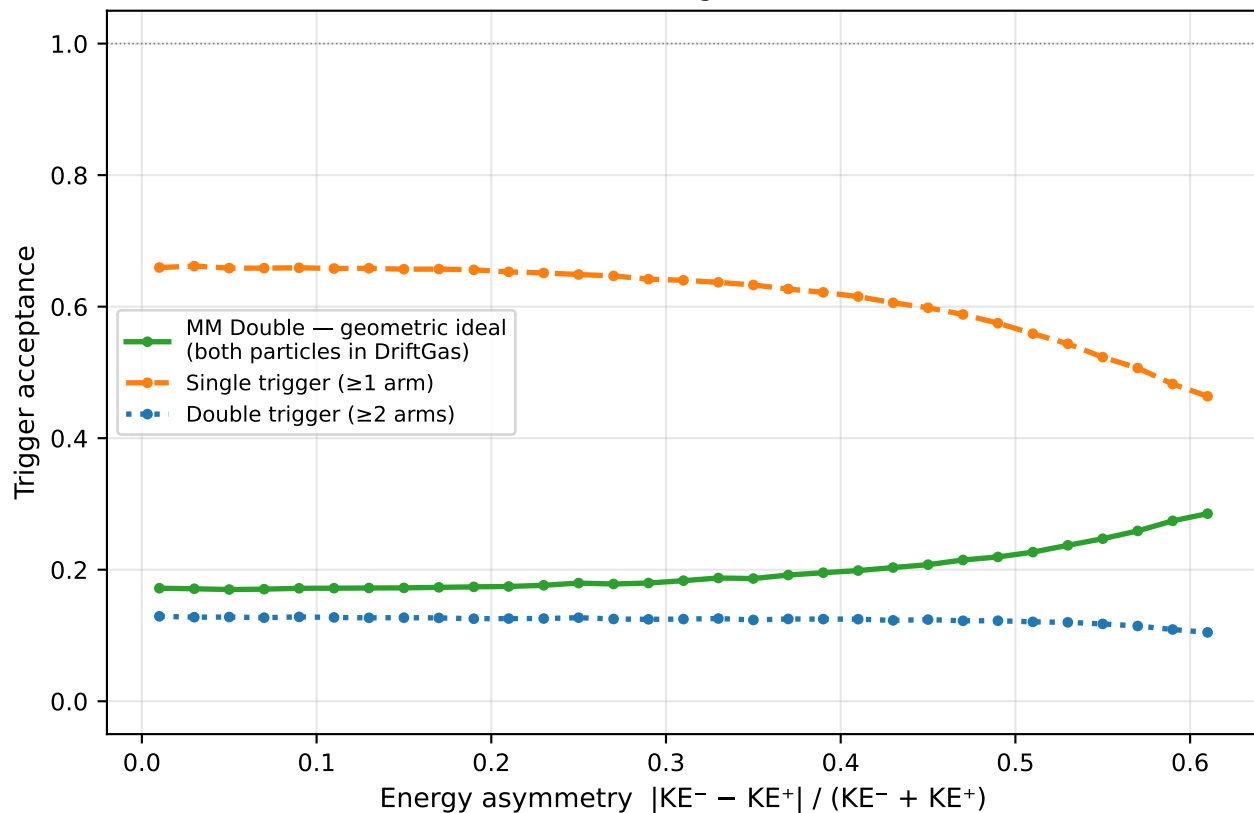


IPC background

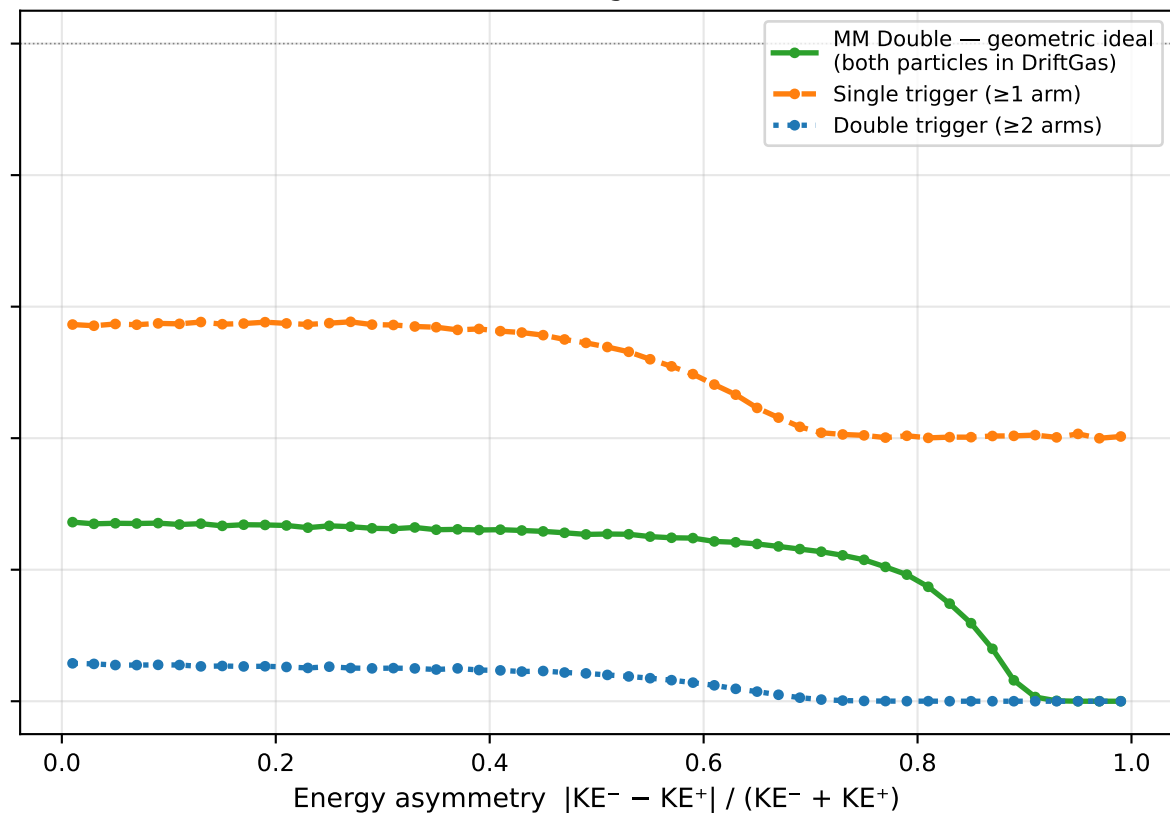


Trigger acceptance vs energy asymmetry

X17 signal



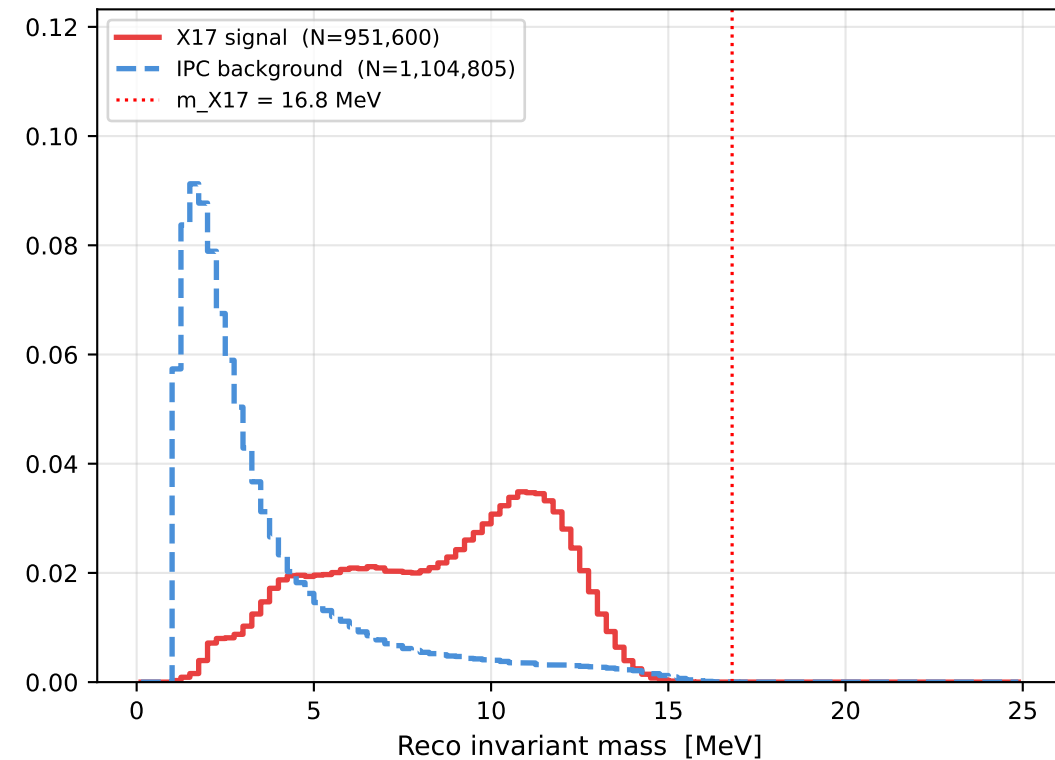
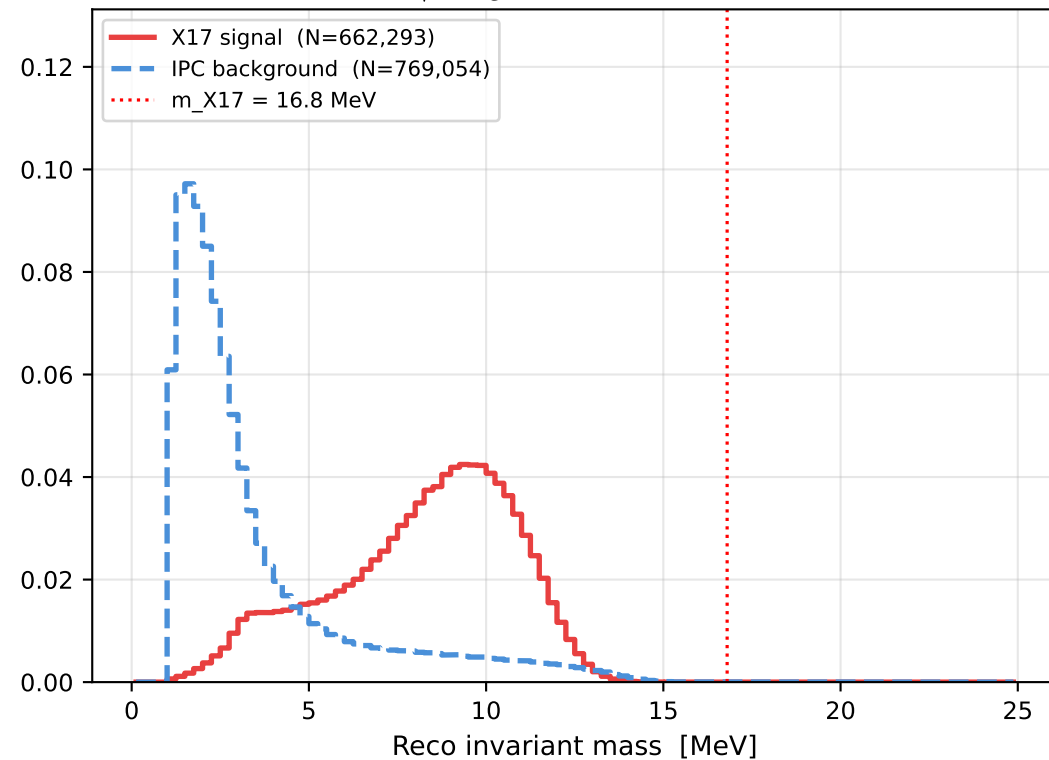
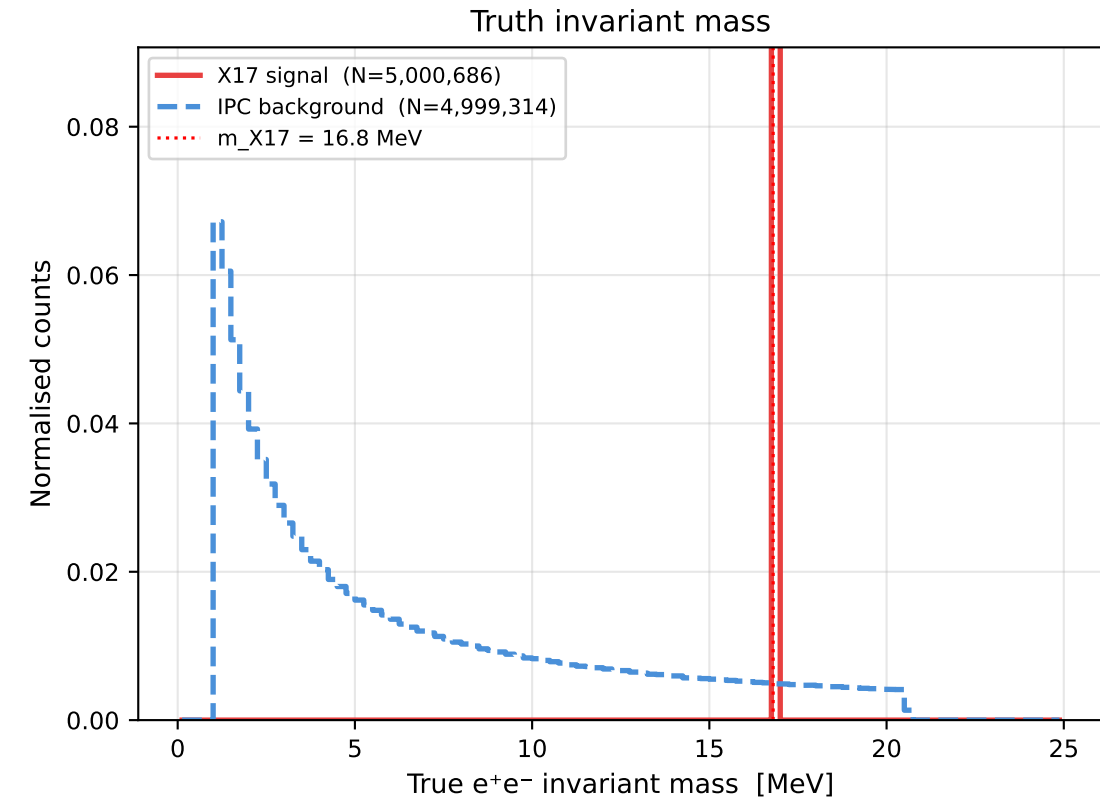
IPC background



Invariant Mass (exact formula including m_e)

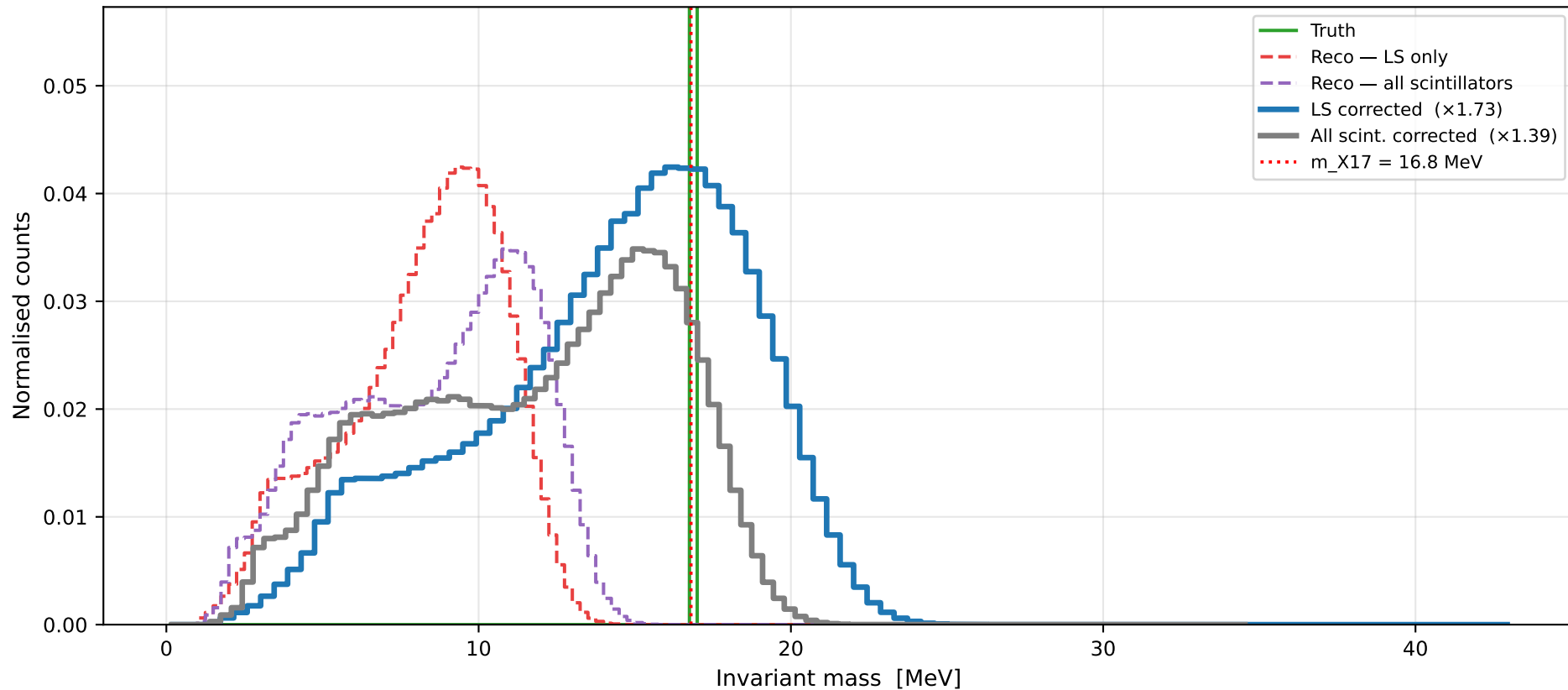
Reco mass — LS only (uncorrected energy)

$$\text{exact: } M = \sqrt{2m_e^2 + 2(E_e - E_{e^+} - p_e - p_{e^+} \cos \theta)}$$

Reco mass — all scintillators
(PIScint + LS + BackScint energy)

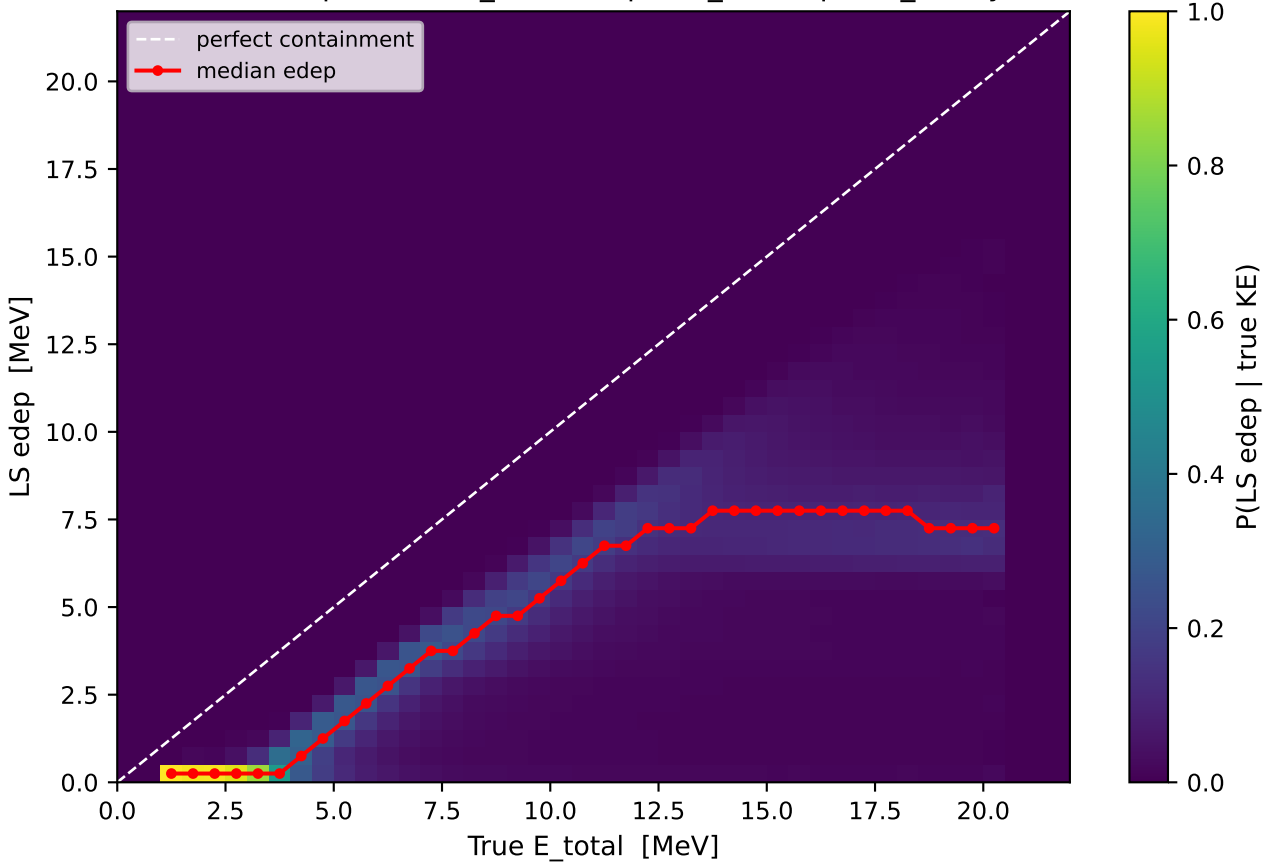
X17 signal: energy-corrected invariant mass

Correction: $E_{est} = (E_{dep} + b)/(1 - a)$ from upstream-loss fit $\Rightarrow M_{corr} \approx M_{reco}/\sqrt{(1 - a_{e^-})(1 - a_{e^+})}$ [large-E approx]

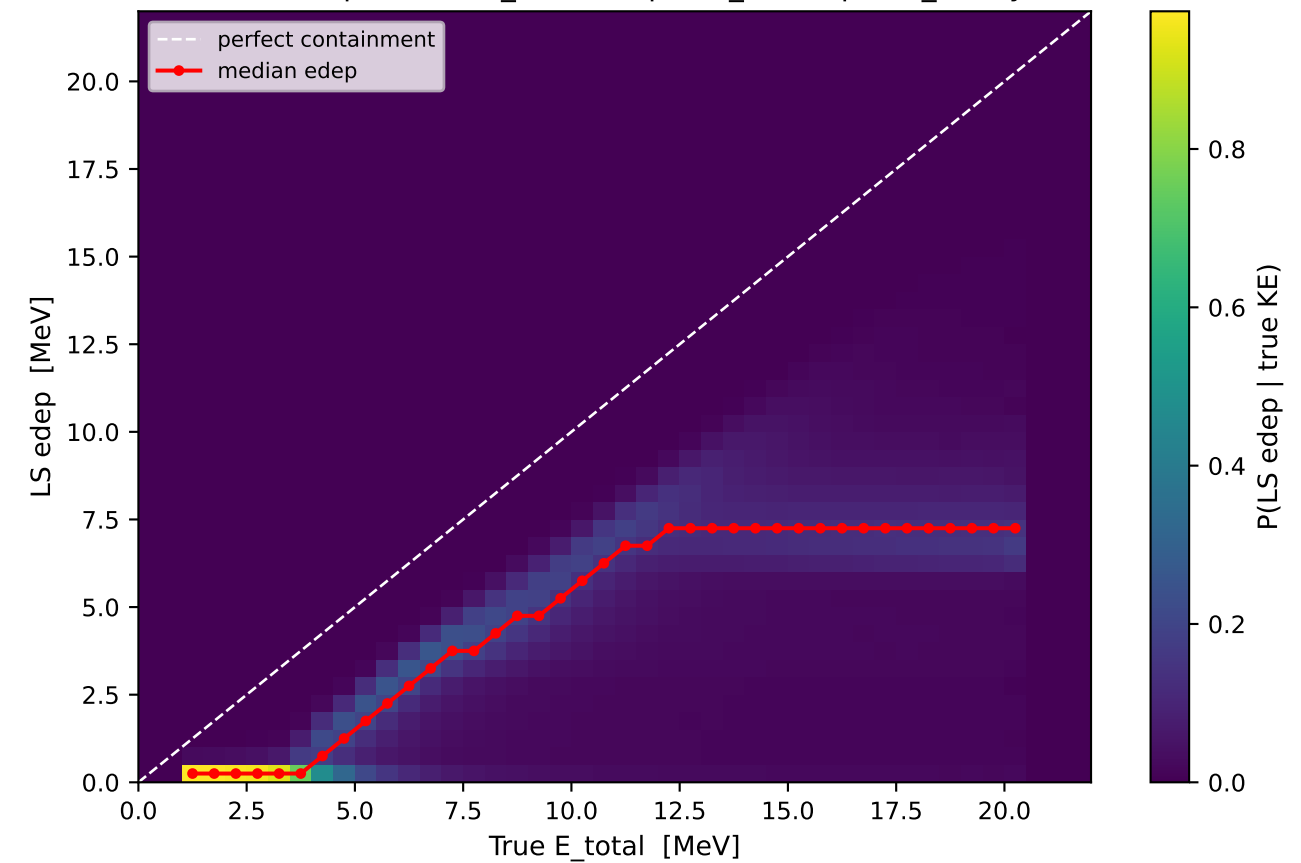


Calorimeter QA: LS energy containment — X17 + IPC combined

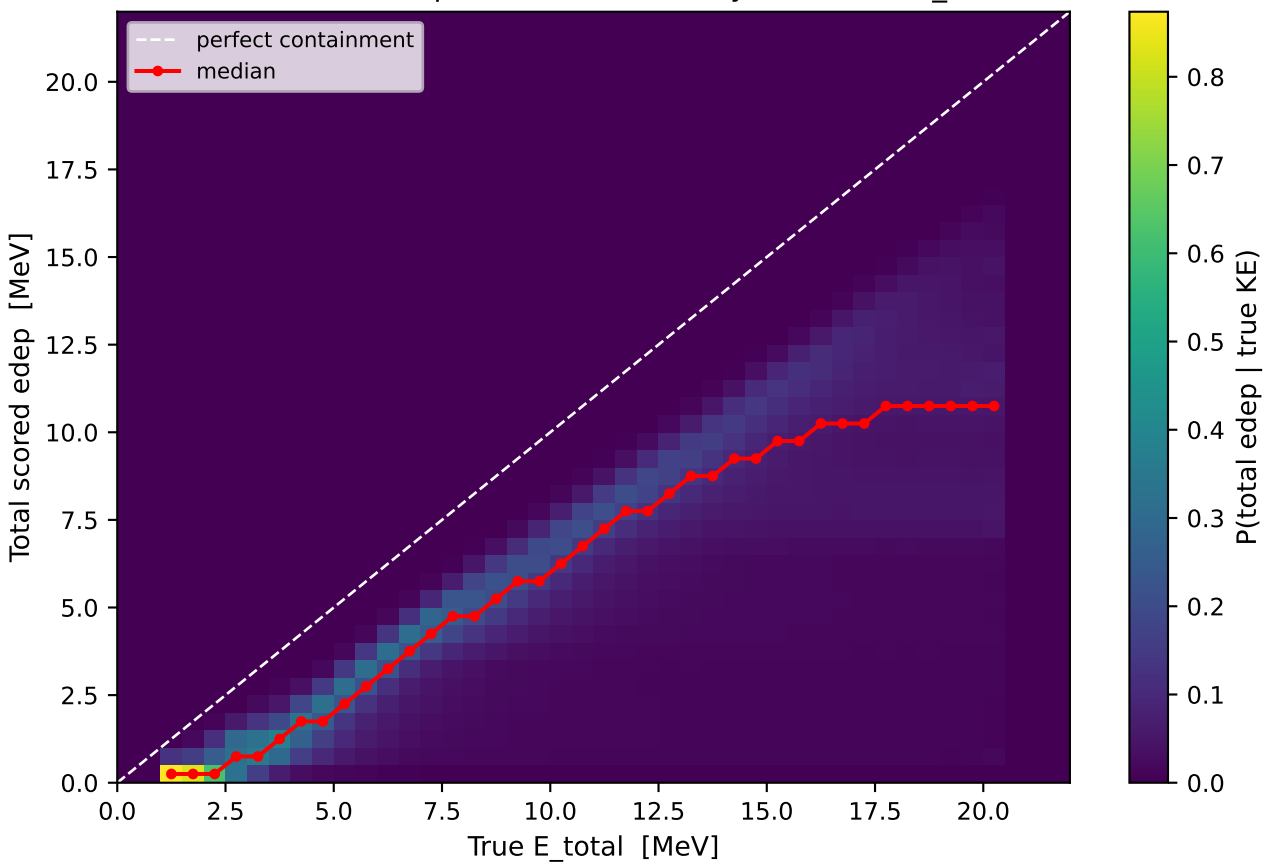
e⁻: LS edep vs true E_{total} (LiqScint_1 + LiqScint_2 only)



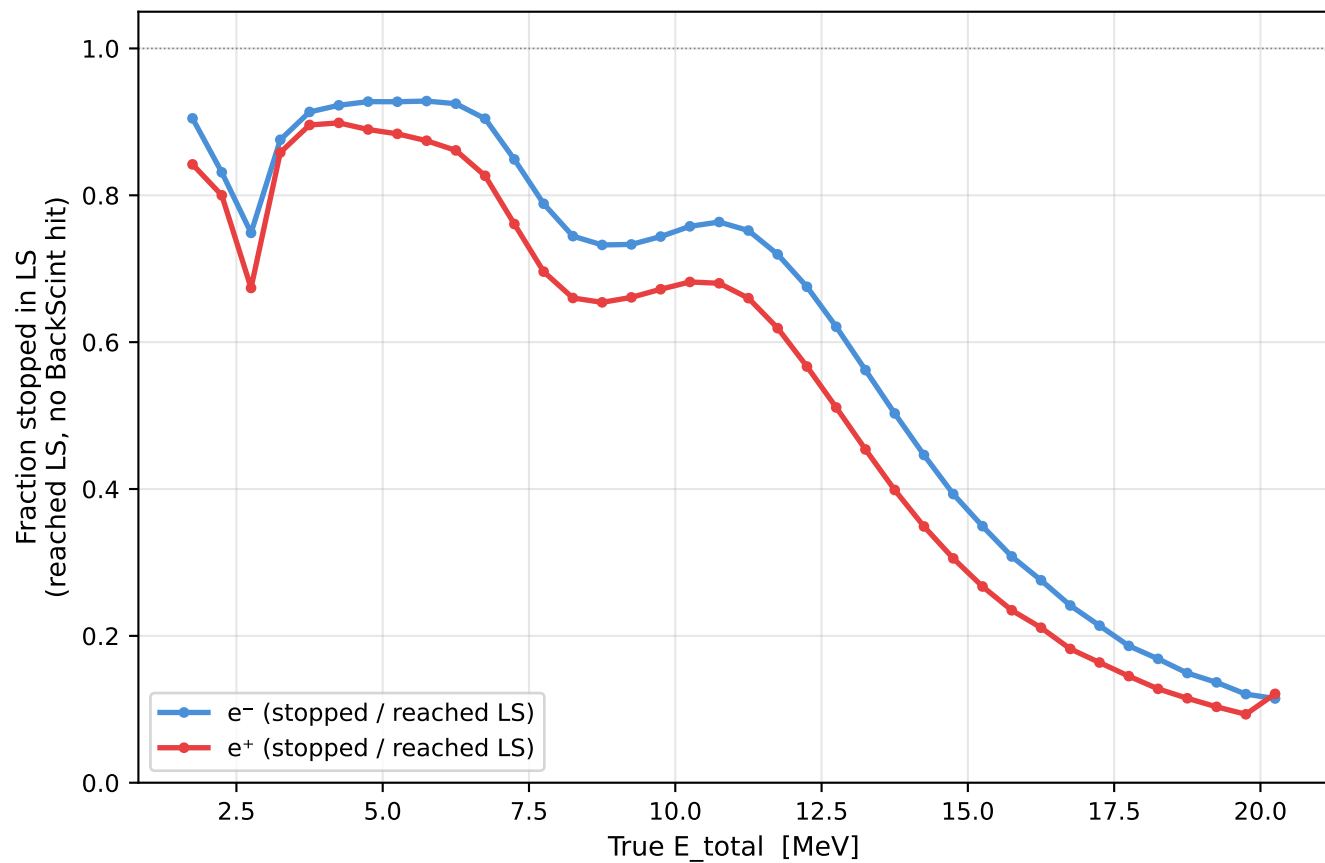
e⁺: LS edep vs true E_{total} (LiqScint_1 + LiqScint_2 only)



e⁻ + e⁺: total edep in all scintillator layers vs true E_{total}

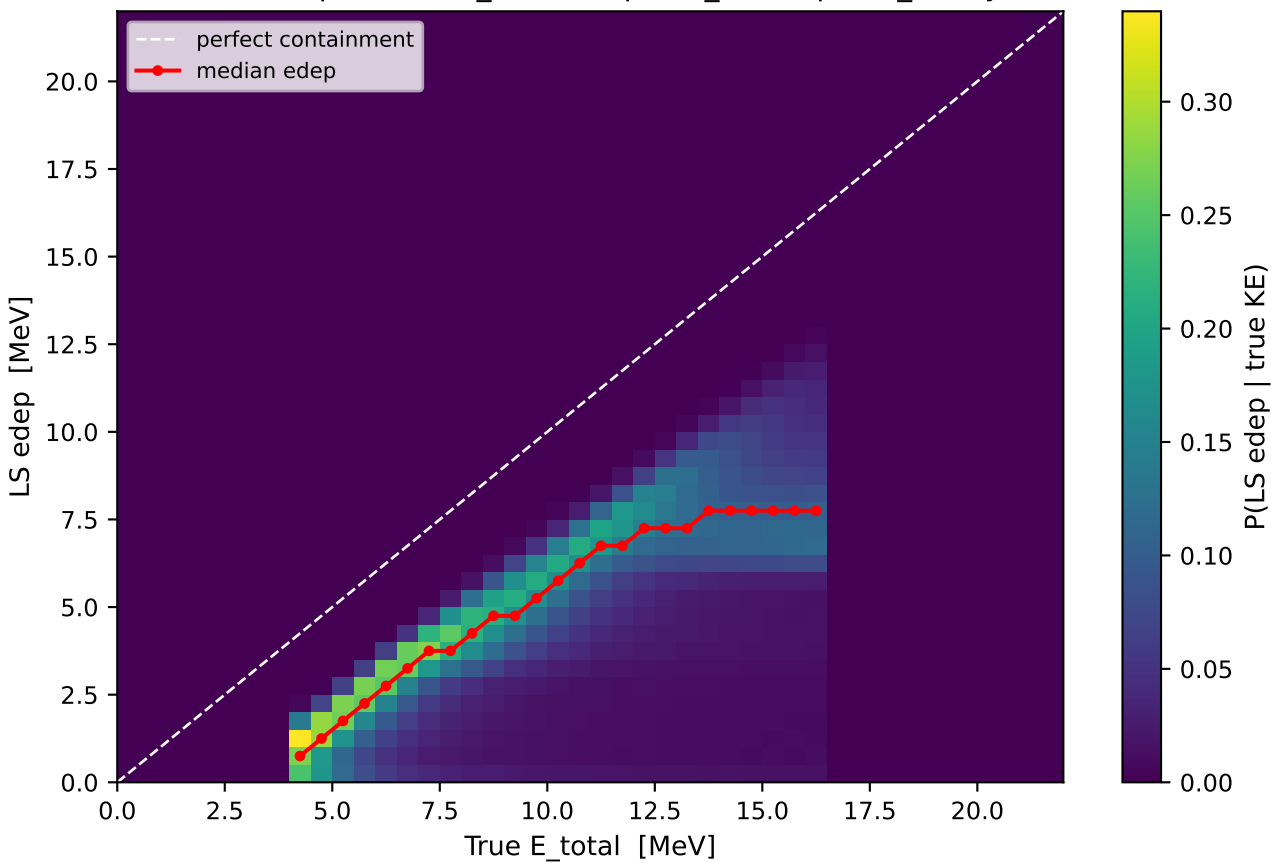


Containment fraction vs KE

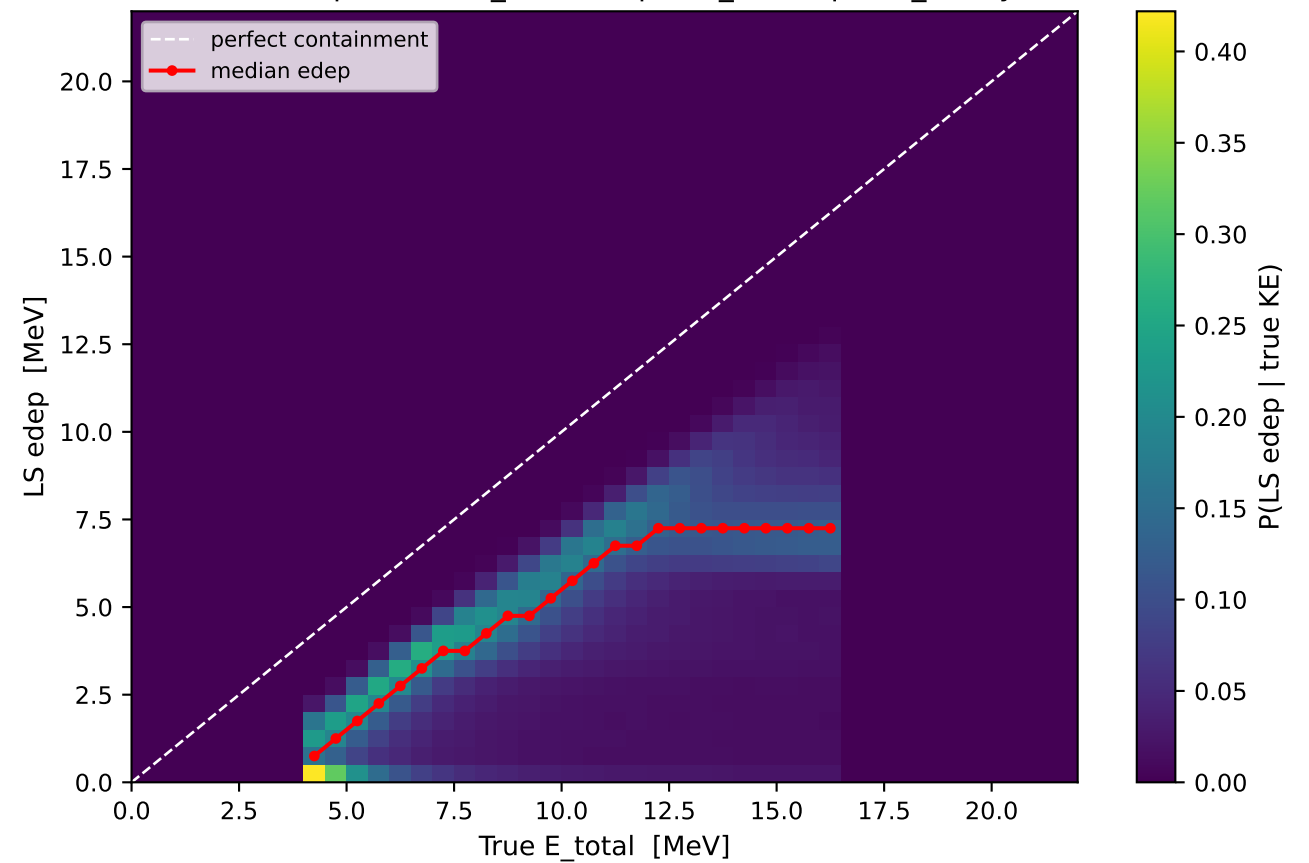


Calorimeter QA: LS energy containment — X17 signal only

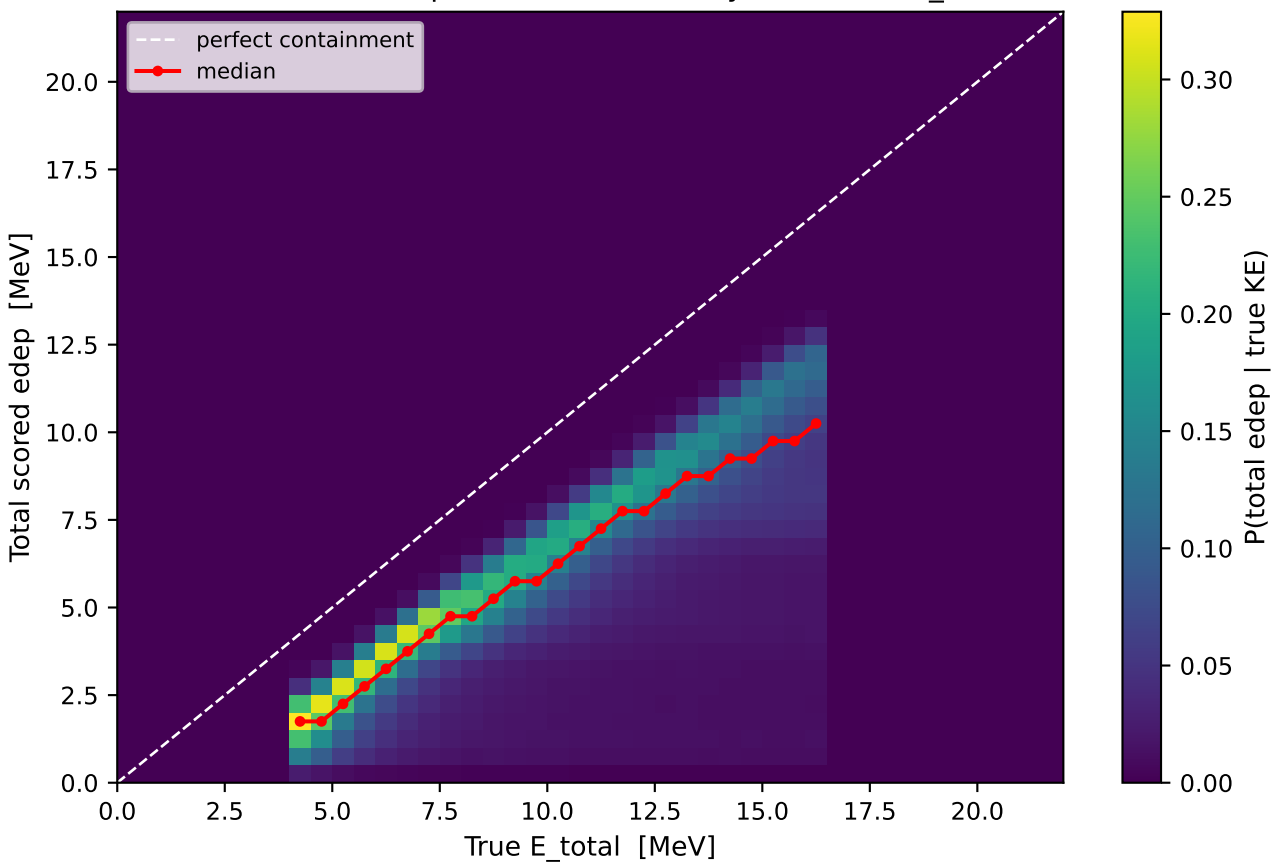
e⁻: LS edep vs true E_{total} (LiqScint_1 + LiqScint_2 only)



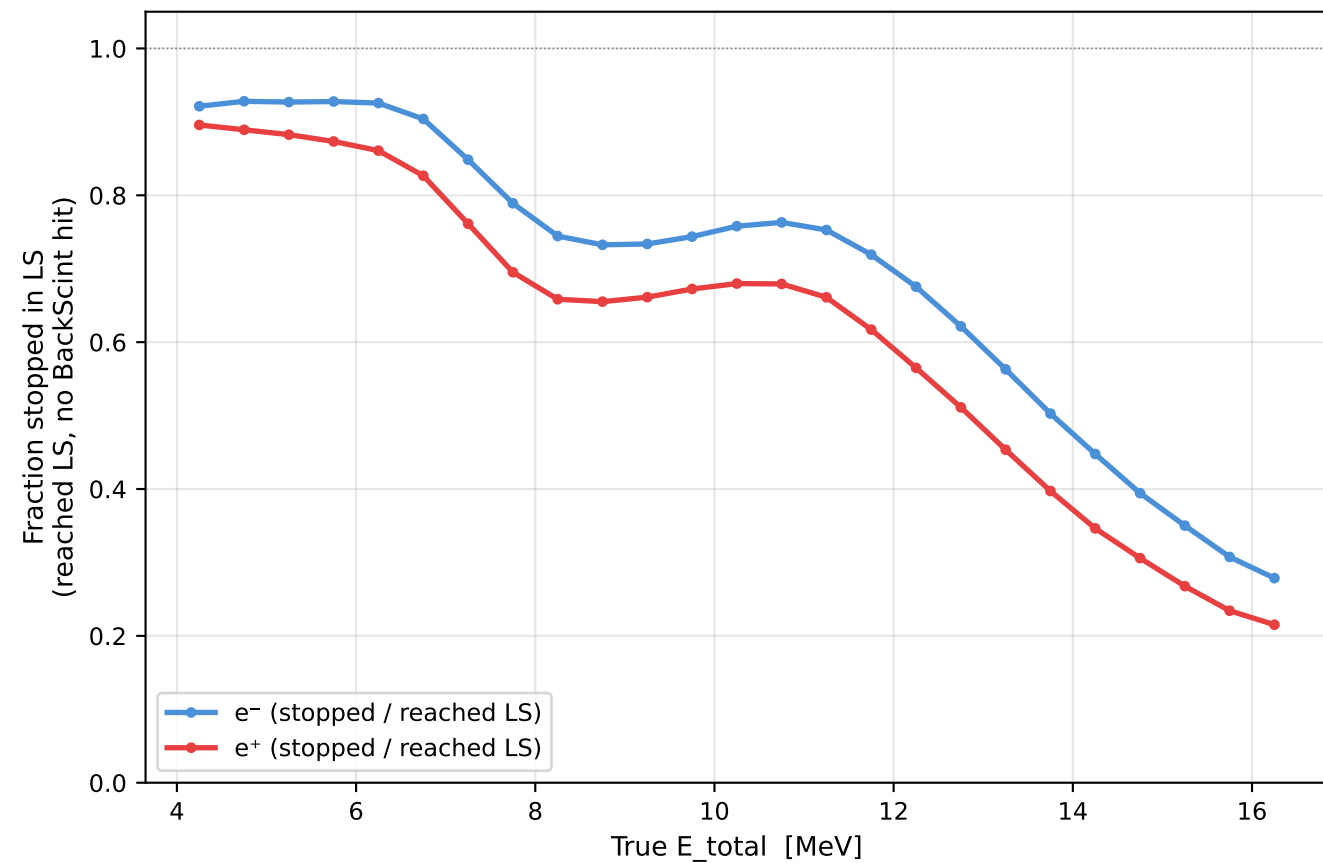
e⁺: LS edep vs true E_{total} (LiqScint_1 + LiqScint_2 only)



e⁻ + e⁺: total edep in all scintillator layers vs true E_{total}

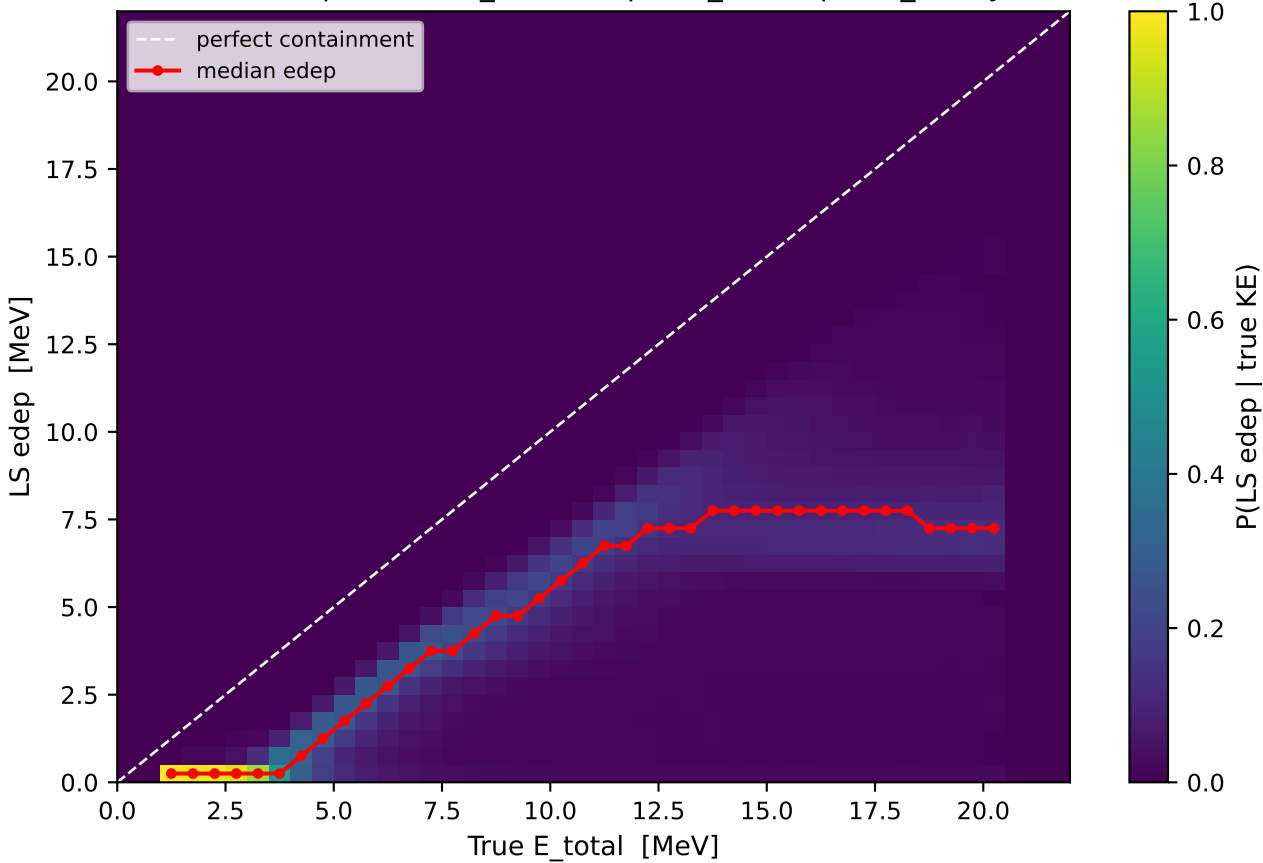


Containment fraction vs KE

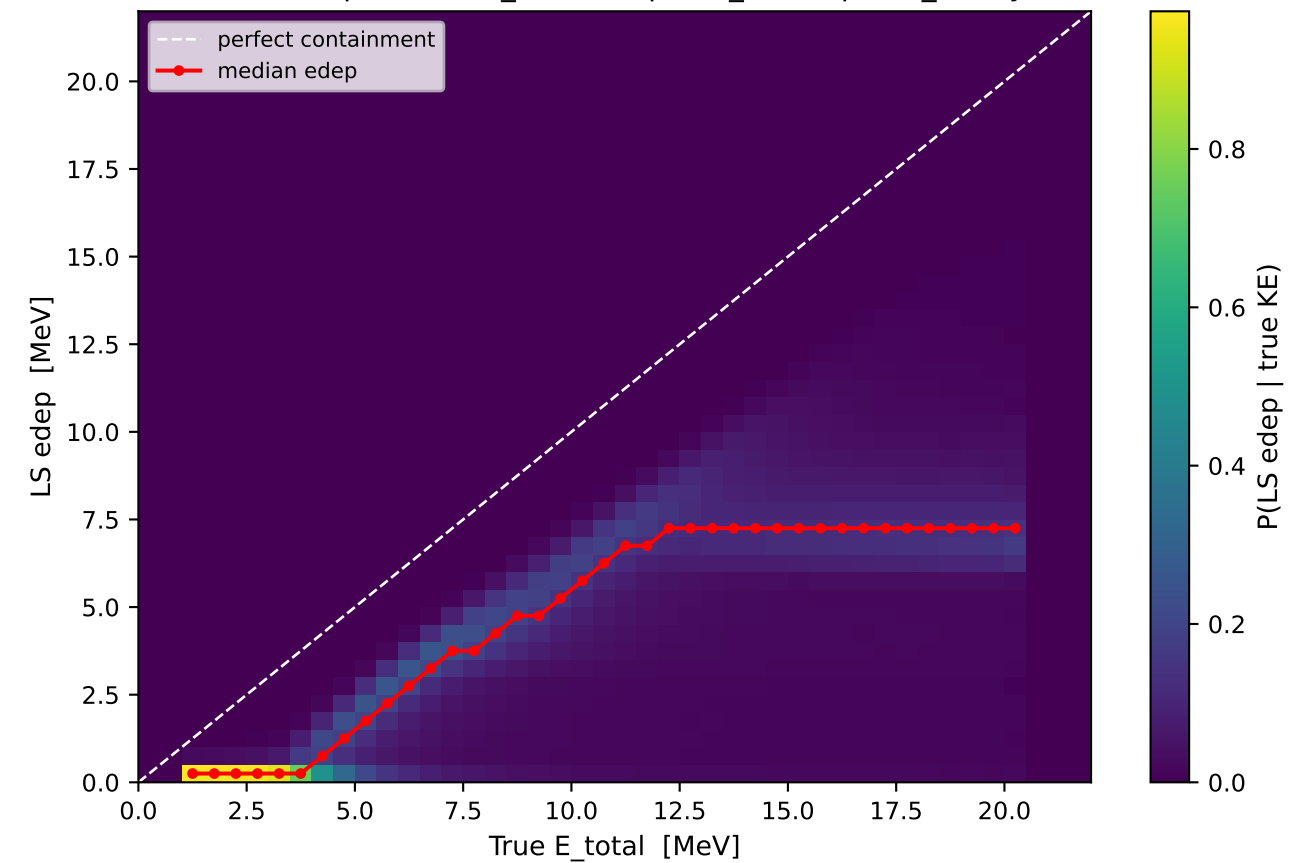


Calorimeter QA: LS energy containment — IPC background only

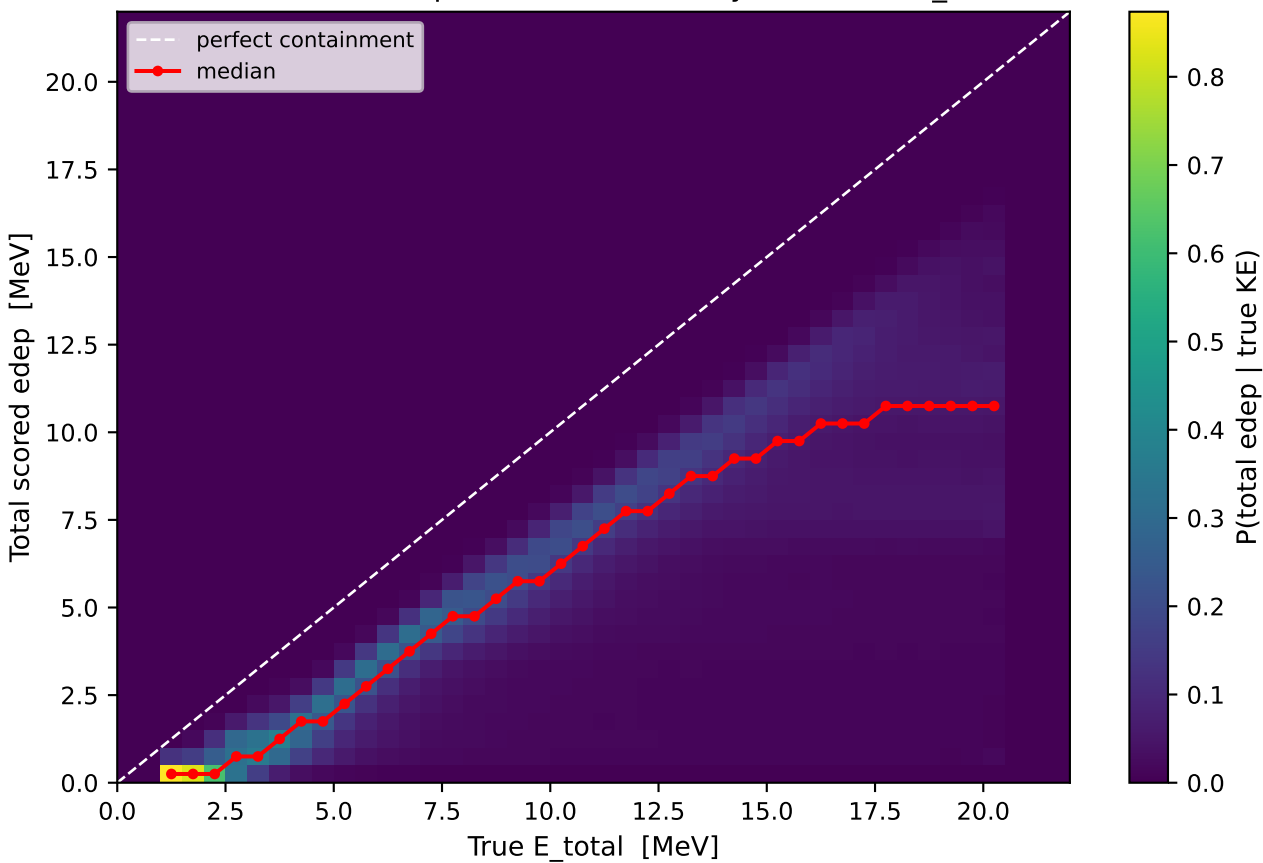
e⁻: LS edep vs true E_{total} (LiqScint_1 + LiqScint_2 only)



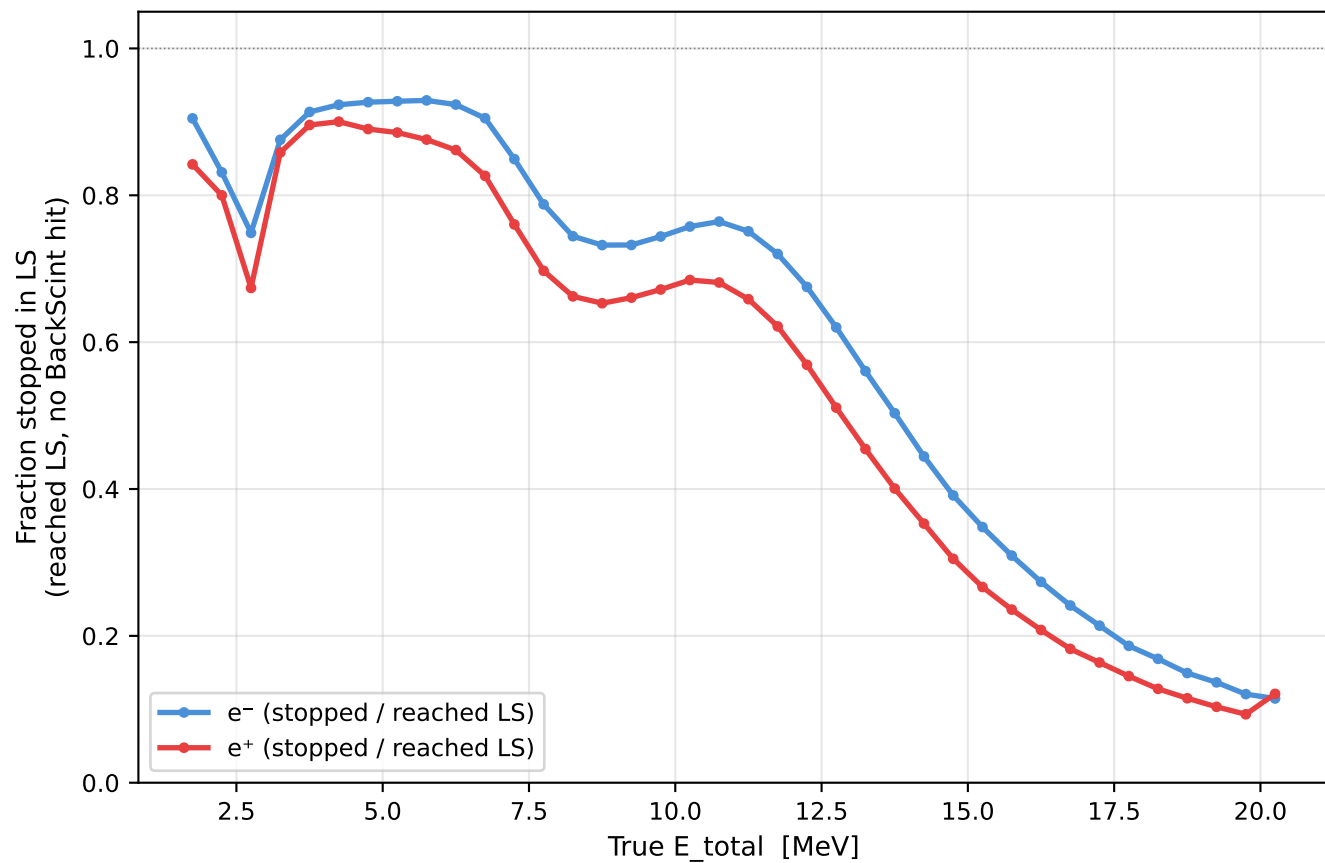
e⁺: LS edep vs true E_{total} (LiqScint_1 + LiqScint_2 only)



e⁻ + e⁺: total edep in all scintillator layers vs true E_{total}

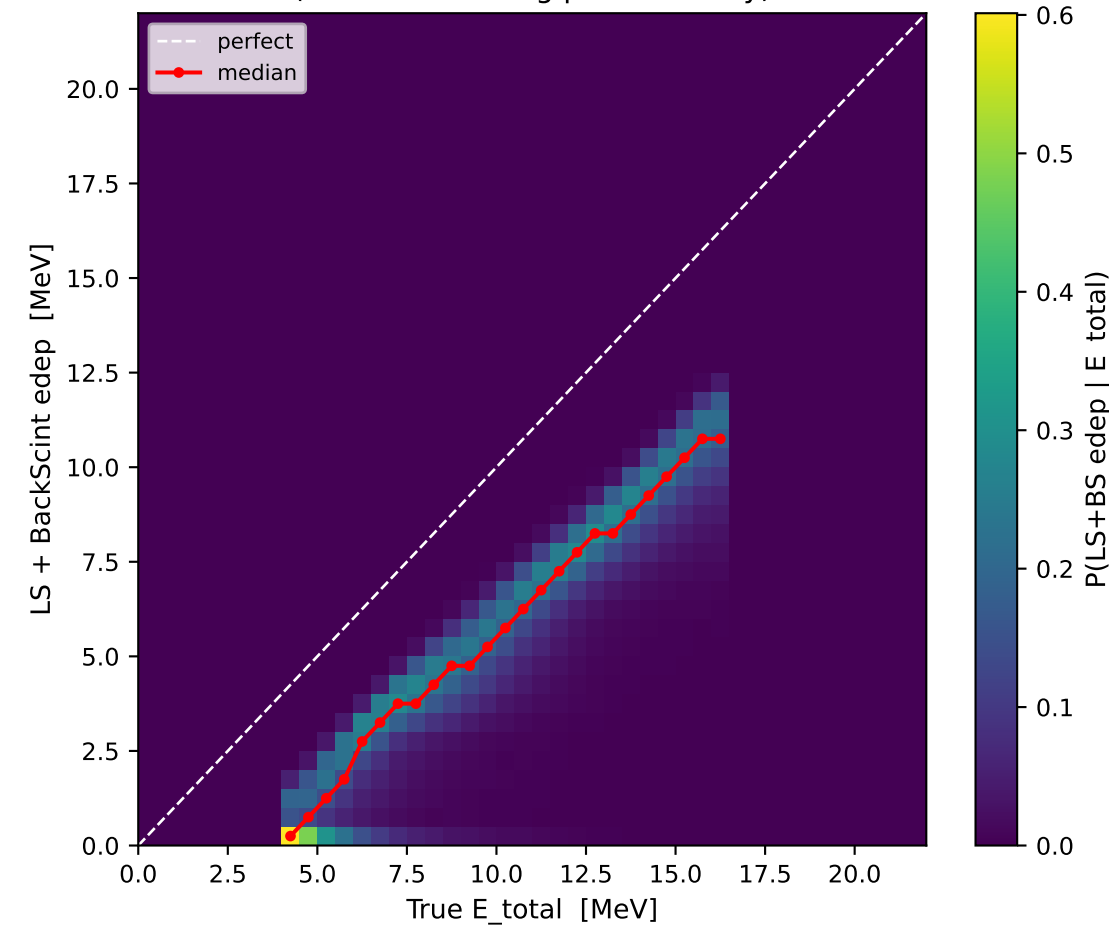


Containment fraction vs KE

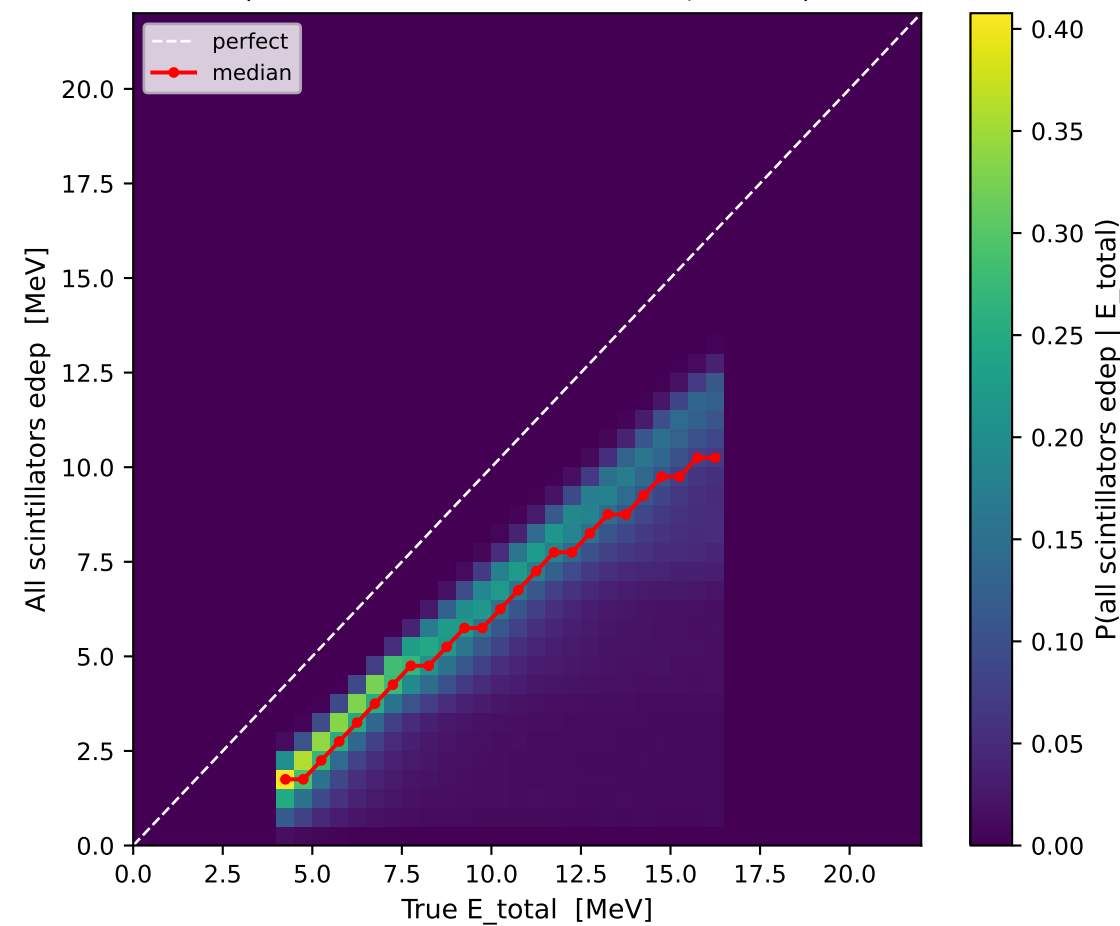


Extended Calorimeter Containment — X17 signal only

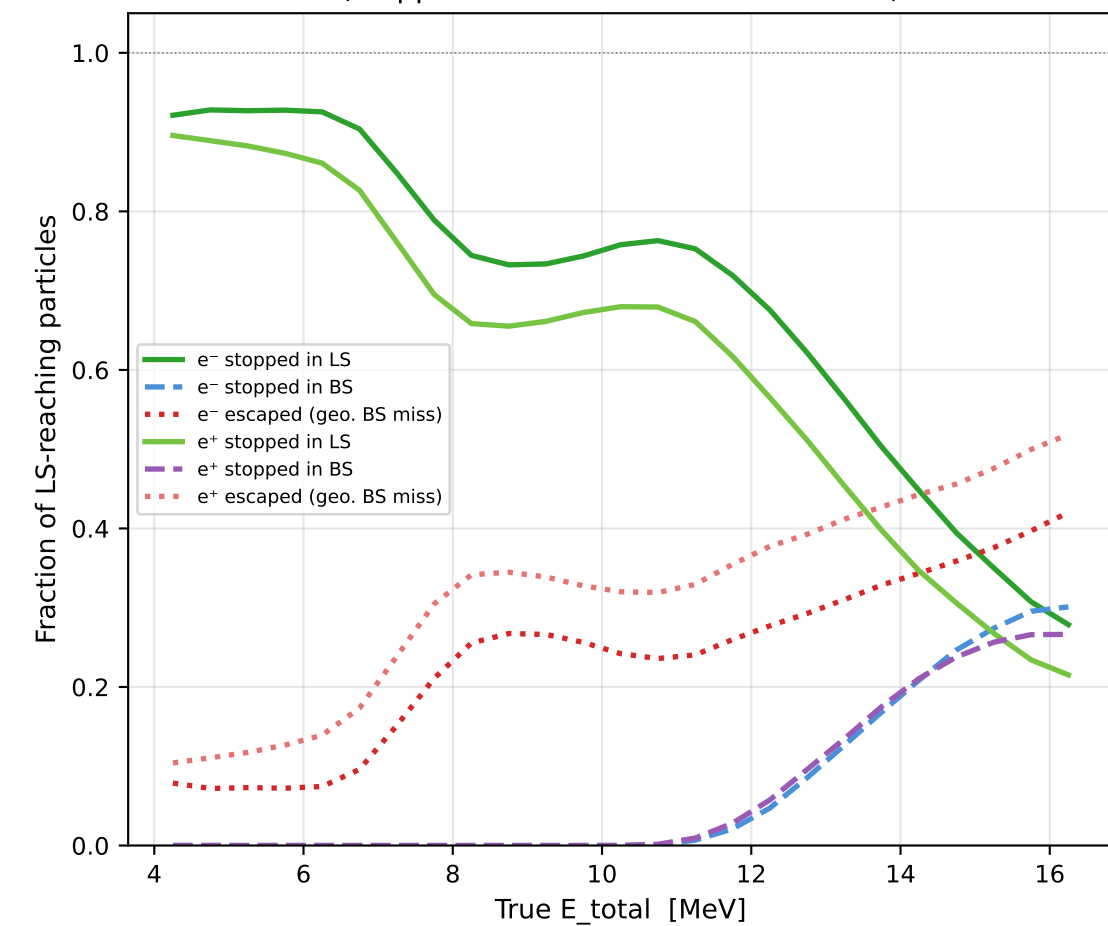
e⁻: LS + BackScint edep
(BackScint-hitting particles only)



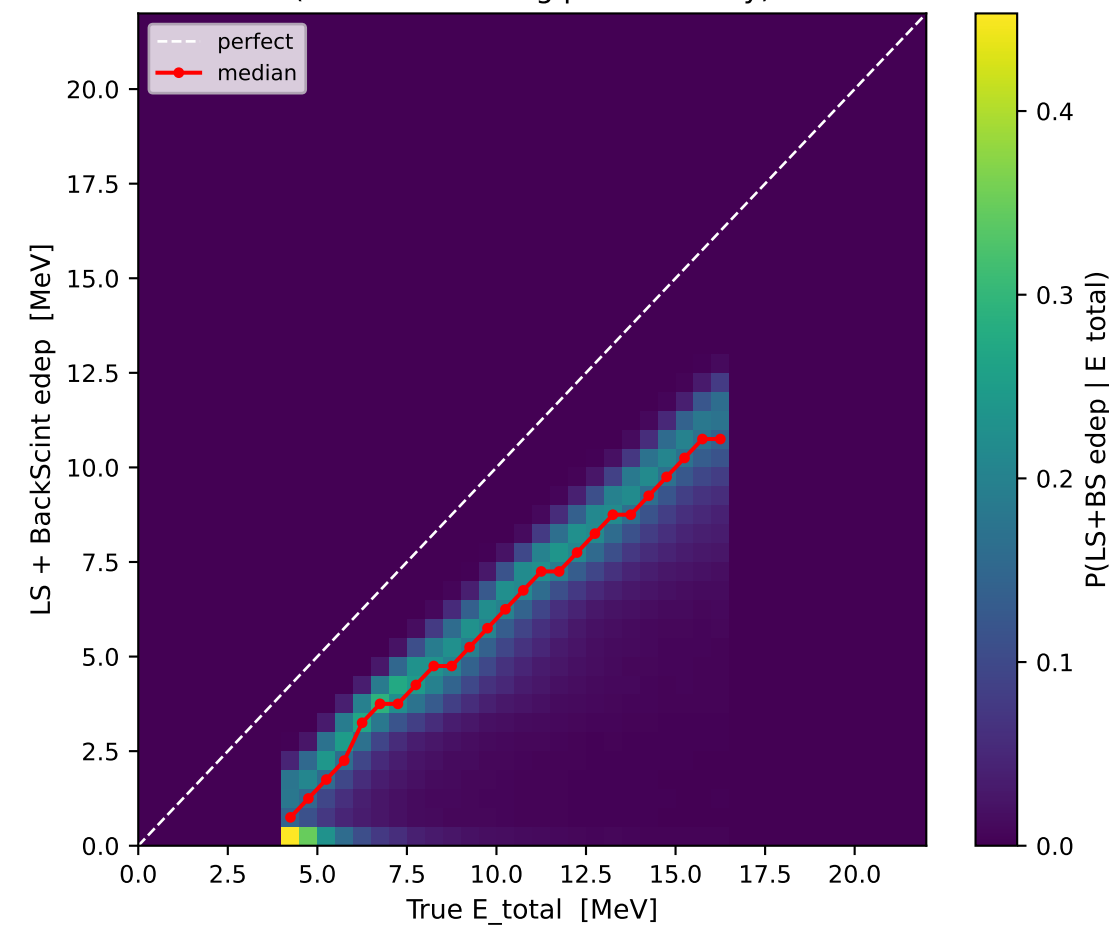
e⁻: all scintillators edep
(PlasticScint + LS + BackScint; no MM)



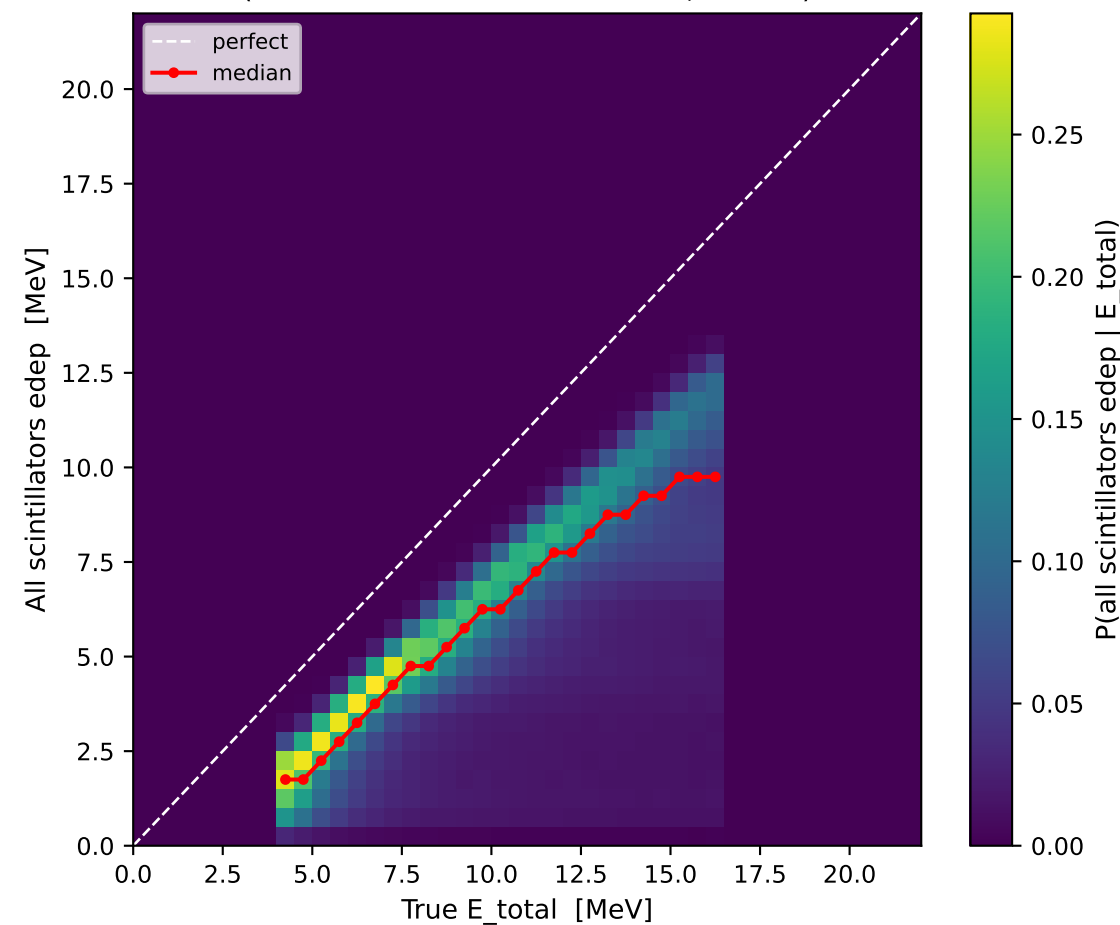
BackScint coverage — 3 categories
(stopped = last min-KE hit < 0.1 MeV)



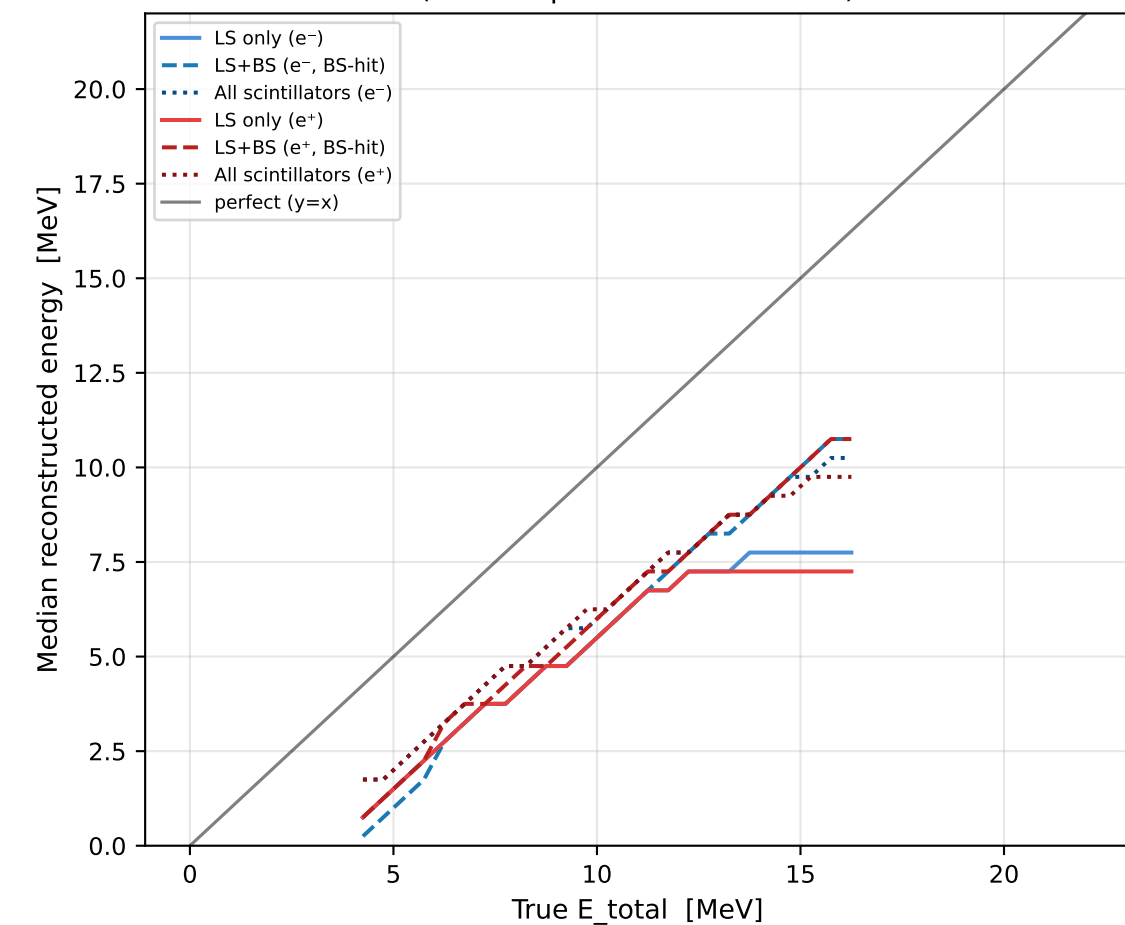
e⁺: LS + BackScint edep
(BackScint-hitting particles only)



e⁺: all scintillators edep
(PlasticScint + LS + BackScint; no MM)



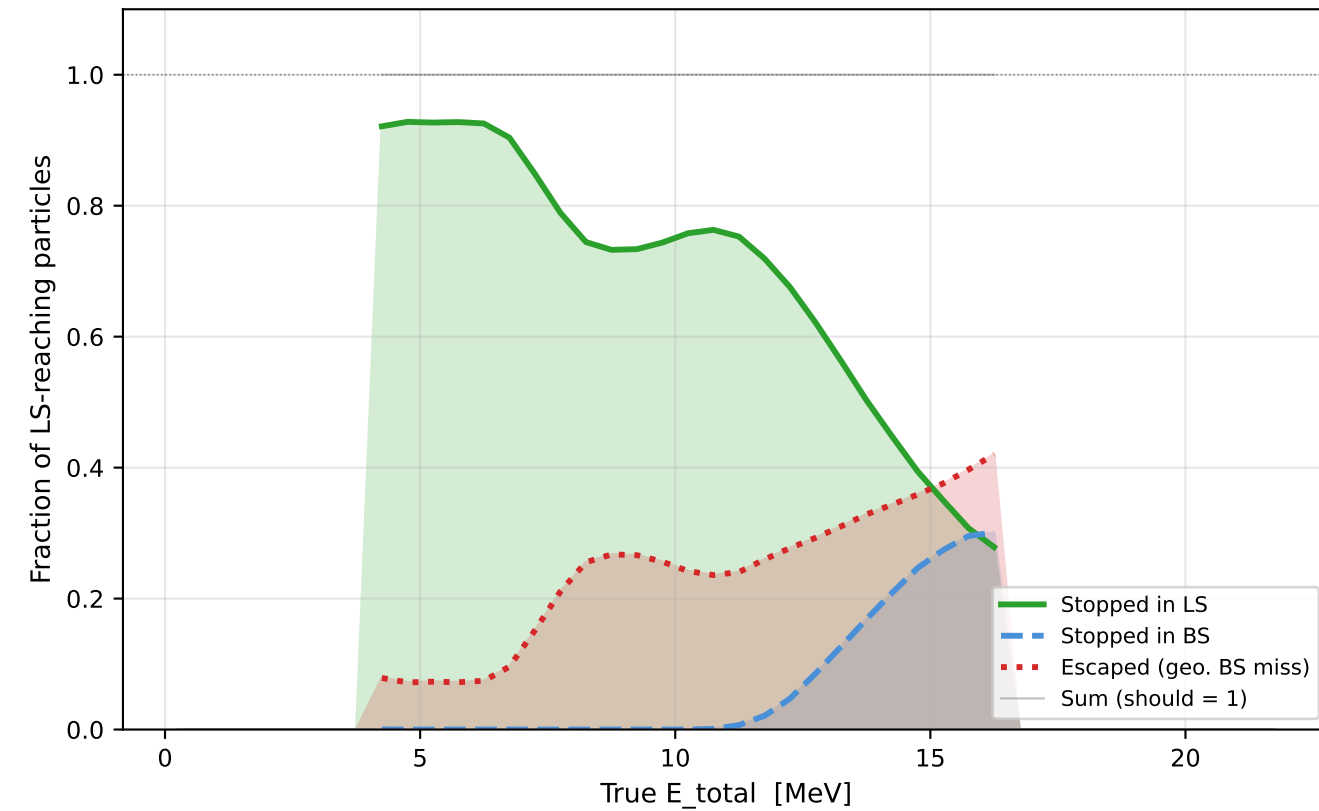
Energy estimate comparison
(median per method vs truth)



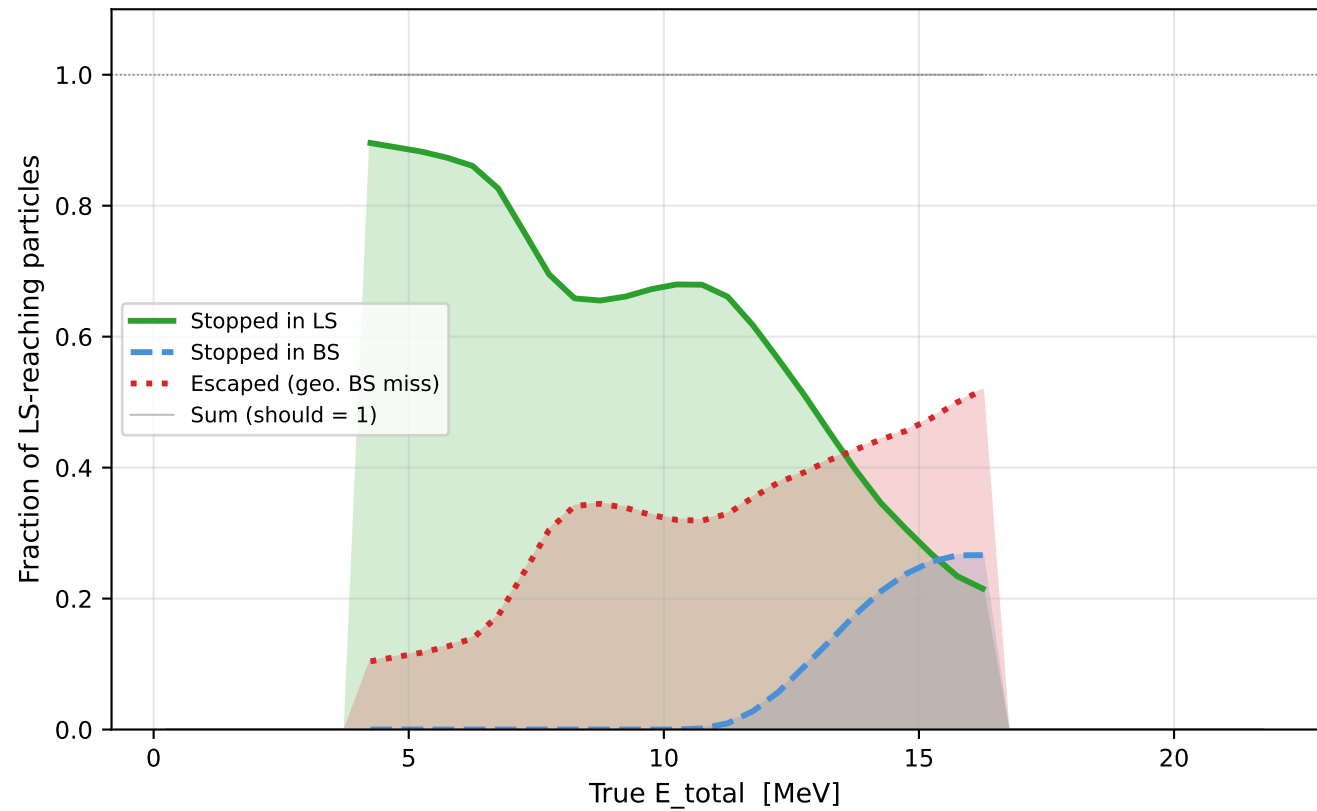
BackScint Coverage — Per Particle and Event Type

Stopped = last min-KE hit < 0.1 MeV | Escaped = reached LS, not stopped in LS or BS

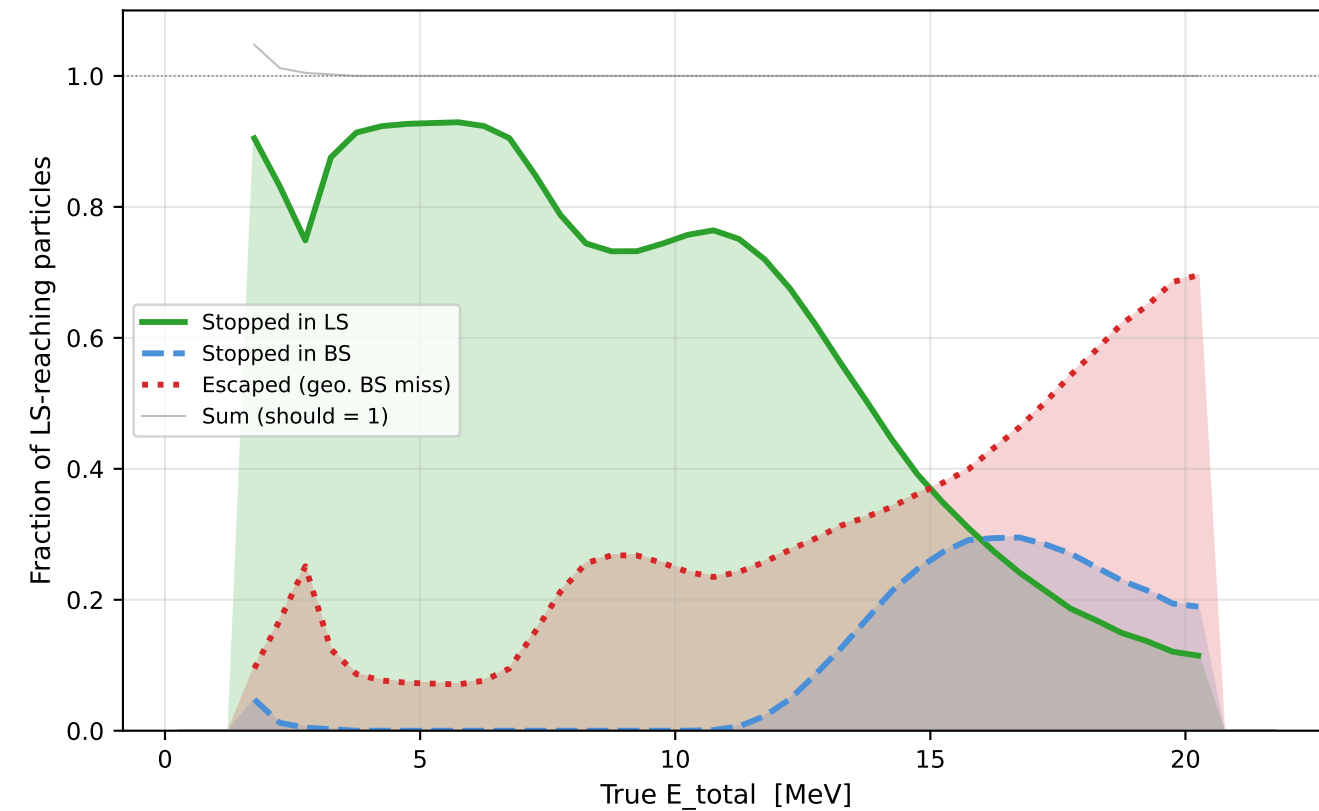
X17 signal — e⁻
(N reach LS = 1,874,797)



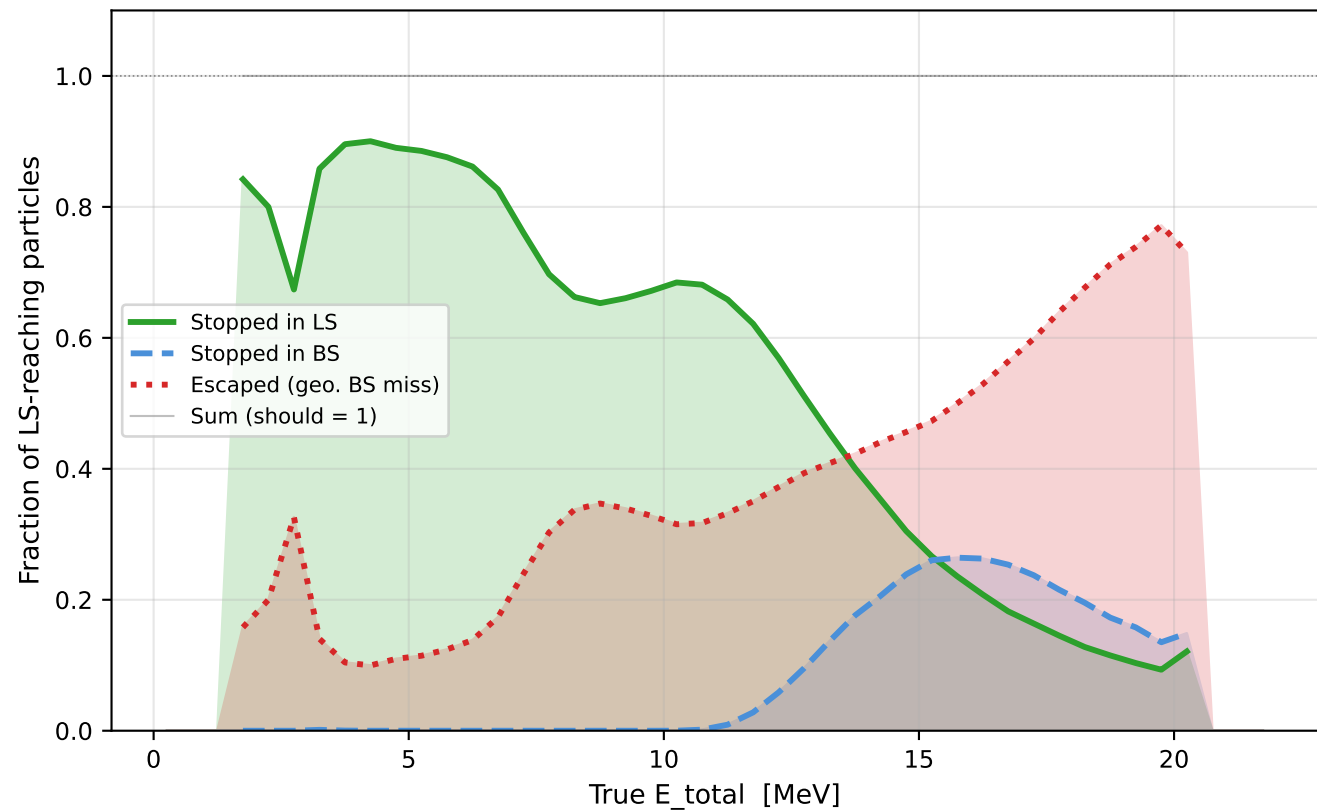
X17 signal — e⁺
(N reach LS = 1,838,979)



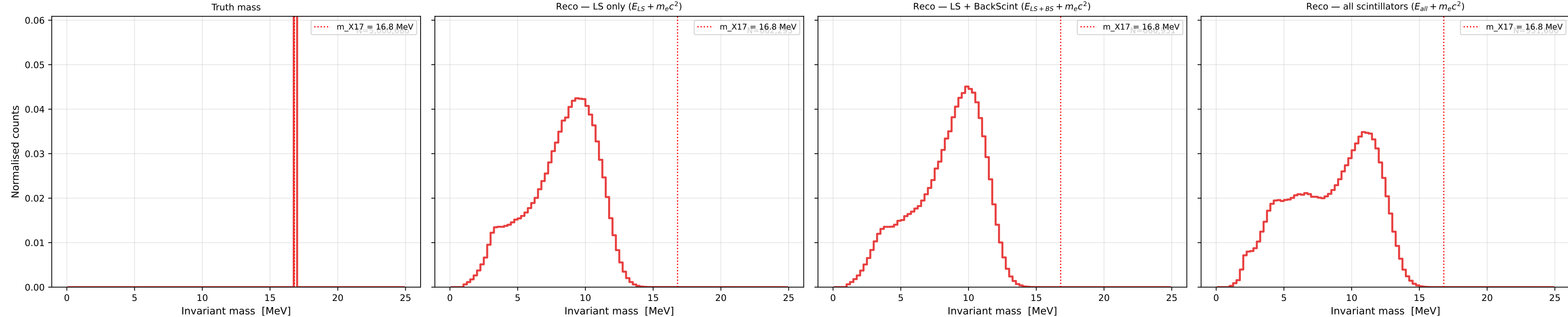
IPC background — e⁻
(N reach LS = 1,668,056)



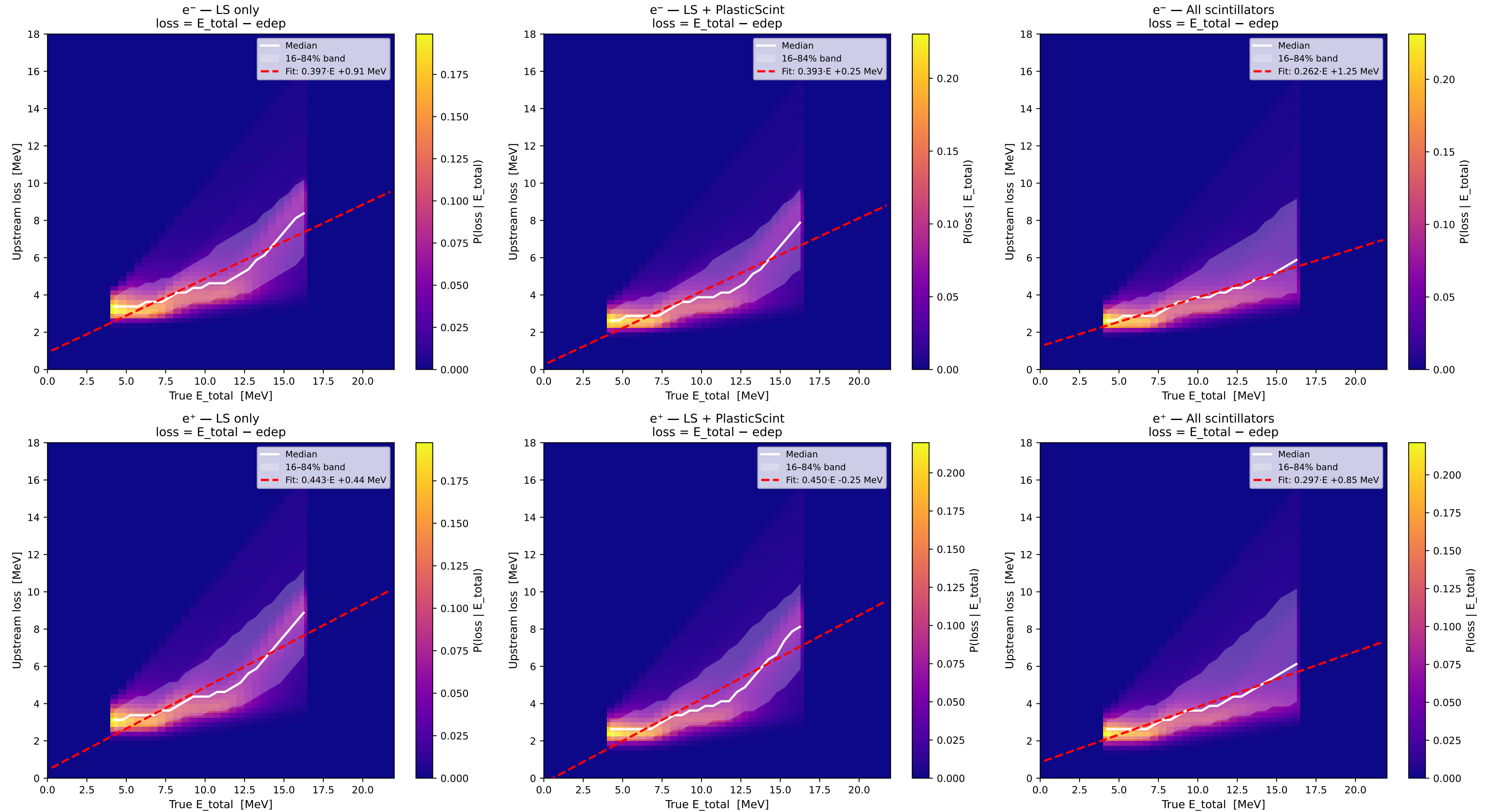
IPC background — e⁺
(N reach LS = 1,637,399)



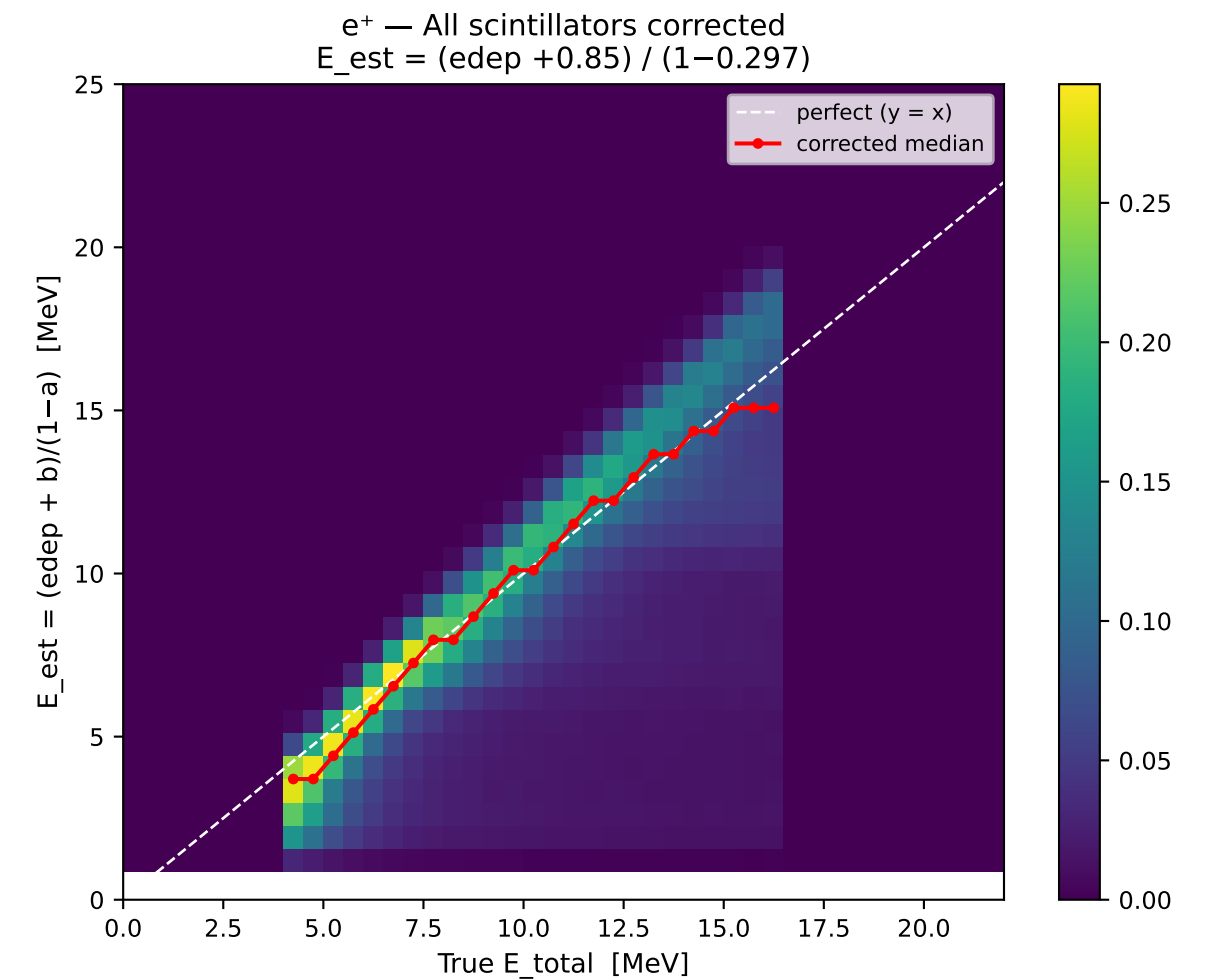
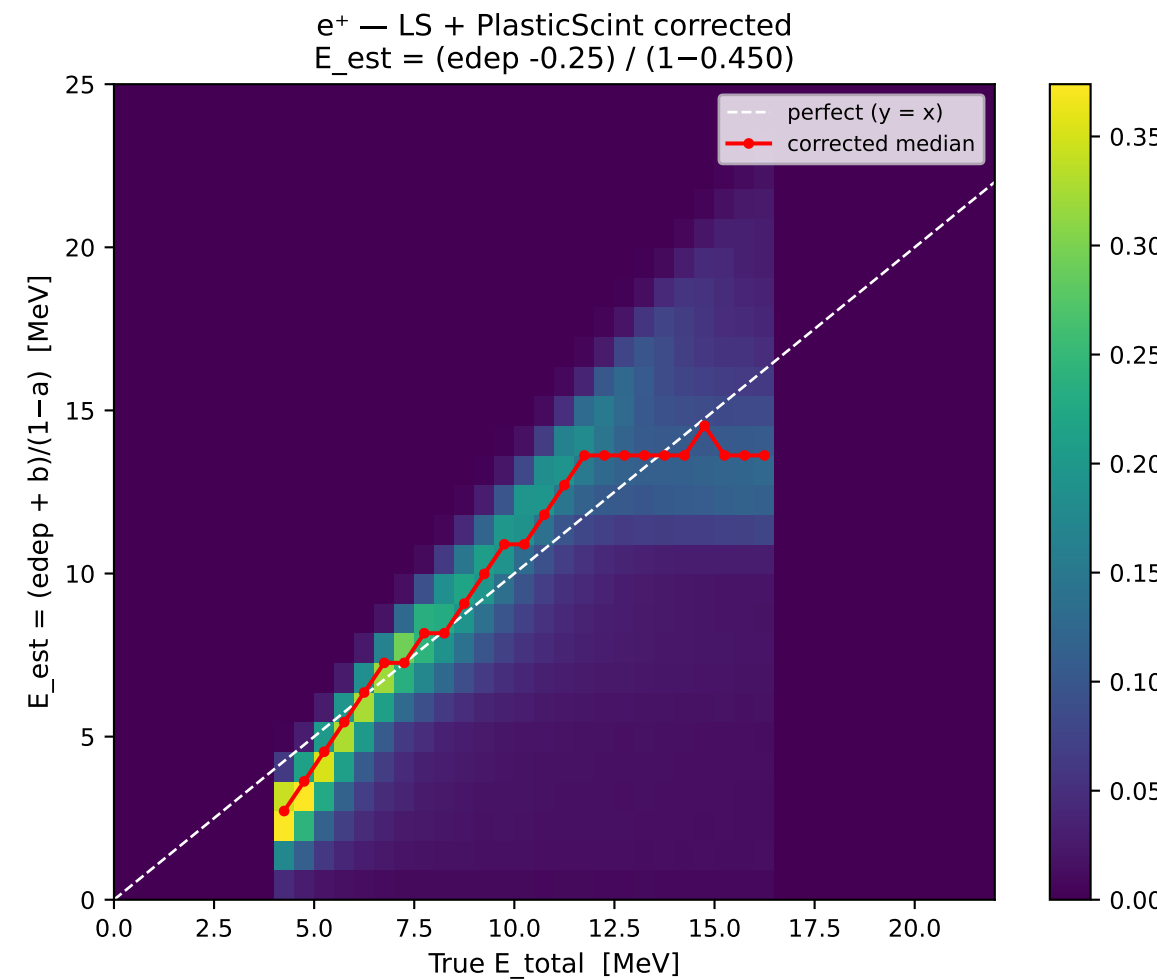
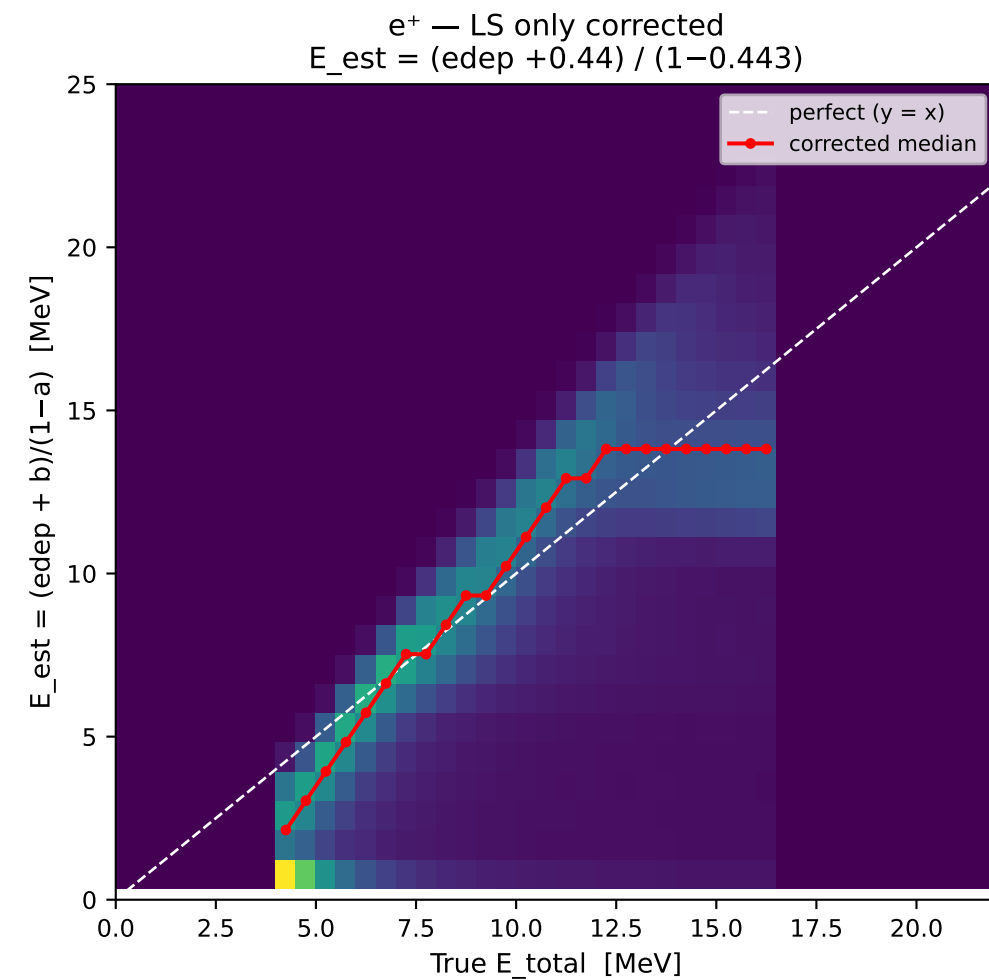
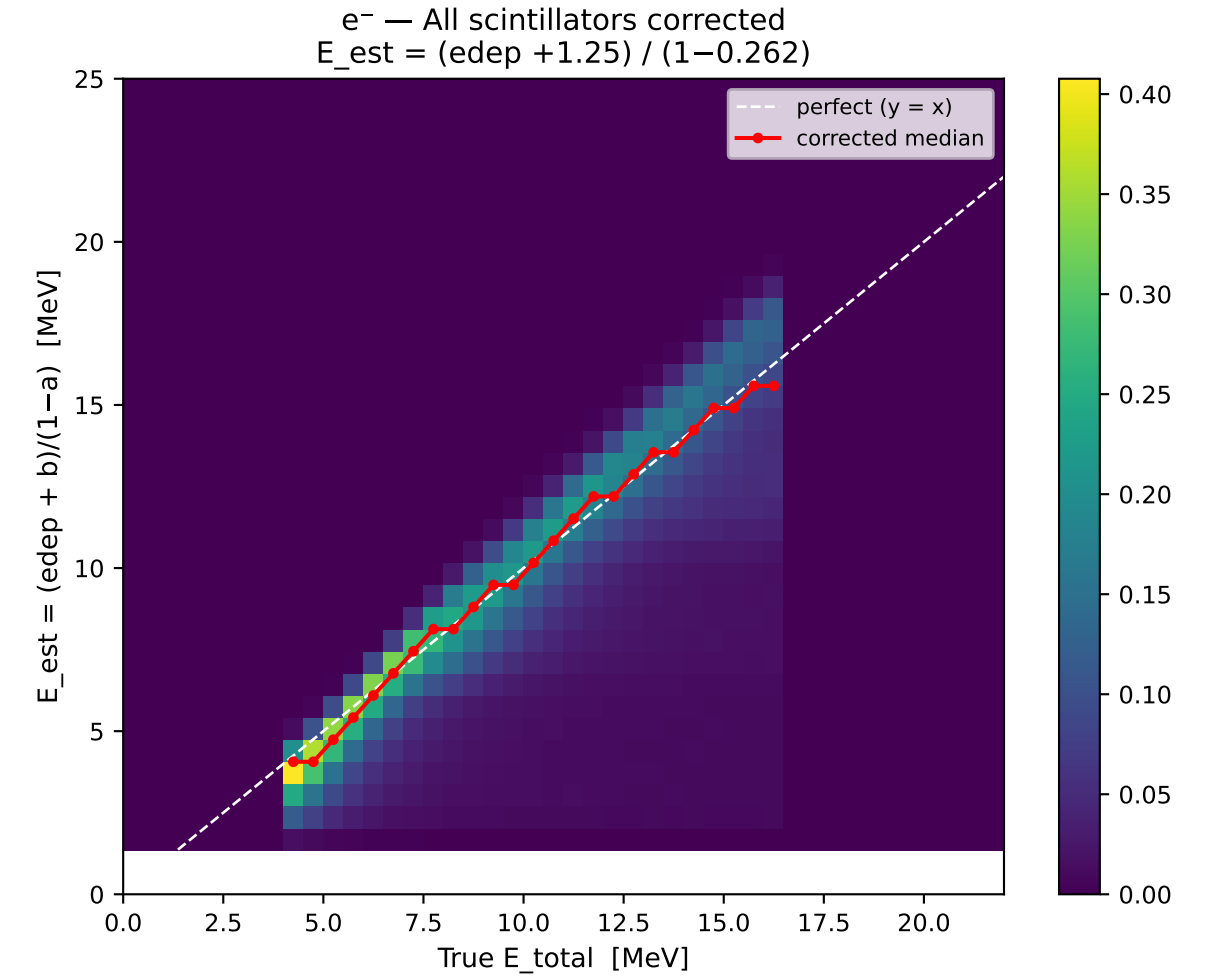
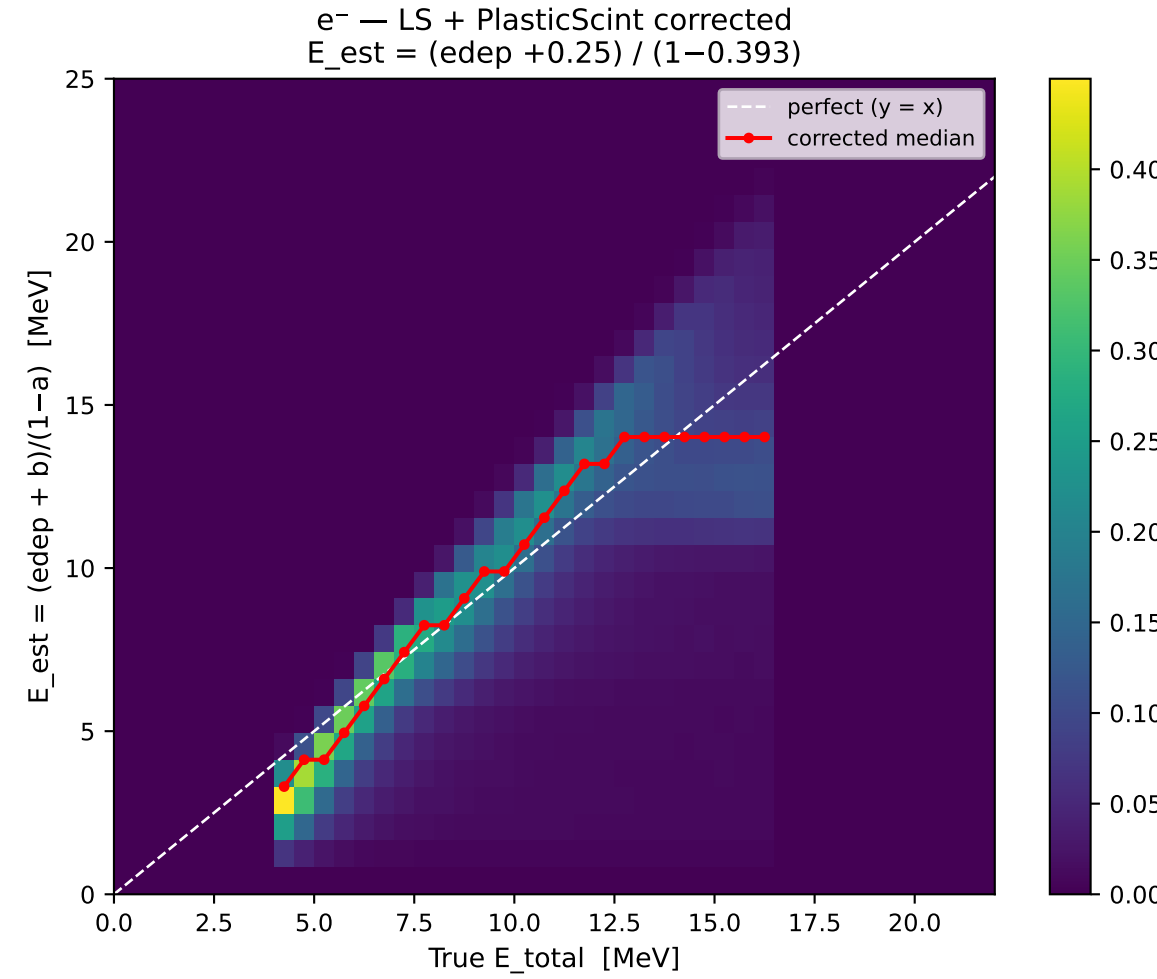
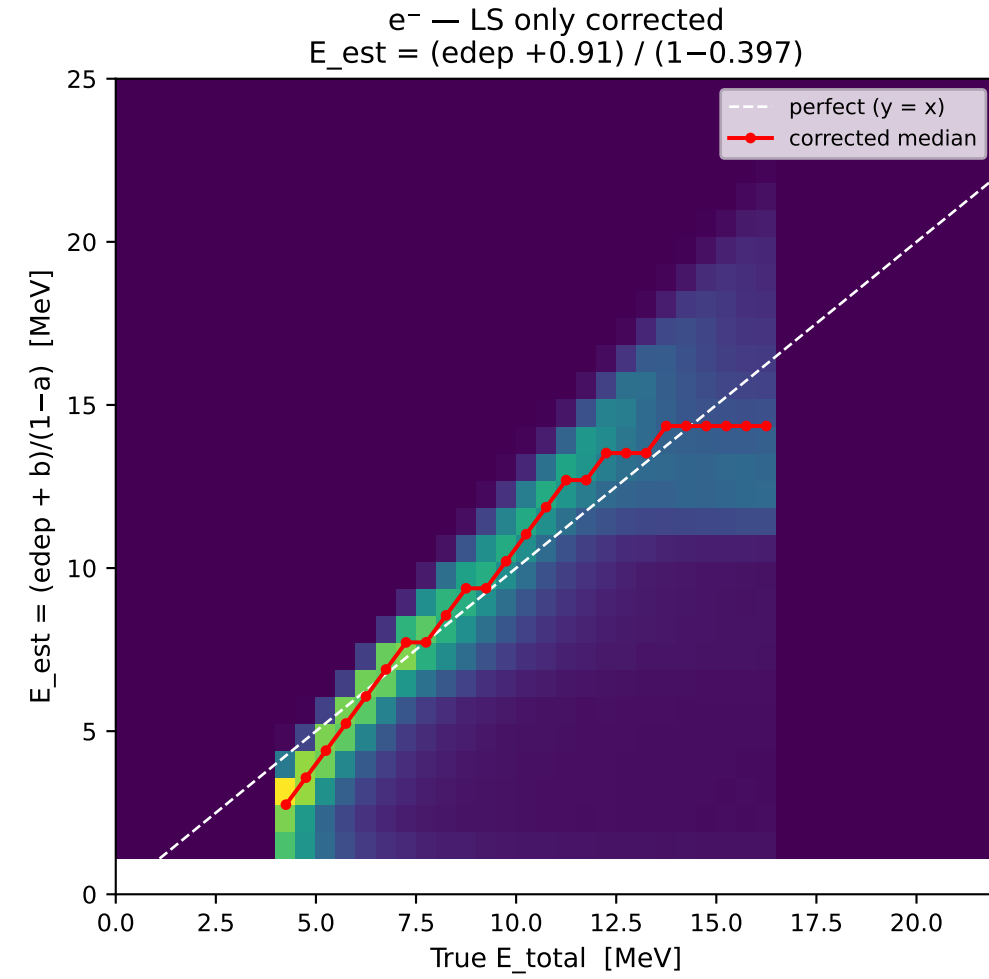
Reco mass — energy estimation method comparison (X17 signal)



Upstream Energy Loss (X17 signal) — $\text{loss} = E_{\text{total}} - \text{deposited energy}$
columns: LS only | LS + PlasticScint | all scintillators (PIScint+LS+BS)

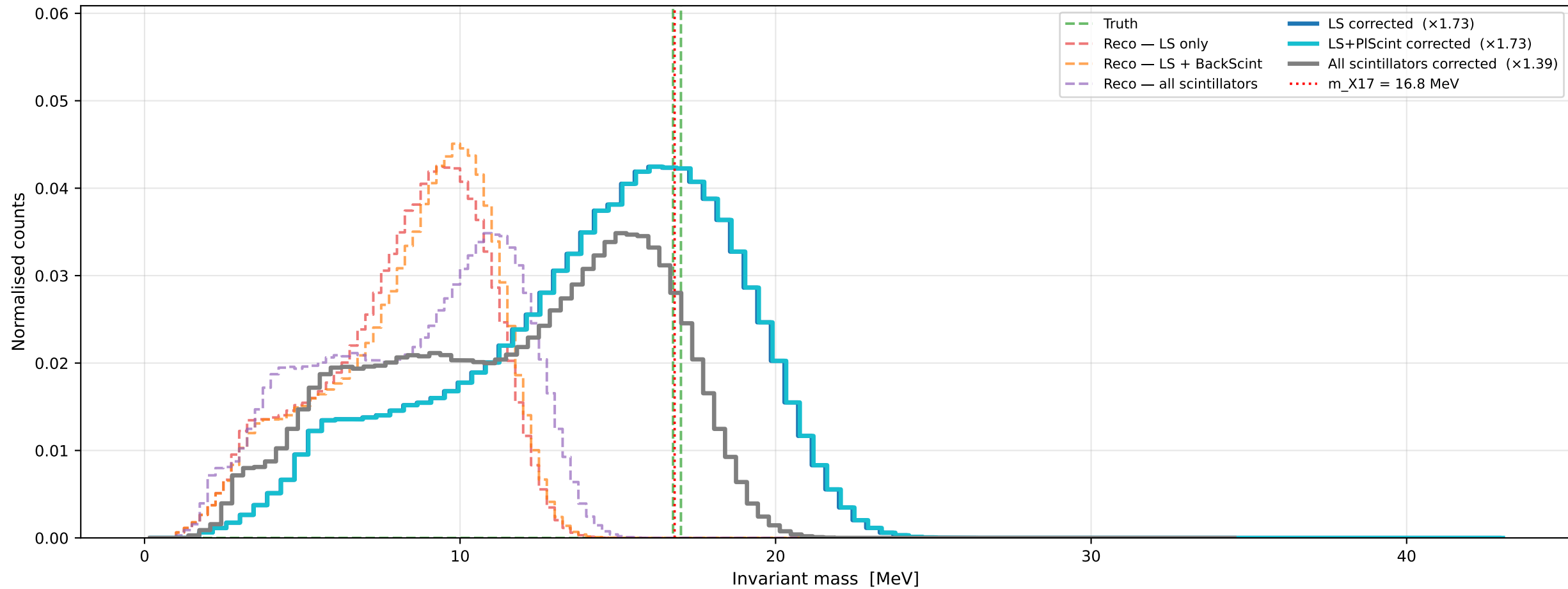


Corrected E estimate vs True E_total (X17 signal) $E_{\text{est}} = (\text{edep} + b) / (1 - a)$
columns: LS | LS + PlasticScint | all scintillators



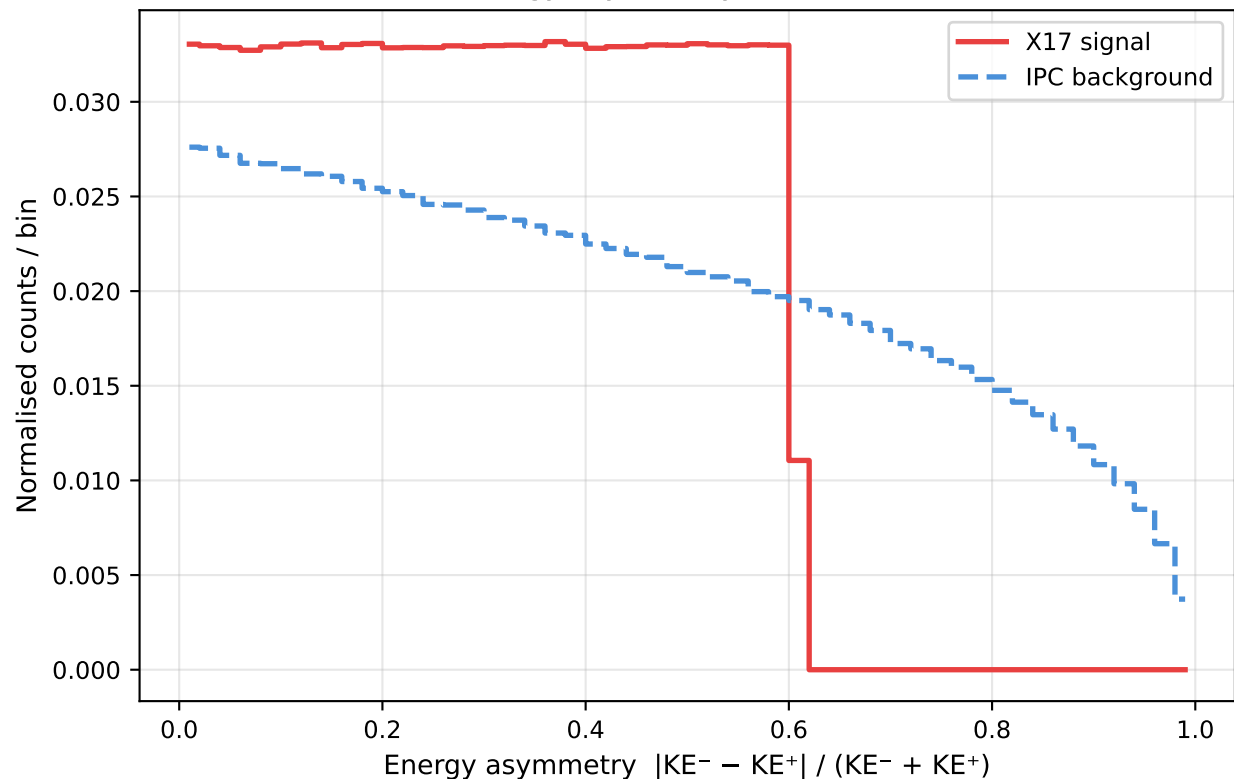
X17 signal: invariant mass — uncorrected (dashed) vs corrected (solid)

Correction: rescale M by $\sqrt{1/((1-a_{e-})(1-a_{e+}))}$ — exact at peak for symmetric pairs; approximate in asymmetric tails

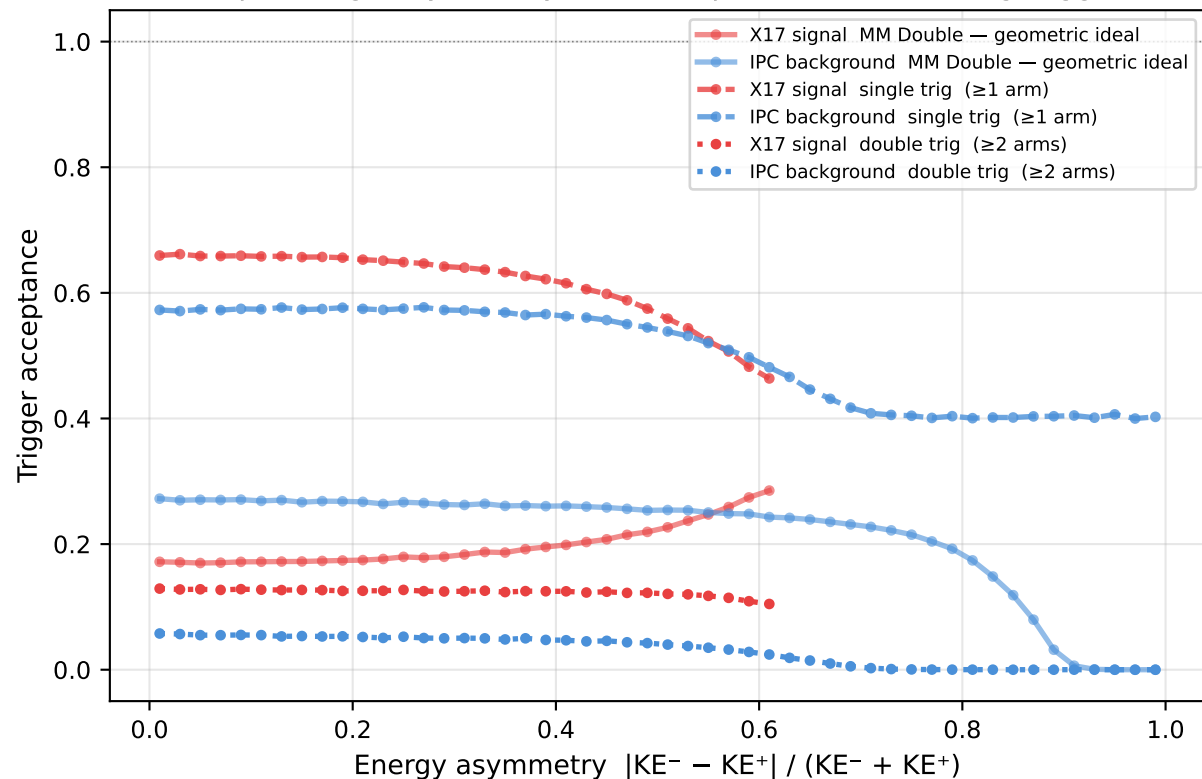


Asymmetric Pair Acceptance

Energy asymmetry distribution

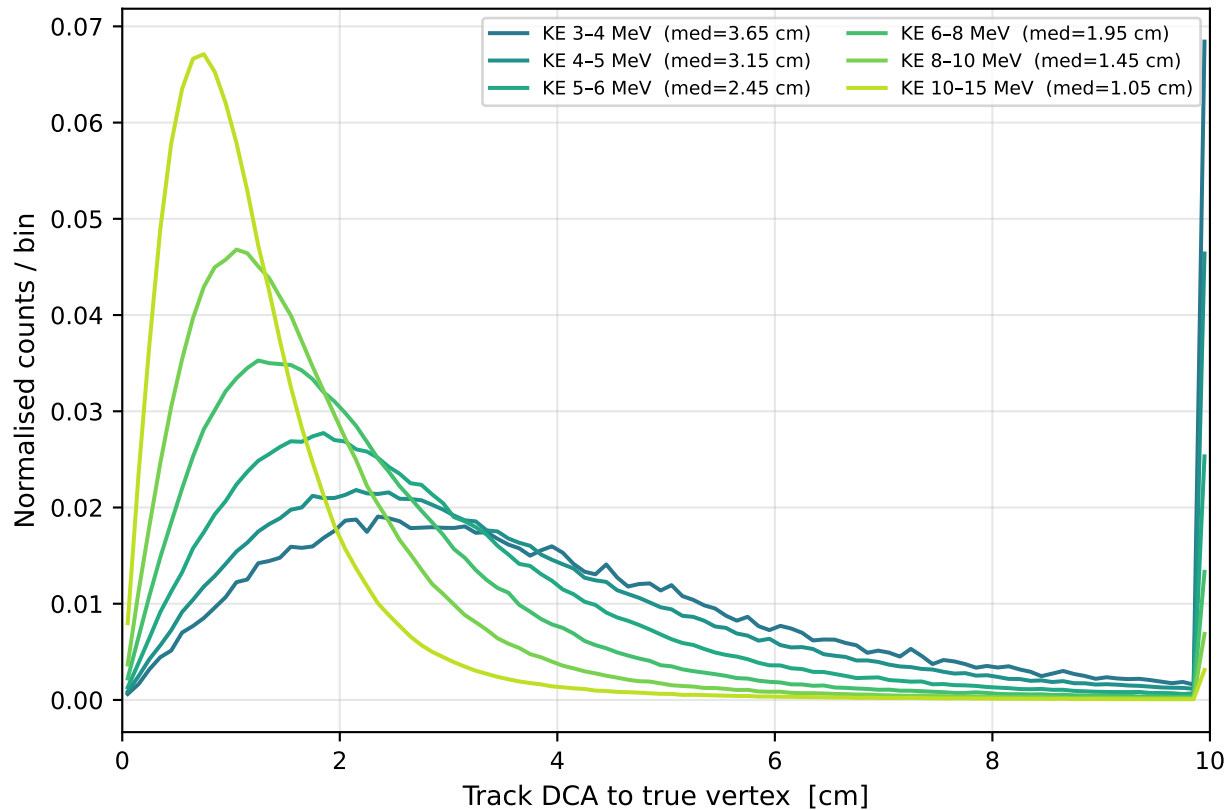


Trigger acceptance vs asymmetry (drops at high asymmetry = low-KE particle not reaching trigger)

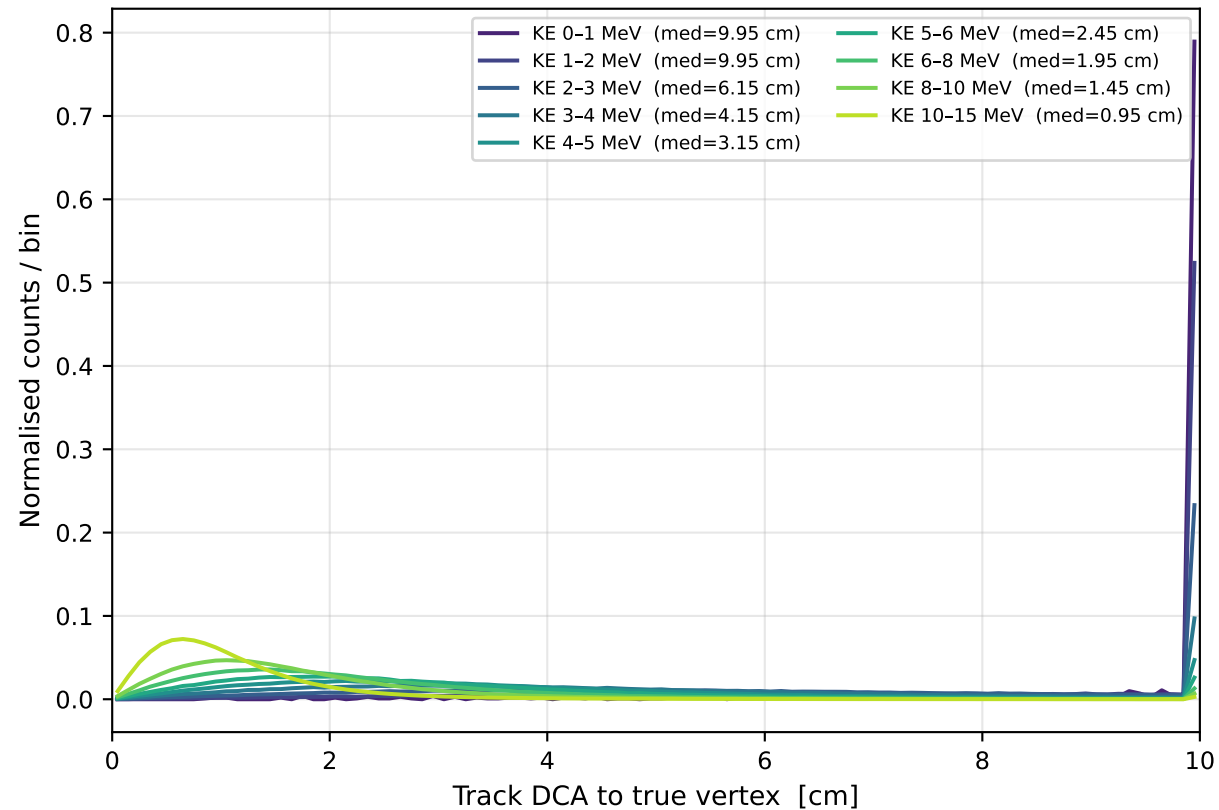


MM Track Pointing Resolution

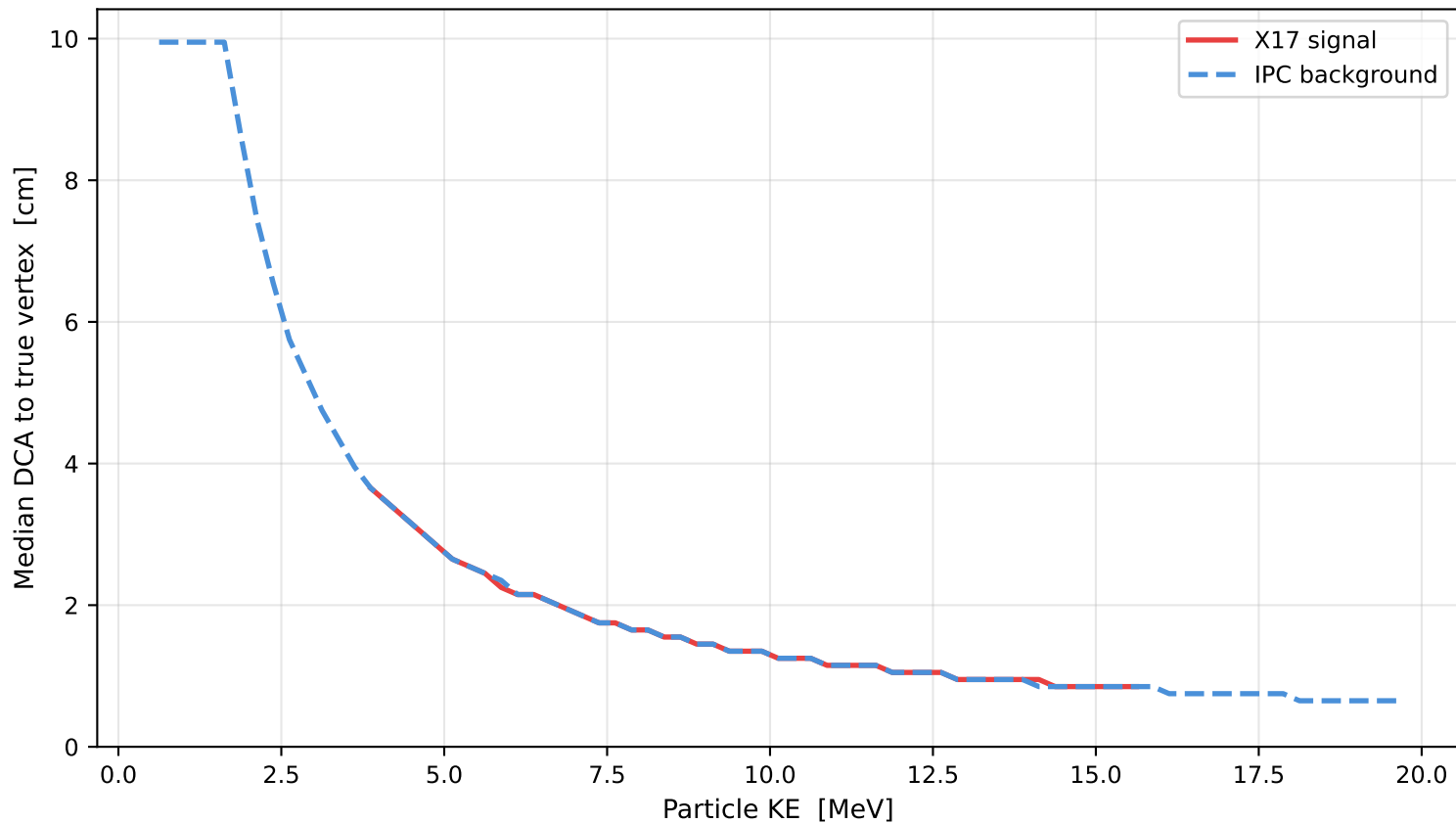
MM track pointing — X17 signal



MM track pointing — IPC background



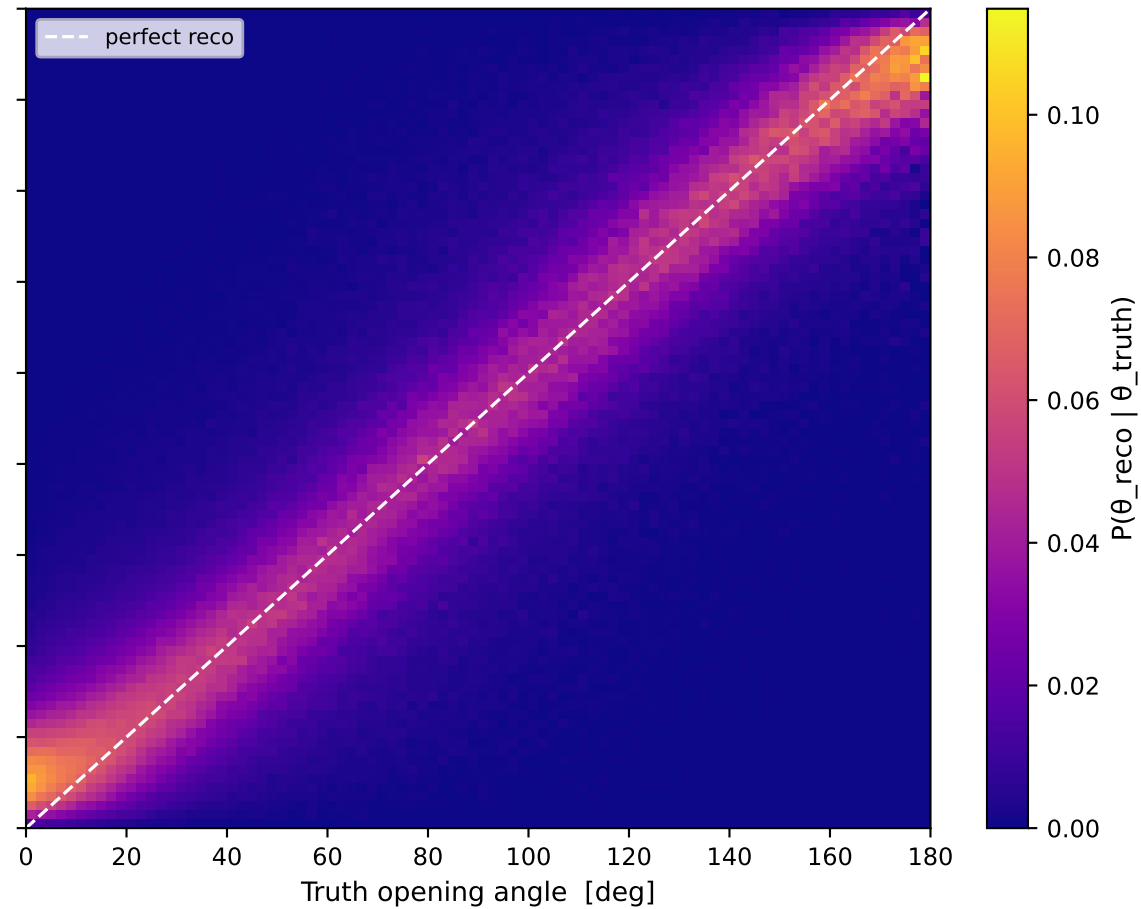
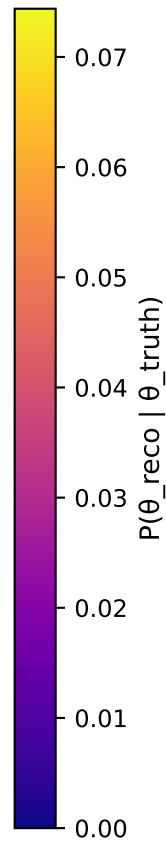
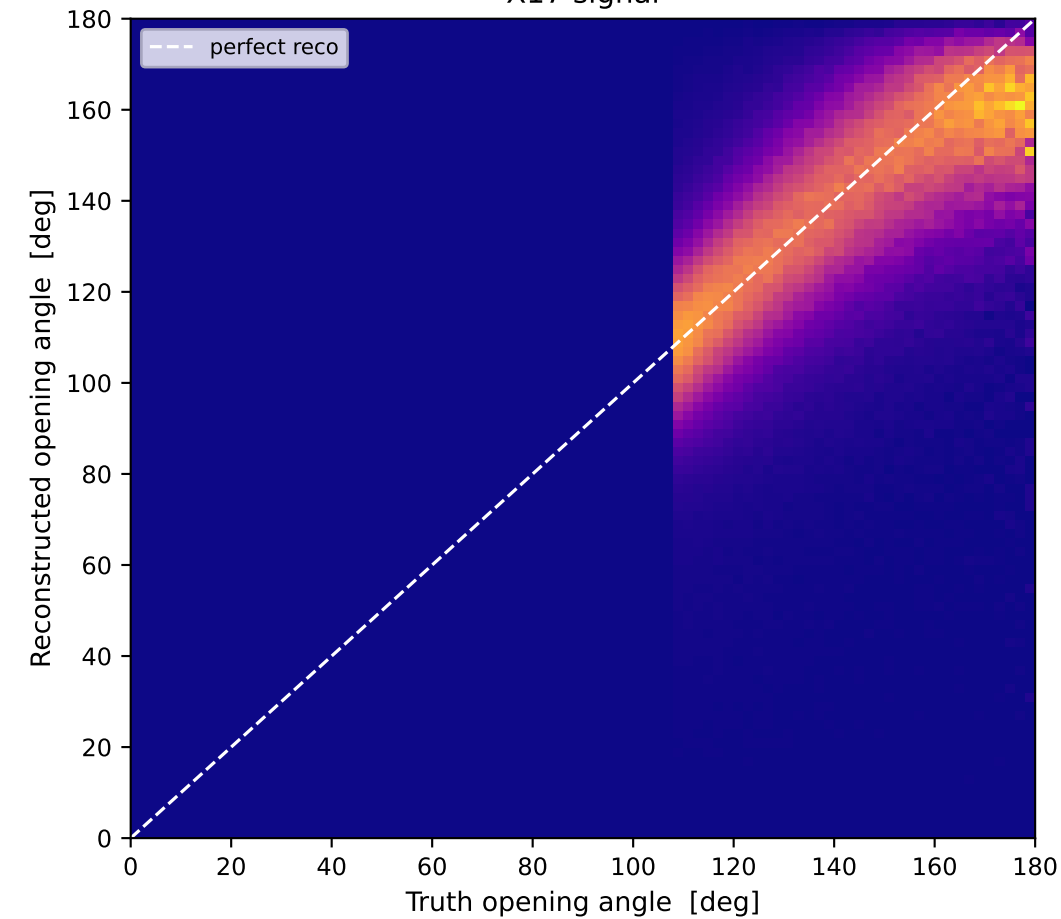
Track-to-vertex DCA vs energy
(dominated by multiple scattering in upstream material)



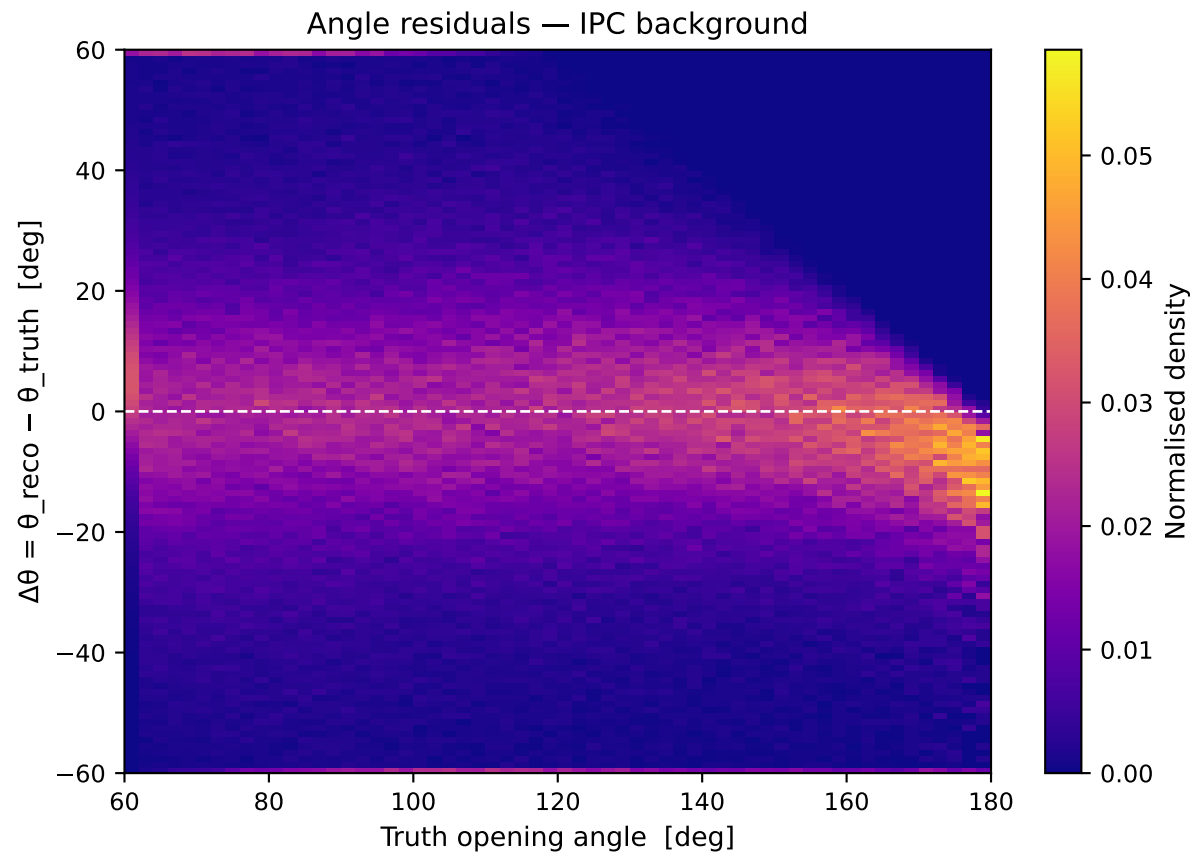
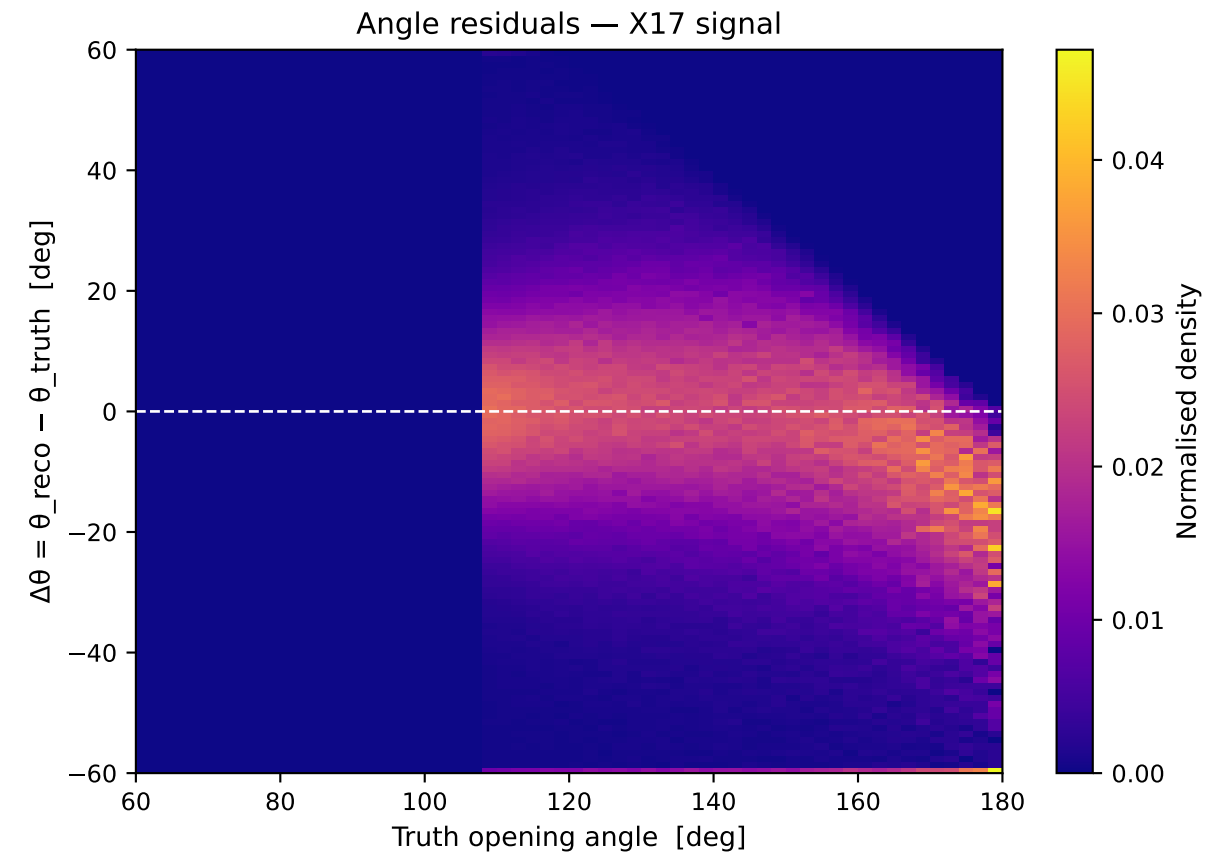
Truth vs Reconstructed Opening Angle

X17 signal

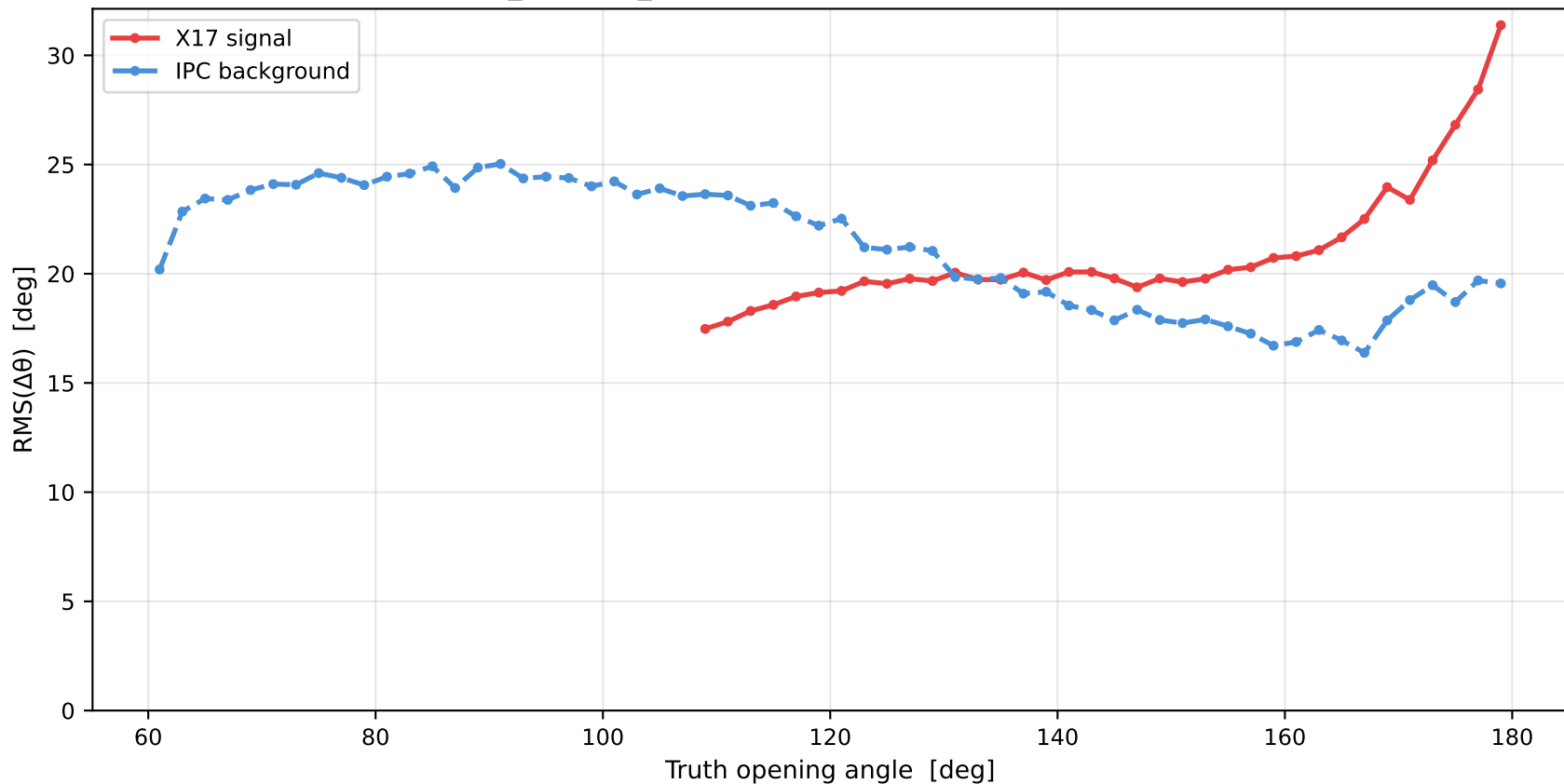
IPC background



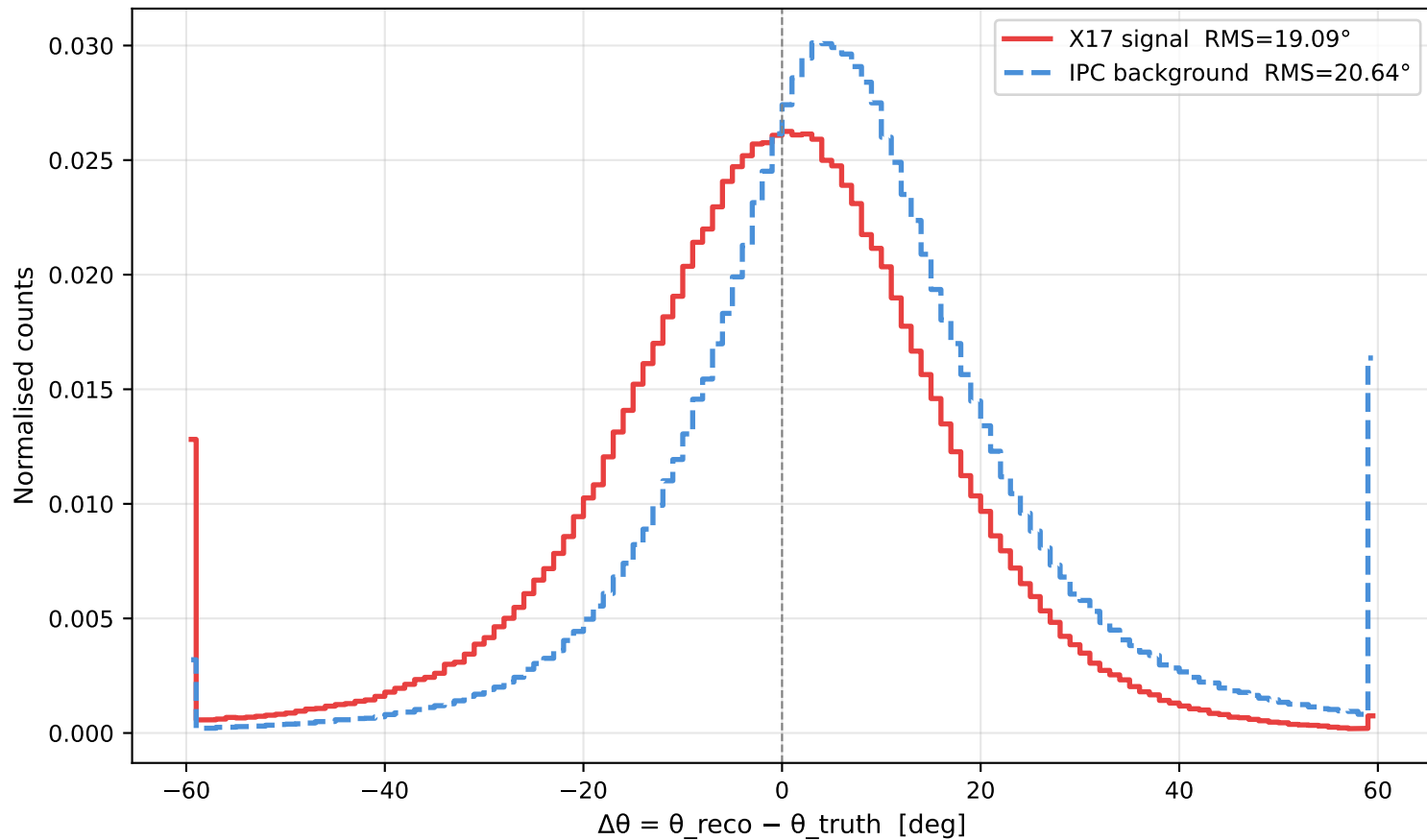
Opening Angle Reconstruction



Opening angle resolution vs truth angle
(RMS of $\theta_{\text{reco}} - \theta_{\text{truth}}$; dominated by MS in He-3 capsule + air)

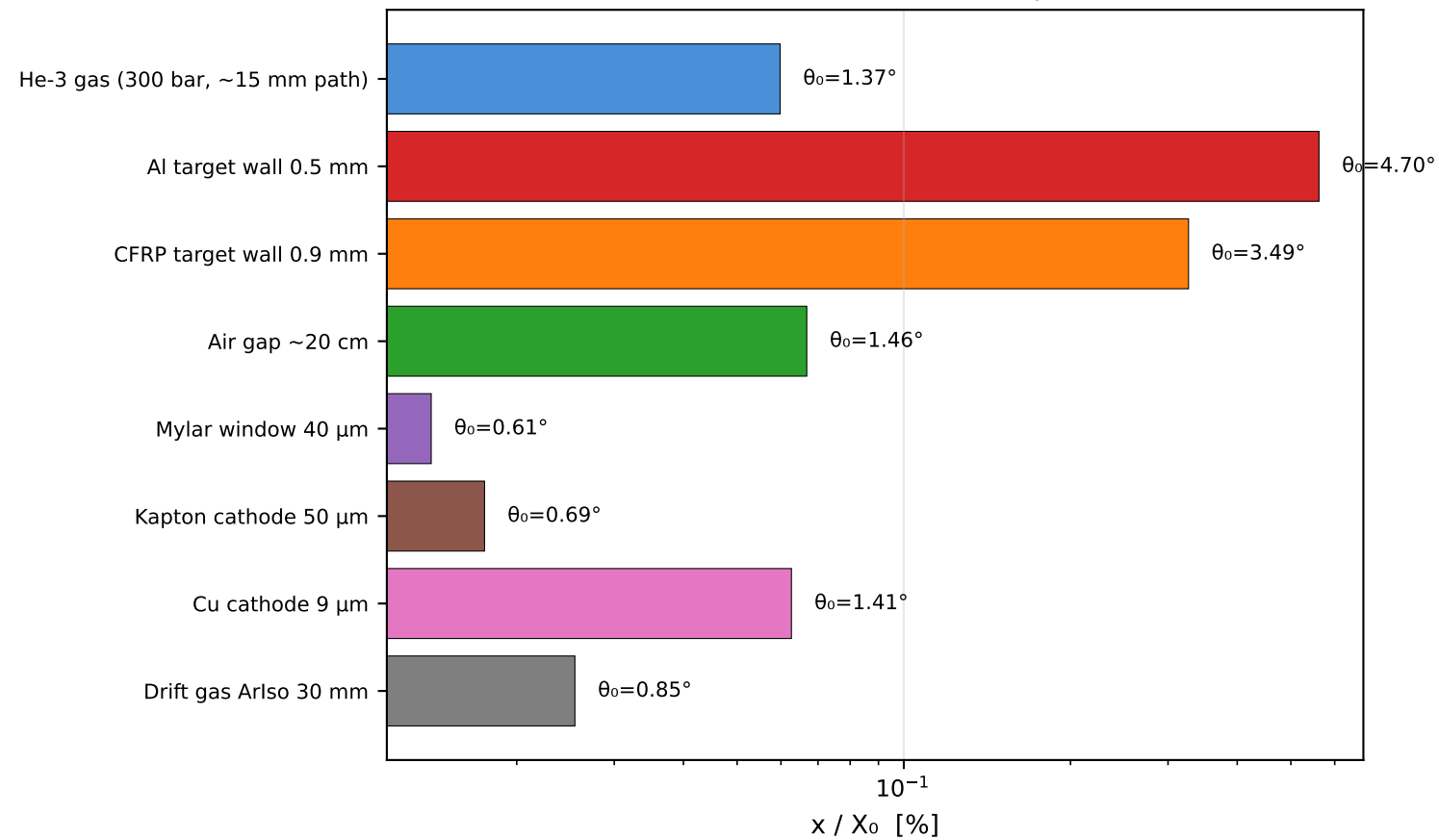


Opening angle residual (all events with MM hits)

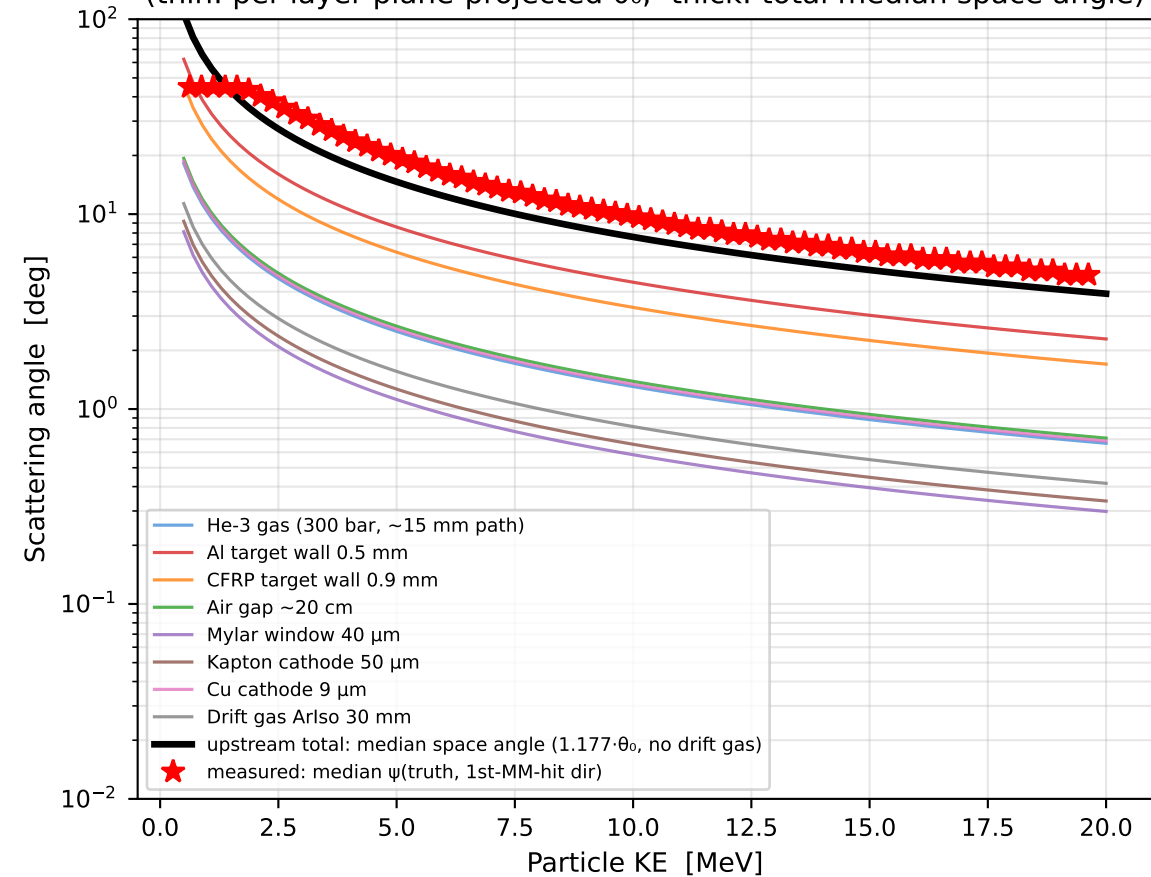


Multiple-Scattering Budget — where the angular resolution is lost

Material budget (old-data geometry)
total $x/X_0 = 1.13\%$ ($\theta_0 = 6.90^\circ$ at $p = 10$ MeV)

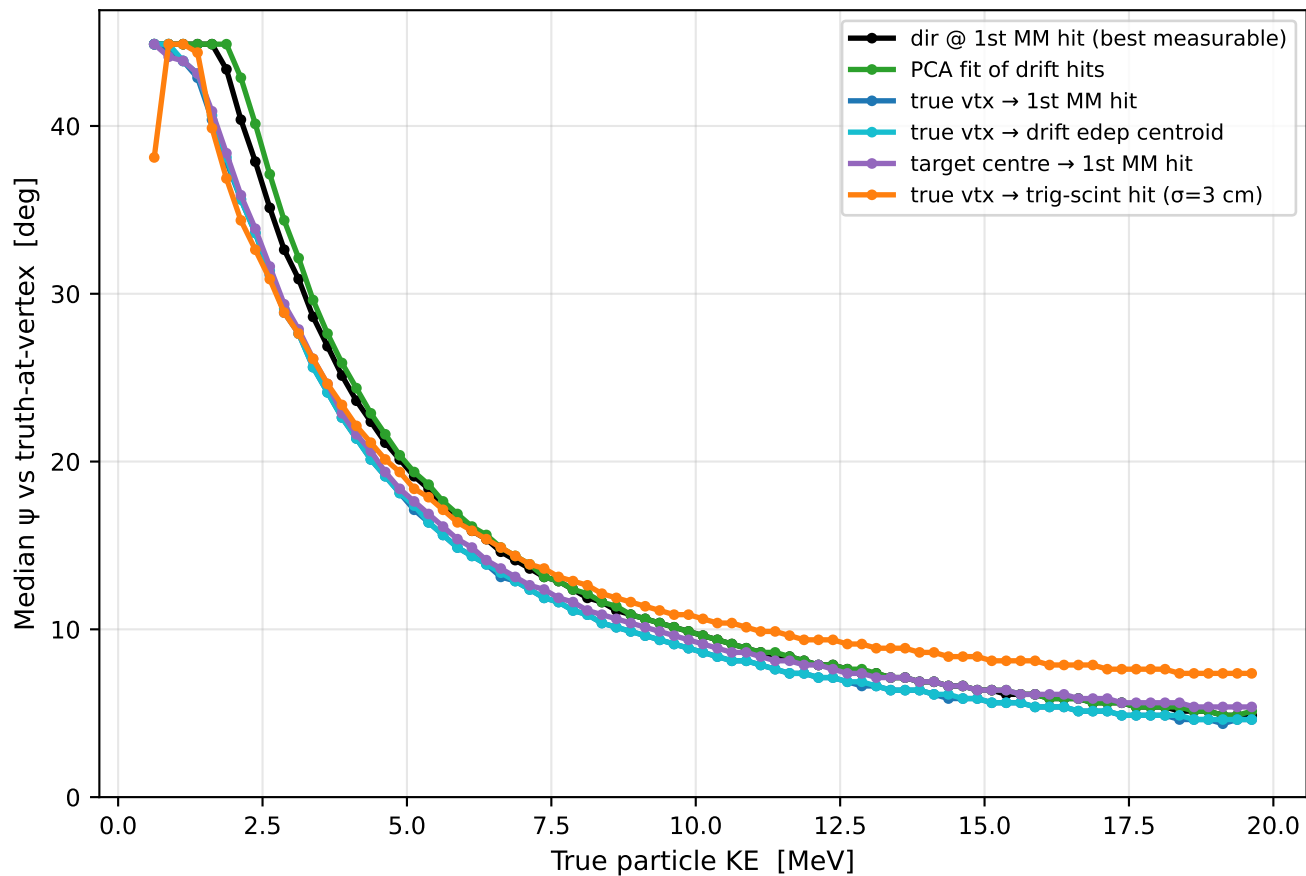


Highland prediction per layer vs measured upstream error
(thin: per-layer plane-projected θ_0 ; thick: total median space angle)

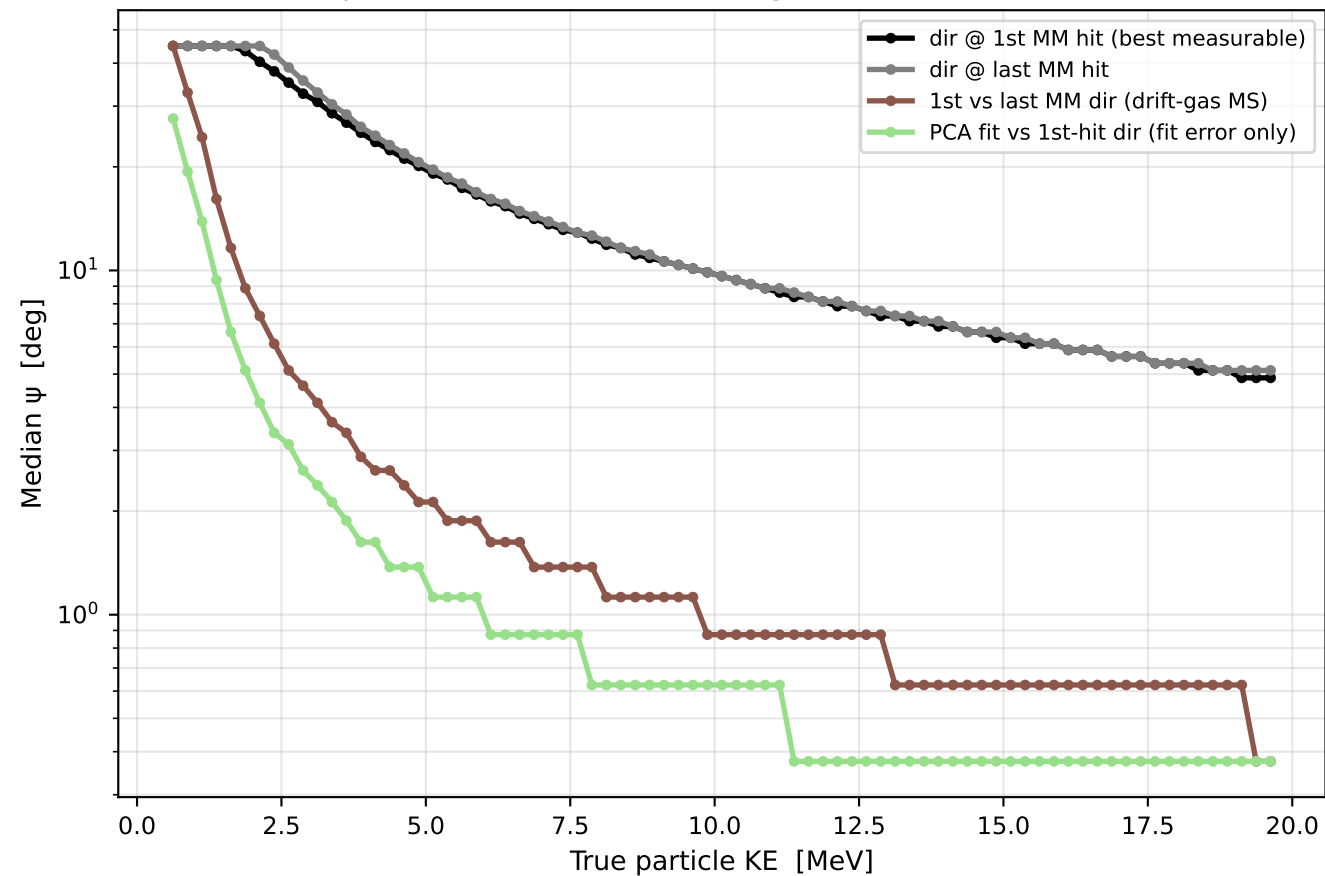


Per-Particle Direction Errors ($e^- + e^+$, X17 + IPC)

Direction estimators vs truth
(lower = better measurement of the direction at the vertex)

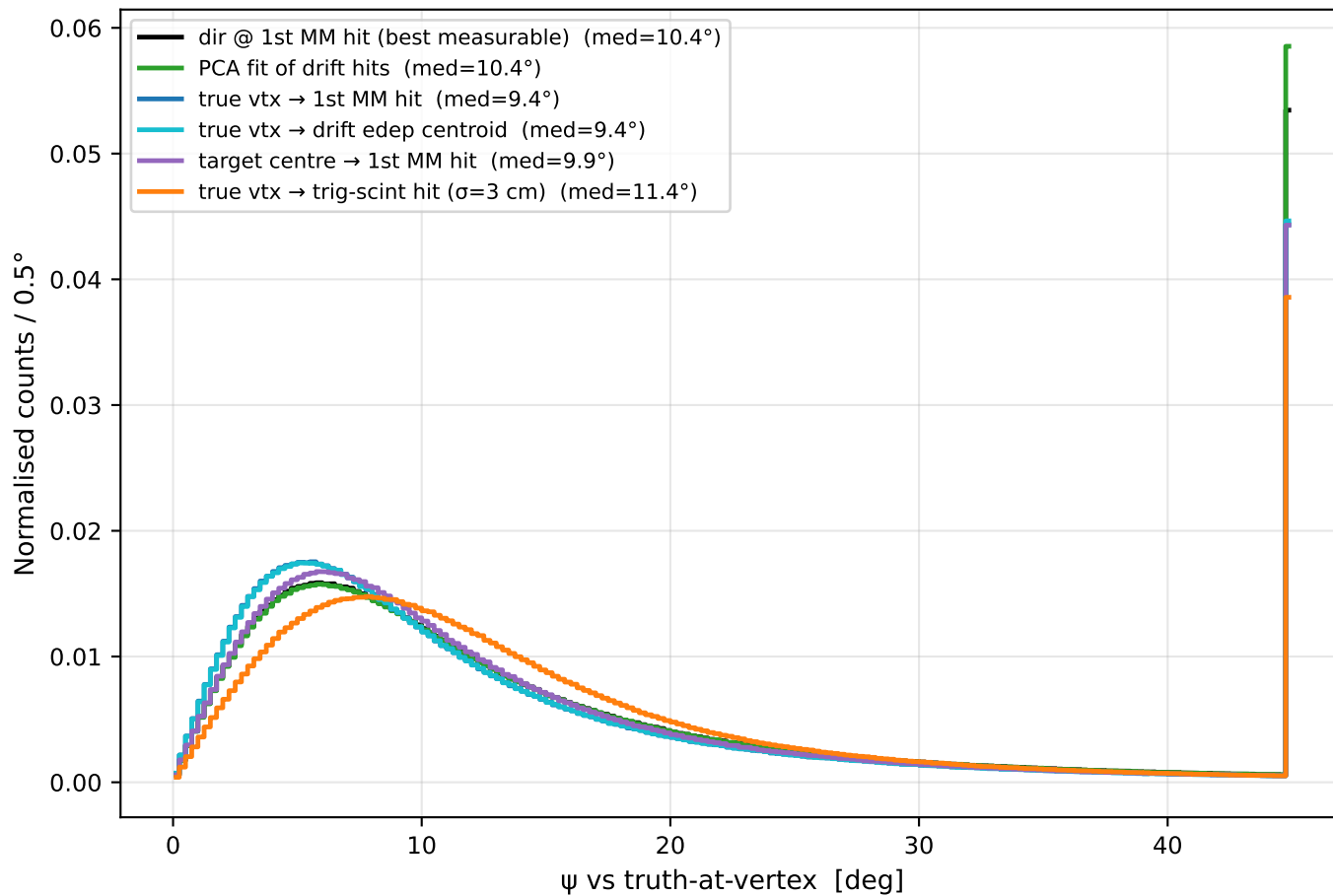


Error budget decomposition
upstream MS (black) vs drift-gas MS vs PCA fit error

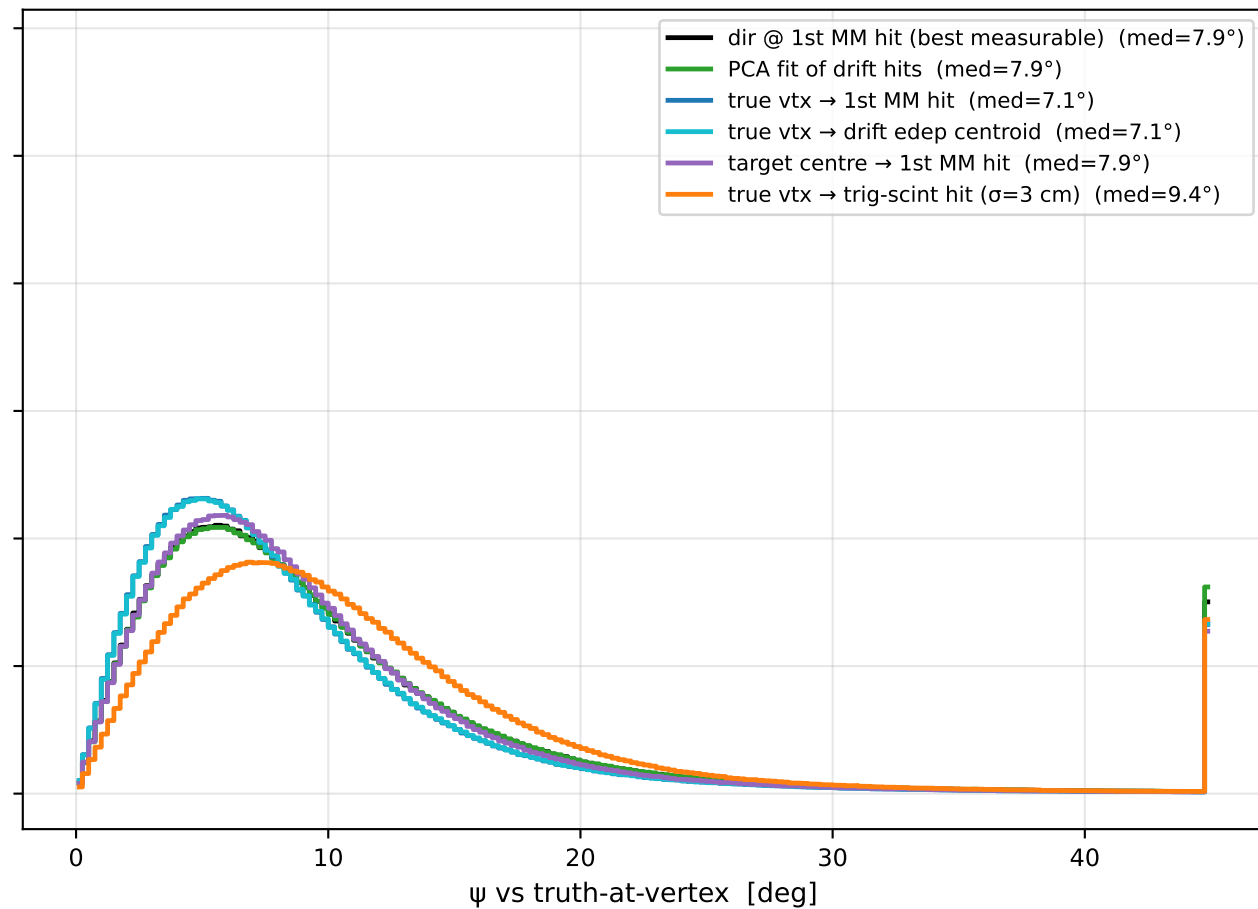


Direction-Error Distributions ($e^- + e^+$, X17 + IPC)

all KE

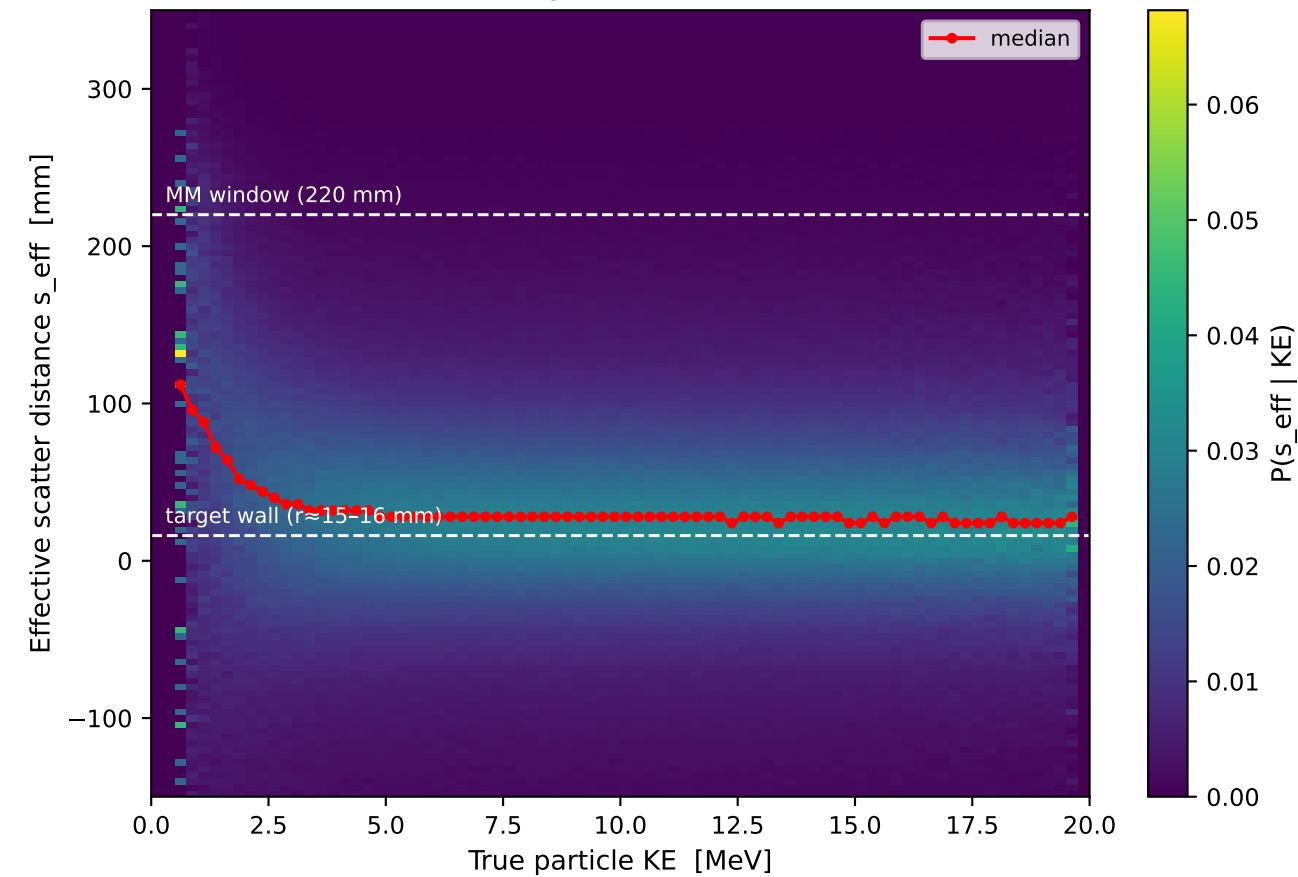


KE > 8 MeV

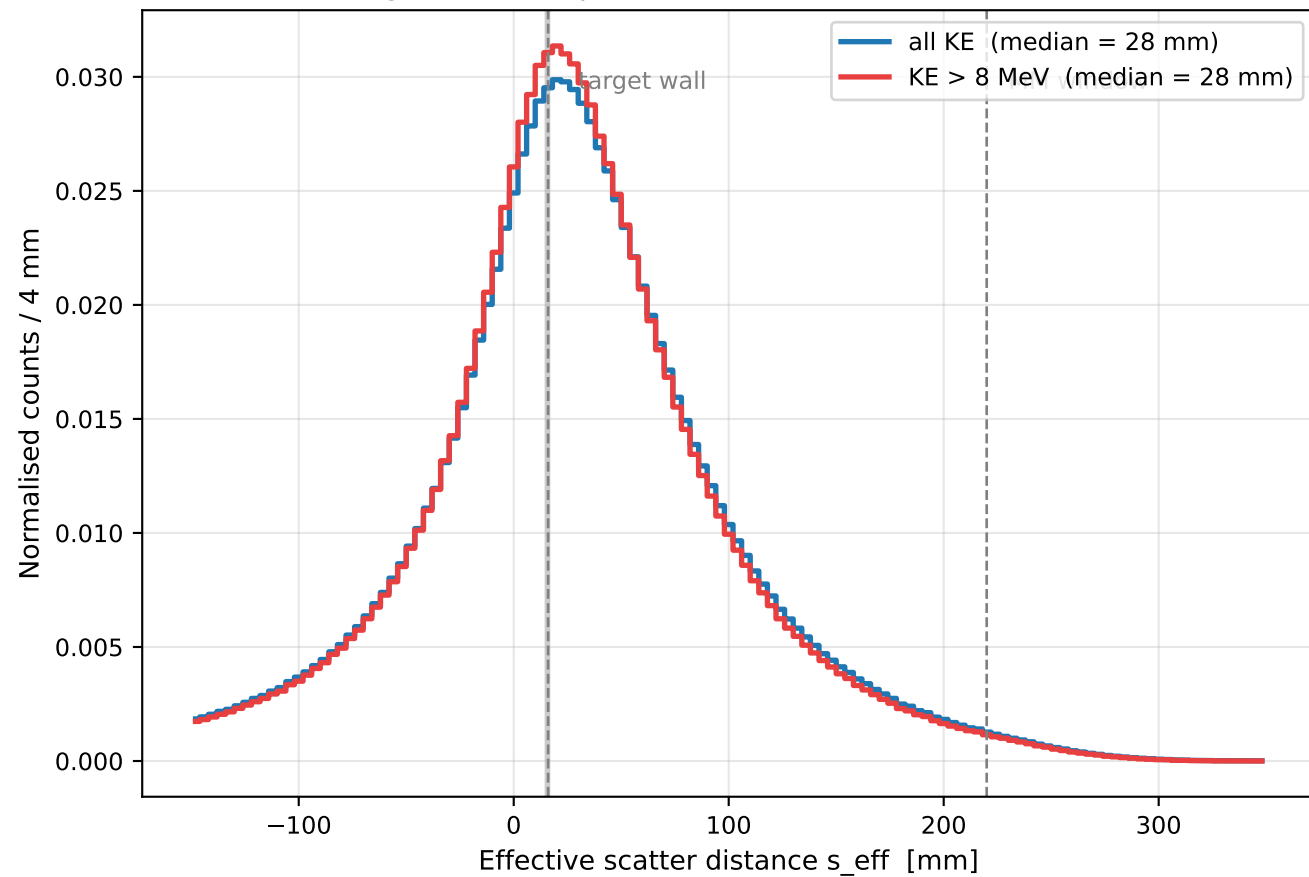


Scattering Localization ($s_{\text{eff}} = t - \delta/\tan\psi$; requires $\psi > 1.5^\circ$)

Scattering localization vs KE

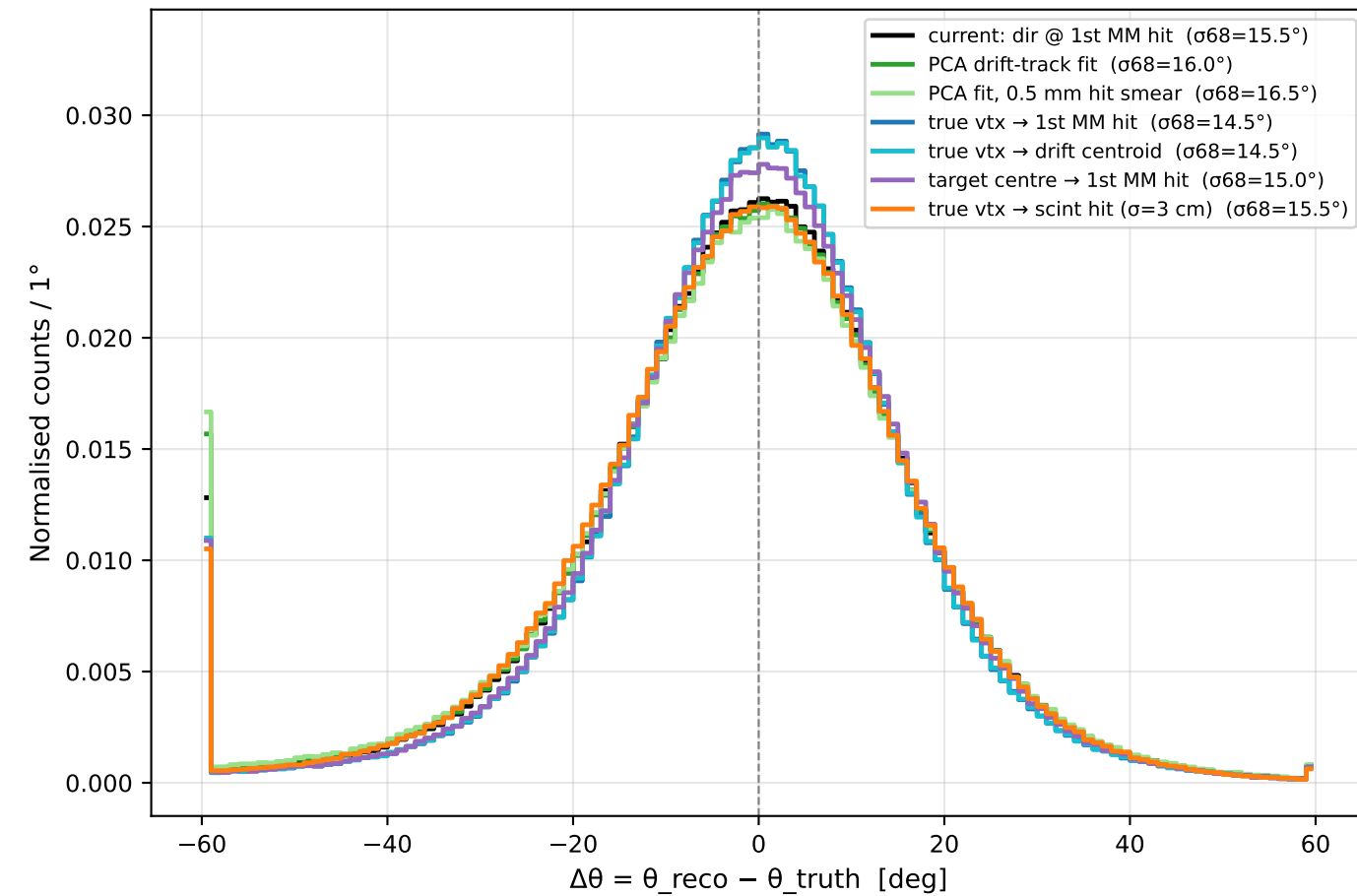


Effective scatter location
(single-scatter equivalent; distributed MS smears this)

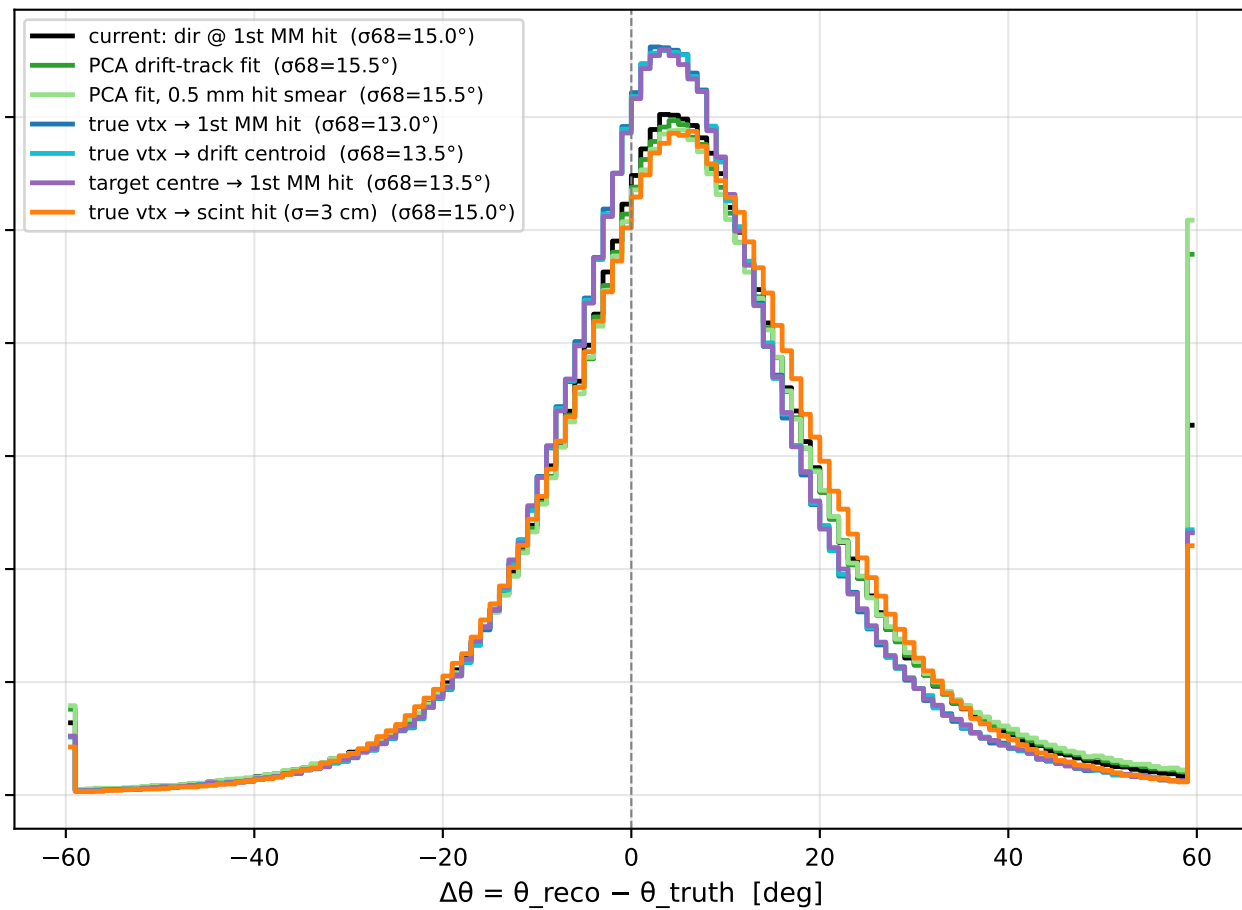


Opening-Angle Residuals by Reconstruction Method

X17 signal

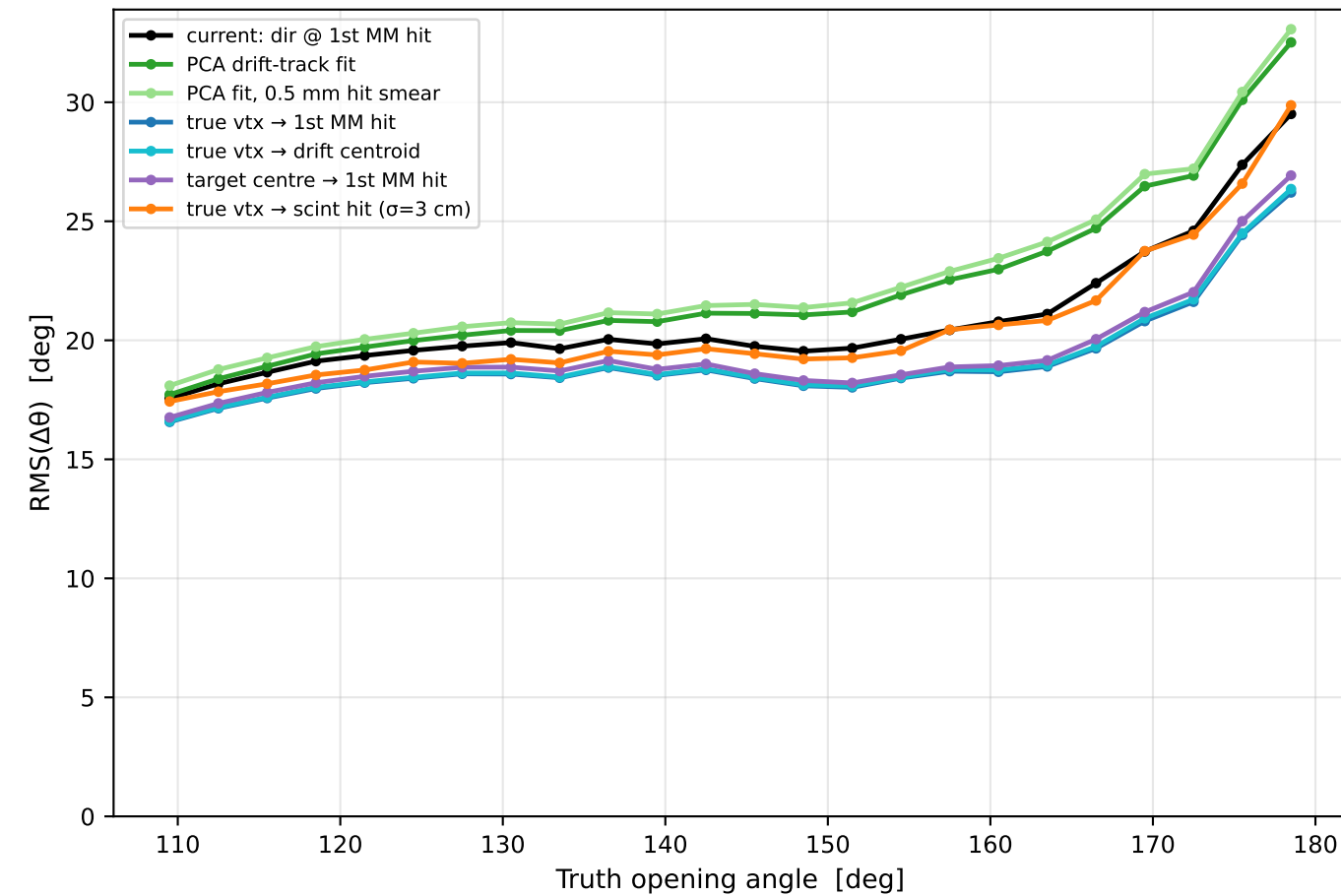


IPC background

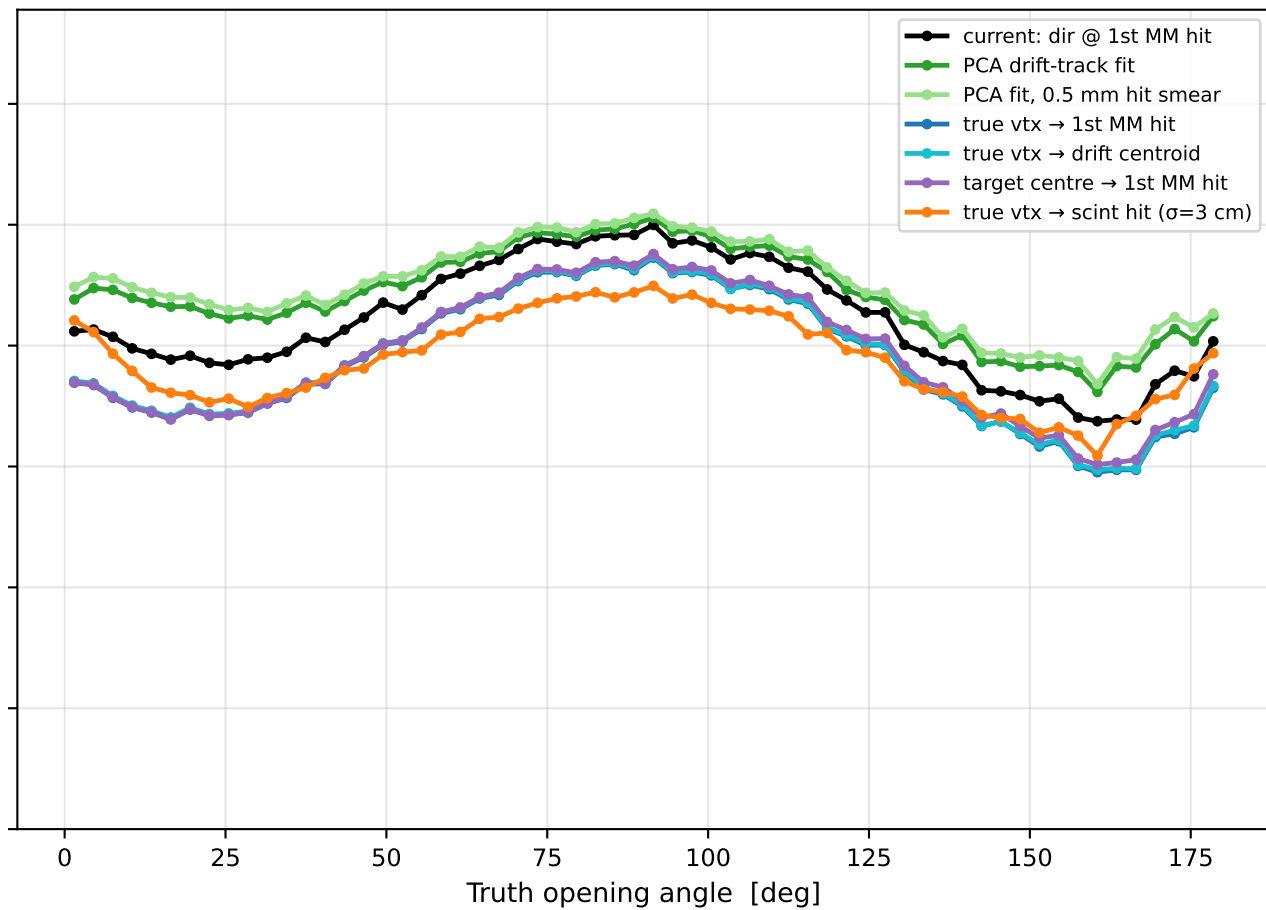


Opening-Angle Resolution vs Truth Angle by Method

X17 signal

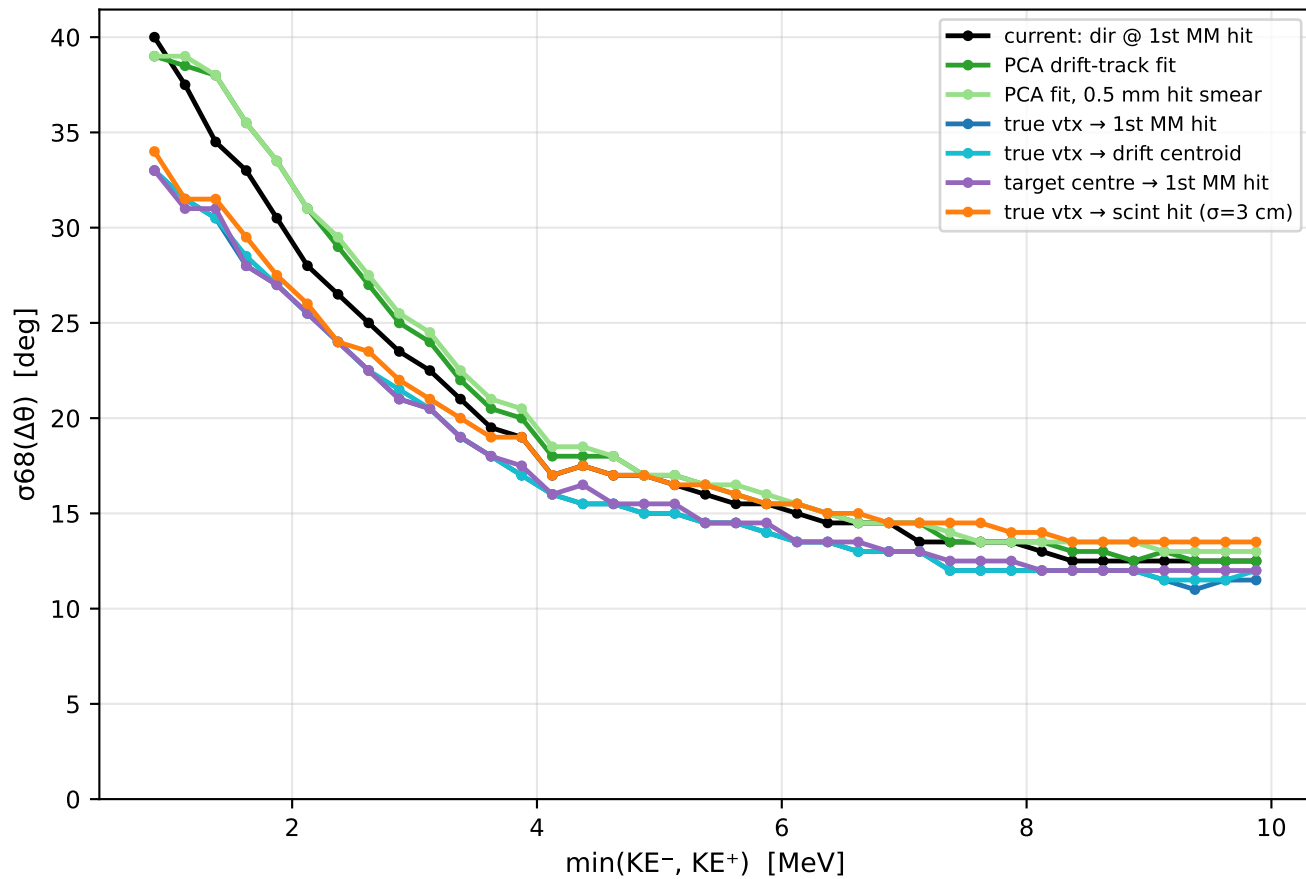


IPC background

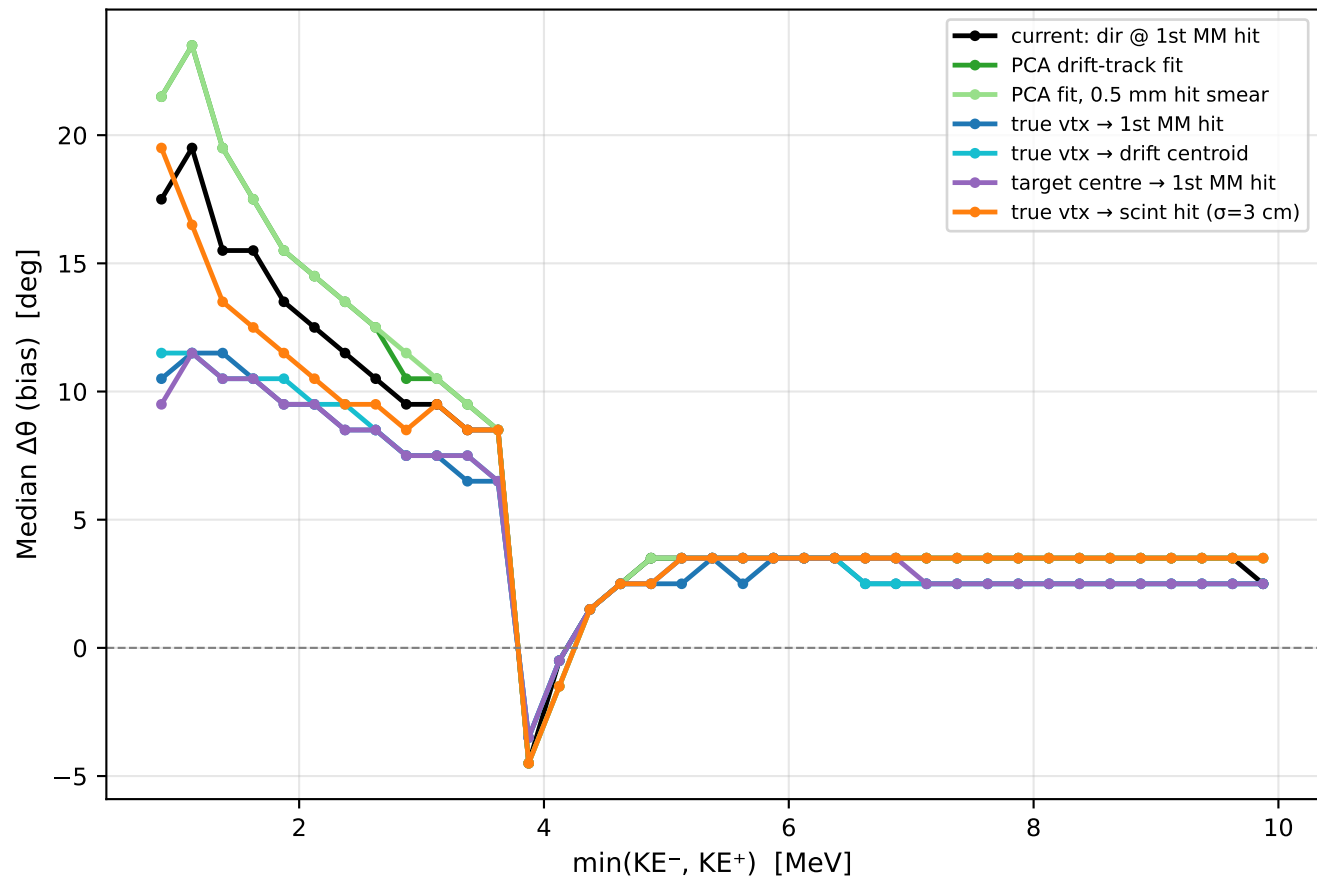


Opening-Angle Performance vs Pair Energy (X17 + IPC combined)
→ an energy cut directly buys angular resolution

Resolution vs min pair energy

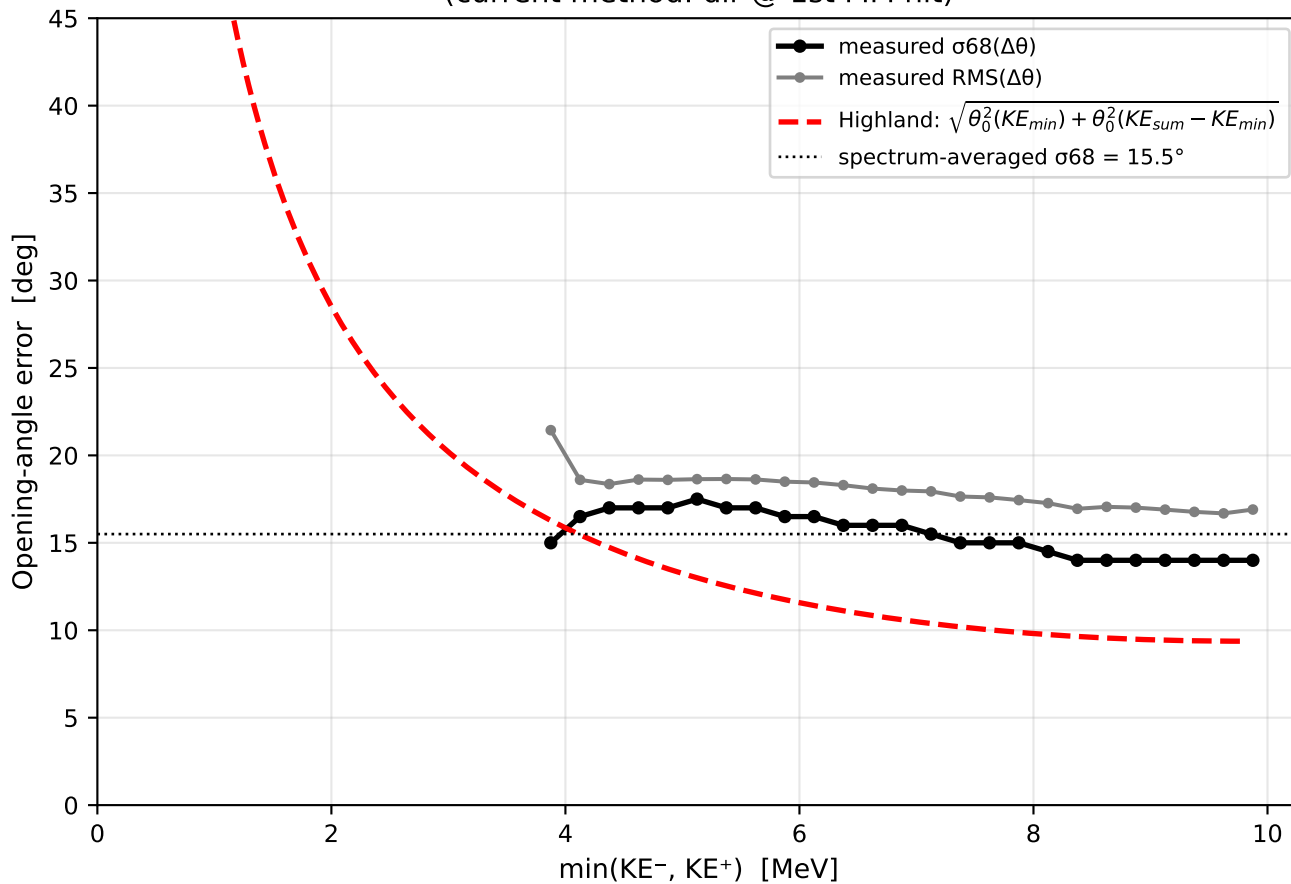


Bias vs min pair energy



Opening-Angle Residual vs Energy — budget reconciliation (X17)

X17: opening-angle residual vs pair energy
(current method: dir @ 1st MM hit)



Why 13° from a 7° budget: energy spectrum weighting

